

Traceable Threshold Encryption without a Trusted Dealer

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Abstract

The fundamental assumption in t -out-of- n threshold encryption is that the adversary can only corrupt fewer than t parties. However, this may be unrealistic in practical scenarios where shareholders could have financial incentives to collude. Boneh, Partap, and Rotem (Crypto'24) addressed the case where t or more shareholders collude, adding a traceability mechanism to identify at least one colluder. Their constructions require a trusted dealer to distribute secret shares, but it is unclear how to achieve traceability without this trusted party. Since threshold encryption aims to avoid a single point of failure, a natural question is whether we can construct an efficient, traceable threshold encryption scheme without relying on a trusted dealer.

This paper presents two dealerless, traceable threshold encryption constructions by extending the PLBE primitive of Boneh et al. (Eurocrypt'06) and combining it with the silent setup threshold encryption construction of Garg et al. (Crypto'24). Our first construction achieves an amortized ciphertext size of $O(1)$ (for $O(n)$ ciphertexts), and the second achieves constant ciphertext size in the worst case but with a less efficient preprocessing phase. Both have constant secret key sizes and require no interaction between parties.

A limitation of Boneh et al.'s constructions is that they only guarantee identifying one colluder, leaving the problem of tracing more traitors unsolved. We address this by applying a technique to our first construction that enables tracing up to t traitors.

Keywords: Threshold Encryption, Traitor Tracing, Silent Setup.

1 Introduction

Threshold encryption [24, 25] is a fundamental tool for secure encryption amongst multiple parties, as it allows generating a *succinct* ciphertext which can be decrypted by a committee of parties. At its core, semantic security holds only as long as less than t -out-of- n parties are compromised. While assuming that an adversary cannot obtain t shares seems reasonable, this model is insufficient in practice. In the real world, committee members might be incentivized to give away or sell their secret keys or decryption shares, potentially allowing an adversary to break the scheme's security. Motivated by this limitation, Boneh, Partap, and Rotem [12] recently introduced the concept of traceability in threshold encryption. They consider a decoder box built by at least t colluders that, given a ciphertext, returns its decryption. A tracer with black-box access to this decoder can identify at least one of the colluders. In [12], three constructions are given, but all rely on a trusted dealer to distribute secret shares. In many real-world uses of threshold encryption, this is a serious weakness. For example, in blockchain encrypted mempools [4, 21, 27], which inspired their work, a trusted dealer could expose private transactions and cause major financial loss.

Replacing the trusted dealer with a standard distributed key generation (DKG) protocol [39, 31, 45] does not work, since any group of t parties can compute another party's secret share. This means they could frame an honest party by including its share in the decoder box. One might try using a general MPC protocol instead, but it is much less efficient than DKG in terms of communication and computation. An open problem from [12] is to design a DKG between the parties and the tracer, where the tracer only learns what is needed for tracing, and the parties only learn what is needed for decryption. A key requirement is that a group of t parties must not be able to learn the secret shares of the remaining $n - t$

parties. Driven by the need to remove the trusted dealer and keep ciphertexts succinct, the main question we study in this paper is:

Question 1.1. *Can we construct an efficient traceable threshold encryption scheme with constant-size ciphertext that does not require a trusted dealer?*

Another limitation of the constructions in [12] comes from the techniques they use. Their approach adapts existing traitor tracing schemes to the threshold encryption setting, specifically by converting the schemes from [11] and [32]. These original schemes are made to catch only one traitor, which is the best possible in the non-threshold case. But in the threshold setting, it is possible to trace up to t traitors, since at least t colluders are needed to build a working decoder. This leads to another key open question left by [12].

Question 1.2. *Can we construct an efficient traceable threshold encryption scheme that guarantees to trace any decoder back to more than one, optimally t , colluders?*

We depart from [12] by constructing a traceable threshold encryption scheme without relying on a non-threshold base. Our approach combines Private Linear Broadcast Encryption (PLBE) [14] with silent setup threshold encryption (STE) [30]. In STE, the setup is non-interactive and dealer-free: each party independently generates a secret key and publishes a public key. Since the keys are fully independent, no coalition can derive another party’s key. We also take a first step toward addressing the second open question by formally defining the problem and presenting a construction that traces up to t colluders, achieving $O(1)$ amortized ciphertext size (and linear in the worst case).

1.1 Settings for Traceable Threshold Encryption

We start with a brief description of our collusion model as well as an introduction to the different settings discussed in this paper.

The Decoder Model. Following [12], we assume colluders create a *stateless* decoder (i.e., resettable), and tracing is performed through *black-box* access to it. The tracing algorithm must identify at least one colluder while guaranteeing that no innocent user is falsely accused. We give two constructions. The first enables *public tracing*, where anyone can trace a decoder without needing a trusted party. The second achieves shorter ciphertext in the worst case but requires *private tracing*, with a tracing key tk , so only the holder of tk can trace. Importantly, the tracer is *not* trusted with confidentiality: leaking tk does not break semantic security unless t or more parties are compromised. This sets our approach apart from [12], which needs both a trusted tracer and dealer.

We now move to describe different three encryption settings in which our constructions operate.

Amortized Setting. This setting captures applications where an intermediary entity collects and bundles ciphertexts from multiple non-mutually trusted users. We define the efficiency of the scheme with respect to the size of the batch. In such a setting, we require that ciphertexts in the batch remain secure against malicious intermediaries and other encryptors. Furthermore, an important property is that the set of users can be completely dynamic and does not need to be known to the intermediary ahead of time, i.e., any new user joining the system only needs to access some public information and does not need to interact with the intermediary or other encryptors. Informally, we denote this public information as *pub*. While there is no fixed bound on the number of ciphertexts that can use the same *pub*, we note that collusion-resilience decreases with that number, as it opens the door for “hardcoding” attacks (i.e., decoders that hardcode the specific *pub* attached to the batch and do not work for ciphertexts outside the batch). It will become apparent later in Section 2 that the tracing procedure relies on the tracer being able to manipulate *pub* to generate tracing ciphertexts, causing decoders that only work for fixed *pub* to evade tracing. Essentially, the premise is that as long as only a limited number of ciphertexts are encrypted under *pub*, a decoder that avoids being traced by hardcoding said *pub* is not particularly useful since such a decoder would only work for this particular set of ciphertexts. In the extreme, if the same *pub* were to be reused *forever*, a hardcoding decoder can decrypt all ciphertexts without risk of being traced.

In Section 6.2, we present a construction in the amortized setting that allows to reuse and amortize pub (which has size $O(n)$ and would otherwise have to be generated fresh for every ciphertext) across $O(n)$ ciphertexts, leading to an amortized ciphertext size of $O(1)$. Moreover, our scheme can identify t traitors in public tracing.

Epoch-based Setting. The epoch setting can be viewed as a more formal variant of amortization that is carried out in fixed time intervals: A single key is generated at the beginning of each epoch, and similarly, the set of encryptors/ciphertexts in each epoch is entirely dynamic. This setting captures many practical applications, such as threshold-encrypted mempools for MEV prevention [16, 21, 27]. We require that some entity (henceforth called the *epoch leader*) generate an epoch key (essentially a new instance of pub) at the beginning of each epoch. Ciphertexts are encrypted under that epoch key during the epoch only. Security holds even against malicious epoch leaders. This effectively mitigates the hardcoding decoder attack because decoders that hardcode an epoch key and refuse to work for different epoch keys become useless at the start of the next epoch. As long as colluders cannot predict or influence future epoch keys, a decoder that works for future ciphertexts (i.e., ciphertexts that are generated at least one epoch after the decoder’s creation) cannot evade tracing. In other words, decoders that evade tracing by hardcoding the epoch key they work for are only useful until the end of the epoch. We stipulate that updating an illicit decoder every single epoch requires interaction between a hypothetical “buyer” and colluding committee members, which significantly increases the threat of the collusion getting exposed.

We achieve a similar tradeoff between efficiency and the strength of anti-collusion guarantees as in the amortized setting. With longer epoch intervals, leader election and epoch key generation occur less frequently, improving efficiency. However, decoders that hardcode an epoch key are useful for a longer period, weakening anti-collusion guarantees. Conversely, shorter epoch intervals cause decoders that hardcode epoch keys to become useless quickly, strengthening anti-collusion guarantees. This comes at the cost of more frequent leader election and epoch key generation, reducing efficiency. In Section 8, we present a construction in the epoch-based setting with ciphertexts of size $O(1)$ that is able to identify t traitors.

Preprocessing-based Setting. In this setting, after the parties publish their public keys, a non-interactive preprocessing phase takes place, during which an entity (trusted only for traceability) computes a set of encryption keys derived only from the public keys. Once this set is published, any party can encrypt messages independently. This setup occurs only once and can be used for all subsequent encryptions. In Section 7, we present a construction with a non-interactive preprocessing phase in which the size of the ciphertext is $O(1)$ in the worst case and the size of the public key is $O(n^4)$. As a tradeoff, this construction achieves only private tracing.

1.2 Summary of Our Contributions

Our contributions can be summarized as follows.

- *Traceable Silent Threshold Encryption (TSTE):* We propose a new approach for traceable threshold encryption that avoids reliance on a trusted dealer or interaction between parties. We also extend this model with a preprocessing phase to enable shorter ciphertexts.
- *New Building Block:* We define *Oblivious Silent Threshold Encryption (OSTE)*, a new cryptographic primitive that extends Silent Threshold Encryption (STE) [30] to support tracing. Our definition relaxes certain requirements compared to a prior primitive, which is *Private Linear Broadcast Encryption (PLBE)* [14], to achieve more efficient constructions. We show how these relaxed guarantees still allow us to realize a traceable silent threshold encryption scheme (TSTE).
- *New Constructions:* We present two TSTE constructions based on [30]. Unlike [12], our schemes require neither a trusted dealer nor interaction, yet achieve comparable parameters. Most notably, constant-size ciphertexts and secret keys. See Figure 1 for a comparison.
- *Tracing Multiple Traitors:* We introduce a formal technique for tracing up to t colluding parties and apply it to our baseline construction.

Construction	$ \text{sk} $	$ \text{pk} $	$ \text{ct} $	No trusted dealer?	Public tracing?	Traitors found
Naive [12]	$O(1)$	$O(n)$	$O(n)$	✓	✓	t
First [12]	$O(n^2)$	$O(n^2)$	$O(1)$	✗	✗	1
Second [12]	$O(n^2)$	$O(1)$	$O(1)$	✗	✗	1
Third [12]	$O(1)$	$O(n^{1/3})$	$O(n^{1/3})$	✗	✗	1
This paper: Sec. 6.2	$O(1)$	$O(n^2)$	$O(n)$ worst case $O(1)$ amortized ¹	✓	✓	t
This paper: Sec. 7	$O(1)$	$O(n^4)$	$O(1)$	✓	✗	1

Figure 1: Our constructions compared to [12]. The comparatively best results are highlighted in green.

1.3 Additional Related Work

To the best of our knowledge, [12] is the first construction of traceable threshold encryption, which was subsequently extended to achieve CCA-security [7] but without addressing the trusted dealer problem. Other recent works study collusion resistance in secret sharing [13, 26, 35], but it is unclear how to extend them to traceable threshold encryption schemes.

Traitor tracing, introduced by Chor et al. [20], was originally proposed for piracy protection. Since then, extensive research has focused on broadcast encryption, with an emphasis on minimizing ciphertext size (see [46] for a survey). Notably, Zhandry [47] recently constructed a pairing-based traitor tracing scheme with all parameters independent of the number of users. Similar goals have been achieved using LWE [33, 19] and iO-based techniques [15, 34].

The most relevant to our work is the non-threshold traitor tracing scheme by Branco et al. [17], which removes the need for a central authority. Their approach relies on pairing-based registered functional encryption (RFE) [28], similar in spirit to silent setup [30], where parties generate keys independently and publish auxiliary data. They reduce tracing to constructing quadratic RFE with sublinear ciphertext. Related QFE constructions also appear in [23, 48]. Luo [37] also addresses tracing without a central authority, though their scheme relies on iO and is thus less practical.

2 Technical Overview

In this section, we discuss the main ideas and techniques in this paper.

We present a traceable threshold encryption scheme by adapting Private Linear Broadcast Encryption (PLBE) [14] to the silent setup threshold model (STE) of [30]. Like PLBE, our construction supports a tracing mechanism based on revocation, but it must operate in a more challenging dealer-free threshold setting with non-interactive setup. Furthermore, we present a preprocessing-based construction in which proving traceability requires a more careful analysis of the tracing algorithm in [14].

2.1 Recap of PLBE and STE

Traitor Tracing from PLBE. Briefly, PLBE includes two encryption methods: **Enc** and **TrEnc**. The **Enc** method is used for standard encryption, while **TrEnc** produces ciphertexts for tracing. The **TrEnc** algorithm takes an index $i \in \{0, \dots, n\}$ as input and encrypts the message such that only parties with index $> i$ can decrypt it. During tracing, the tracer queries the decoder with $\text{TrEnc}(0), \text{TrEnc}(1), \dots, \text{TrEnc}(n)$ and considers party $i \in [n]$ to be a traitor if the decryption probability drops significantly when i is revoked. Tracing in PLBE relies on three core properties:

- Indistinguishability: **Enc** and **TrEnc**(0) must be indistinguishable except with negligible probability.
- Index hiding: An adversary lacking sk_i can distinguish between $\text{TrEnc}(i-1)$ and $\text{TrEnc}(i)$ only with negligible probability.

¹This amortization cannot be achieved in the naive construction. We elaborate on this in Section 2.3.

- Message privacy: Ciphertexts encrypted via $\text{TrEnc}(n)$ (i.e., all parties are revoked) must be undecryptable except with negligible probability.

These properties enable tracing because initially, for $\text{TrEnc}(0)$, the decoder’s success probability must be as high as for normal decryption due to indistinguishability, but in the end, for $\text{TrEnc}(n)$, it must be negligible due to message privacy. Hence, for at least one i , the success probability must drop significantly. Index hiding ensures it can only drop if the decoder knows sk_i , so the identified party must be a colluder.

Silent Threshold Encryption (STE). In the silent-setup model, parties non-interactively publish their public keys with respect to a common reference string (CRS). Specifically, the i -th party samples a key pair $(\text{sk}_i, \text{pk}_i)$ and computes a hint hint_i using the CRS, publishing $(\text{pk}_i, \text{hint}_i)$. An *encryption key* ek can be computed from the set of public keys $\text{pk}_1, \dots, \text{pk}_n$, allowing any party to encrypt messages. To enable decryption, an *aggregation key* ak is derived from the hints $\text{hint}_1, \dots, \text{hint}_n$. The key ak ensures that decryption shares from any subset of at least t parties (the threshold) can be combined to recover the plaintext.

Importantly, the encryption key ek can also be computed with respect to a specific subset of parties S (i.e., using only the public keys of parties in S). In this case, only the parties in S can participate in the decryption. Thus, there can be many such encryption keys, each corresponding to a particular subset S (referred to as a *universe* in [30]).

This model is referred to as a *silent setup* since parties do not interact during the setup phase, eliminating the trusted dealer or distributed key generation protocol otherwise required in threshold encryption. In [30], a pairing-based STE construction is presented, which requires a CRS (specifically, a powers-of- τ CRS). This construction achieves constant-size ciphertexts and secret keys, with public information size of $O(n^2)$. The security is proven in the Generic Group Model (GGM) [44, 38]. Since our OSTE constructions are based on this STE scheme, our security proofs also rely on the GGM.

Although our results are based on a combination of techniques from [14] and [30], we emphasize that this integration is far from straightforward. It involves several non-trivial challenges, both conceptual and technical. We now provide an overview of our techniques and building blocks, highlighting how we address these challenges.

2.2 A New Cryptographic Primitive: OSTE

As part of our contribution, we introduce *Oblivious Silent Threshold Encryption* (OSTE), which extends STE [30] with revocation functionality in the spirit of Private Linear Broadcast Encryption (PLBE) [14]. A key novel technical contribution in the definition of OSTE is introducing *relaxed* versions of the PLBE guarantees alongside generalizations that allow us to trace t traitors instead of one. We give further details in the following.

In OSTE, a ciphertext ct is encrypted to a set S such that any entity not in possession of the secret key sk_i is oblivious to whether the i -th party belongs to the set S or not. A core property of OSTE is *membership hiding*, which formalizes this notion of obliviousness and generalizes the index hiding property of PLBE. This property is especially useful for tracing, where a tracer can revoke parties and apply a tracing algorithm similar to that of PLBE. The OSTE definition introduces the TrEnc algorithm for handling tracing queries. Informally, TrEnc takes as input a message m and a set of revoked parties $\mathcal{R}$, and encrypts m to the set $[n] \setminus \mathcal{R}$. Additionally, the message privacy requirement is adapted to the threshold setting. Specifically, in OSTE, message privacy requires that a ciphertext is undecryptable if fewer than the threshold number of parties are unrevoked.

A notable difference from PLBE lies in the membership hiding requirement (which generalizes the index hiding property in PLBE). Unlike in PLBE, where index hiding is defined relative to a fixed revocation order $A = \{i_1, i_2, \dots\}$, our setting allows arbitrary revocation orders. This generalization enables tracing of multiple traitors, as elaborated in Section 5.2.

Another notable aspect of our OSTE definition is that it supports *relaxed* versions of the core PLBE guarantees: indistinguishability and index hiding (generalized to membership hiding). We next discuss the *relaxed* indistinguishability and index hiding requirements.

In standard PLBE, indistinguishability and index hiding must hold against any adversary except with negligible probability. However, we show that this is too strong for our second construction (Section 7), which uses preprocessing to reduce ciphertext size. Hence, in OSTE, we relax these conditions to allow an adversary to produce decoders with up to a fixed (non-negligible) value δ . Namely, we define δ -Membership

hiding and δ -Indistinguishability. In the corresponding games, we require that any adversary has only *negligible* probability to produce a decoder with advantage greater than δ . We continue with a more detailed explanation of the nuances of our decoder-based definitions.

Decoder-based Definitions. The original PLBE definitions from [14] are insufficient for proving traceability, as they define traceability using adversarial decoders, but treat indistinguishability, index hiding, and message privacy via stateful adversaries. Goyal et al. [33] identified an attack exploiting this semantic gap and introduced decoder-based definitions to align these properties with traceability. Following their approach, we extend our OSTE definitions to include decoder-based variants and use them to prove traceability.

Specifically, in our δ -indistinguishability definition, the adversary must construct a stateless δ -decoder that distinguishes between **Enc** and **TrEnc** ciphertexts with advantage at least δ . The adversary chooses the message and all parties' keys. The decoder then receives a ciphertext and must guess whether it was generated by **Enc** or **TrEnc** with an empty revoked set.

For δ -membership hiding, the adversary constructs a stateless δ -decoder that distinguishes whether a ciphertext was encrypted with or without revoking a specific party i , again with advantage at least δ . The adversary selects i , the message, and all secret keys, except for sk_i .

Note that an adversary wins the δ -indistinguishability or δ -membership hiding games if and only if it produces a valid δ -decoder with *non-negligible* probability.

We formally define OSTE in Section 4.

A Traceable Silent Threshold Encryption from OSTE. In Section 5.1, we show how to compile any OSTE scheme into a traceable threshold encryption scheme (TSTE), following the transformation from PLBE to traitor tracing. The main technical challenge lies in proving traceability under the *relaxed* PLBE properties. In Theorem 5.7, we prove that despite these relaxations, the resulting scheme still achieves standard traceability, i.e., the adversary's advantage in the traceability game remains negligible. Specifically, we show that any adversary breaking traceability with *non-negligible* advantage must also succeed in the δ -membership hiding or δ -indistinguishability games with non-negligible advantage for some δ defined in the theorem. This is achieved via a more refined analysis of the tracing algorithm of [14] with respect to the relaxed definitions and elevated to the silent-setup threshold setting.

2.3 Π_{OSTE} : A Baseline Construction

We now present our baseline OSTE construction. This scheme achieves an $O(1)$ *amortized* ciphertext size, though ciphertexts may be $O(n)$ in the worst case.

Starting Point. Recall that a feature of the STE model is that an encryptor can dynamically select the decryption committee S (so-called *universe* in [30]) by computing an ek that corresponds to S . We leverage this flexibility to implement **TrEnc** by excluding parties (i.e., revoking them) from the decryption set.

The Challenge. However, a central limitation is that the aggregation key ak , which is specific to S , can be computed only if S is known. In our context, for tracing purposes, we require *membership hiding*: the ciphertext must not reveal whether a particular party i is in S or not. Thus, a core challenge is to construct **TrEnc** without leaking which parties are revoked.

Modifications to the STE of [30]. To achieve this, we present modifications to the STE scheme in Section 6. Specifically, we rerandomize all public keys: for parties in S , we set $\text{pk}'_i \leftarrow \alpha \text{pk}_i$; for parties not in S , we set $\text{pk}'_i \leftarrow \bar{\alpha}_i \text{pk}_i$. In other words, all unrevoked parties share the same masking factor α , while revoked parties receive distinct random masks. The ek and ak are then computed with respect to these rerandomized keys. To allow decryption, the ak is included in the ciphertext. We carefully adjust the encryption of [30], accounting for the rerandomization such that any party in S can still use its secret key sk_i to compute its partial decryption, while parties outside of S cannot participate in decryption. Crucially, an adversary without access to sk_i cannot distinguish between αpk_i and $\bar{\alpha}_i \text{pk}_i$, and thus cannot distinguish whether the i -th party was revoked or not. Intuitively, this trick conceals whether a party has been revoked since a decryptor that does not know sk_i cannot distinguish αpk_i from $\bar{\alpha}_i \text{pk}_i$.

While rerandomization is a standard technique, applying it in a threshold setting with tracing while proving membership hiding and message privacy under realistic adversarial models is non-trivial. In

particular, we address practical challenges such as rushing adversary attacks against membership hiding, where adversaries choose their public key depending on an honest party’s public key. This allows them to check whether the honest party is revoked or not by observing whether the honest key and the revoked key were rerandomized by the same value α (see Section 6.2). We formally establish membership hiding via a reduction to the external Diffie-Hellman (XDH) assumption.

Amortization. A limitation of this construction is that the ciphertext includes the aggregation key $\mathbf{ak}$, which is linear in n . However, this cost can be amortized: when encrypting multiple messages to the same set S , we include $\mathbf{ak}$ only once. Thus, the amortized ciphertext size is $O(1)$.

Beyond Amortized Setting. To further address ciphertext size in the worst case, we propose two models where ciphertexts are $O(1)$ without amortization. The *preprocessing-based* model, discussed in Section 7 and the *epoch-based* model, discussed in Section 8. We next briefly discuss the epoch-based construction and discuss the preprocessing-based construction in the following section.

Epoch-based Traceable Threshold Encryption. In many practical applications, such as threshold-encrypted mempools for MEV prevention [16, 21, 27], encryption is organized around epochs: each epoch has a public key, and all ciphertexts during that epoch are encrypted under it. At the epoch’s end, the committee decrypts all messages. We can exploit this structure to generate and publish a *single* rerandomization $([\alpha]_1, \mathbf{ek}, \mathbf{ak})$ per epoch, used by all ciphertexts within that epoch. This reduces ciphertext size to be constant in the number of parties. While decoders that only decrypt for hardcoded epoch keys can evade being traced, they are useless to decrypt ciphertexts generated during any future epoch. We formalize this setting and its implications in Section 8.

Remark 2.1 A naive traceable silent threshold encryption scheme is known: each party samples its own key pair $(\mathbf{sk}_i, \mathbf{pk}_i)$, and encryption proceeds by t -out-of- n secret-sharing the message m , with the i -th share encrypted under $\mathbf{pk}_i$. Traceability follows from PLBE: in TrEnc , the tracer revokes party i by encrypting a random value under $\mathbf{pk}_i$ instead of the correct share. This yields ciphertexts of size $O(n)$ in the worst case. Amortization is possible if the encryptor chooses a secret s , encrypts s , and then encrypts messages under s . However, since s cannot be shared with other users (who could then decrypt all messages), this naive scheme is inherently incompatible with amortized and epoch-based settings, where a public key is used by potentially distrustful users. Moreover, the preprocessing technique introduced next does not apply to this naive scheme.

2.4 Π_{OSTEP} : A Preprocessing-based Construction

In Section 7, we present a second construction of OSTE that achieves constant-size ciphertexts even in the worst case, through a carefully designed *preprocessing phase* with private tracing. The core idea is to precompute a polynomial-size set of encryption and aggregation keys, allowing the encryptor to avoid recomputing or attaching large $\mathbf{ak}$ values during each encryption. However, this approach raises new challenges for tracing:

- Normal encryption and tracing queries need to remain indistinguishable, even though some precomputed keys revoke parties.
- Membership hiding needs to be preserved, even though the adversary has access to all precomputed keys and can, in principle, hardcode behavior for a subset of them in the decoder.

A Set of Precomputed Keys. To support preprocessing, after all parties have published their public keys, we generate *polynomial-size* set of precomputed encryption and aggregation keys $\mathcal{K} = (\mathbf{ek}_1, \mathbf{ak}_1), \dots, (\mathbf{ek}_N, \mathbf{ak}_N)$, where each $(\mathbf{ek}_i, \mathbf{ak}_i)$ encrypts to some subset $S \subseteq [n]$. In particular, $\mathcal{K}$ includes keys that:

- Allow decryption by all parties, i.e., $R = \emptyset$.
- Exclude specific subsets $\mathcal{R}_i = \{1, \dots, i\}$, by setting $S_i = [n] \setminus \mathcal{R}_i$. Denote with $\mathcal{K}_i$ the set of keys that revoke $\mathcal{R}_i$

This setup enables revocation by choosing a key targeting S_i . However, to preserve indistinguishability and membership hiding, it must be computationally infeasible to determine which subset each key targets, except for the tracer. To encrypt a message, the encryptor picks a set of keys at random from $\mathcal{K}$ and

encrypts under them. Encrypting under multiple keys is required to amplify decryption correctness, since parties may be revoked by the random choice of keys.

Naive TrEnc Breaks Indistinguishability. A naive $\text{TrEnc}(\mathcal{R}_i)$ algorithm might select keys uniformly at random from $\mathcal{K}_i$ to revoke $\mathcal{R}_i$ and encrypt under them. Consider that, if party i is a colluder, the decoder could check which keys revoke party i . Also, observe that for normal encryption, we would expect that with a certain probability, some of sampled keys revoke party i . A $\text{TrEnc}(\emptyset)$ ciphertext, however, would never revoke party i . Thus, the decoder could distinguish with high probability between Enc and $\text{TrEnc}(\emptyset)$ ciphertexts by checking if any of the colluders are revoked, which clearly breaks indistinguishability.

Our Solution: Distribution Matching. To mitigate this, we ensure that the probability of sampling a key that revokes party i in Enc matches that of sampling a key that revokes party i in $\text{TrEnc}(\emptyset)$, up to negligible error. While this solves the indistinguishability issue, it introduces a similar problem for membership hiding. We address this by ensuring that the distribution of keys in ciphertexts generated by $\text{TrEnc}(\mathcal{R}_i)$ is computationally indistinguishable from those generated by $\text{TrEnc}(\mathcal{R}_{i-1})$ for adversaries not holding sk_i . Specifically, in $\text{TrEnc}(\mathcal{R}_i)$, the key is sampled from the sets $\mathcal{K}_j$ for $j \geq i$, according to a specific probability distribution: a key is sampled from $\mathcal{K}_i$ with probability $1/2 + (i+1)/(2n+2)$ and from $\bigcup_{v \in \{i+1, \dots, n\}} \mathcal{K}_{\text{TR}}[v]$ with probability $(n-i)/(2n+2)$. This distribution is designed to mitigate the aforementioned attack; we justify it in Section 7.1.

Traceability Intuition and Proof. An adversary could try to win the membership hiding game for index i by guessing a key $k \in \mathcal{K}$ as belonging to $\mathcal{K}_{i-1}$, and then outputting 1 if k appears in the challenge ciphertext and 0 otherwise. Since the total number of keys is polynomial and keys in $\text{TrEnc}(\mathcal{R}_i)$ are never drawn from $\mathcal{K}_{i-1}$, this strategy yields a non-negligible advantage. Our key technical insight is that by carefully designing the sampling distribution and selecting appropriate parameters, we can upper-bound the adversary's advantage by a fixed value δ , which is small enough to preserve traceability. This is established in Theorem 5.7. Specifically, in Lemmas 7.3 and 7.4, we prove that the *relaxed* PLBE guarantees hold in our construction for this specific δ . At a high level, we apply the Chernoff bound to argue that when $|\mathcal{K}| = O(n^3)$, an adversary cannot sample a subset of keys $\mathcal{L} \subseteq \mathcal{K}$ for which the probability of sampling a key from $\mathcal{L}$ in $\text{TrEnc}(\mathcal{R}_i)$ deviates by more than δ from the probability in $\text{TrEnc}(\mathcal{R}_{i-1})$. This proof is based on the argument that $\mathcal{K}_i$ is *computationally* indistinguishable from $\mathcal{K}_{i-1}$ for any adversary lacking sk_i , which we show via a reduction to the membership-hiding property of Π_{OSTE} .

Semantic Security. The tracing authority is trusted only for tracing, not for semantic security. The tracing key tk reveals which keys revoke which parties, but does not assist in decryption. In fact, our IND-CPA security game assumes that the adversary generates the entire preprocessed key set $\mathcal{K}$.

3 Preliminaries

Notation. We denote the security parameter by λ and the correctness parameter by λ_c . We write $\text{negl}(\lambda)$ for functions that are negligible in λ and $\text{poly}(\lambda)$ for polynomials in λ . We denote the number of parties by n . In the O notation we ignore factors other than n . For integers $a, b \in \mathbb{N}$, we denote the set $\{1, \dots, b\}$ by $[b]$ and the set $\{a, \dots, b\}$ by $[a, b]$. We write $y \leftarrow A(x; r)$ to execute the algorithm A on input x with randomness r and assign the result to y . If A is executed with uniform randomness, we abbreviate $y \xleftarrow{\$} A(x)$ to indicate that A is probabilistic. Further, we use $\approx_c$ to denote that two random variables are computationally indistinguishable. For a security game Game , we denote by $\text{Adv}_{\mathcal{A}}^{\text{Game}}(x)$ the success probability of the adversary $\mathcal{A}$ when running in the security game $\text{Game}(x)$ ($\text{Adv}_{\mathcal{A}}^{\text{Game}}(x) = \Pr[\text{Game}_{\mathcal{A}}(x) = 1]$). If Game is a distinguishing game (i.e. if the adversary is tasked with guessing a random bit b), then the advantage is the success probability over guessing the bit $\text{Adv}_{\mathcal{A}}^{\text{Game}}(x) = \Pr[\text{Game}_{\mathcal{A}}(x) = 1] - 1/2$. We denote by $\mathbb{H} = \{\omega, \omega^2, \dots, \omega^l\}$ the multiplicative subgroup generated by the l -th root of unity $\omega \in \mathbb{Z}_p$ ($\omega^l = 1$) for $l = |\mathbb{H}| = n+1$. Further, we denote by $L_i(x)$ the Lagrange basis polynomial of element i with respect to $\mathbb{H}$. By $Z(x)$, we denote the vanishing polynomial on $\mathbb{H}$, i.e., $Z(x) = x^l - 1$.

Further, we rely on a univariate formulation of the sumcheck lemma [6, 40].

Lemma 3.1 (Univariate Sumcheck [6, 40]). Let $n = \text{poly}(\lambda)$, $A(x) = \sum_{i=1}^{|\mathbb{H}|} a_i \cdot L_i(x)$ and $B(x) = \sum_{i=1}^{|\mathbb{H}|} b_i \cdot L_i(x)$. It holds that

$$A(x) \cdot B(x) = \frac{\sum_i a_i b_i}{|\mathbb{H}|} + Q_x(x) \cdot x + Q_Z(x) \cdot Z(x),$$

where both Q_x and Q_Z are polynomials with degree $\leq |\mathbb{H}| - 2$ defined as

$$Q_x(x) = \sum_i a_i b_i \frac{L_i(x) - L_i(0)}{x}$$

$$Q_Z(x) = \sum_i a_i b_i \frac{L_i^2(x) - L_i(x)}{Z(x)} + \sum_{i \neq j} a_i b_j \frac{L_i(x) L_j(x)}{Z(x)}$$

The original sumcheck is concretely stated for $b_i = 1$. In this case, we treat general inner products as a straightforward generalization (see [18]).

Non-Interactive Zero-Knowledge Proofs of Knowledge. Our constructions rely on Non-Interactive Zero-Knowledge Proofs of Knowledge (NIZK-PoKs). A NIZK-PoK proof system PS_R is a tuple of algorithms $\text{PS}_R = (\text{Setup}, \text{Prove}, \text{Verify}, \text{Sim}, \text{Ext})$ to prove knowledge of witnesses w to statements χ in the corresponding relation R . A detailed definition can be found in Appendix A.

Bilinear Pairings. A bilinear pairing ensemble $E = (\mathbb{G}_1, [1]_1, \mathbb{G}_2, [1]_2, \mathbb{G}_T, p, \circ)$ is an ensemble of cyclic groups $\mathbb{G}_1$ with generator $[1]_1$, $\mathbb{G}_2$ with generator $[1]_2$ and $\mathbb{G}_T$ of prime order p with a pairing operation $\circ: \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ that satisfies bilinearity and non-degeneracy. We usually write $\circ$ infix.

We rely on the external Diffie-Hellman assumption [2, 8].

Definition 3.2 (XDH). Let $\mathcal{G}$ be an algorithm to generate a pairing ensemble E . The external Diffie-Hellman problem (XDH) is hard for $\mathcal{G}$ if for all PPT adversaries $\mathcal{A}$ there exists a negligible function negl such that

$$|\Pr[\mathcal{A}(1^\lambda, E, [x]_1, [y]_1, [xy]_1) = 1] - \Pr[\mathcal{A}(1^\lambda, E, [x]_1, [y]_1, [v]_1) = 1]| \leq \text{negl}(\lambda),$$

where $E \xleftarrow{\$} \mathcal{G}(1^\lambda)$ and $x, y, v \xleftarrow{\$} \mathbb{Z}_p$.

Some of our security proofs rely on the Generic Group Model (GGM) [44]. We also stress that the XDH assumption holds in the GGM as well as the decisional AGM [29, 42, 3, 41]. We list the necessary definitions and theorems in Appendix A.

4 Oblivious Silent Threshold Encryption

In this section, we define *oblivious silent threshold encryption* (OSTE), extending the definition of *silent threshold encryption* [30].

Definition 4.1 (Oblivious Silent Threshold Encryption). An oblivious silent threshold encryption scheme consists of a tuple of algorithms $(\text{Setup}, \text{KGen}, \text{IsValid}, \text{Prep}, \text{Enc}, \text{TrEnc}, \text{PartDec}, \text{PartVfy}, \text{DecAggr})$ with the following syntax:

- $\text{CRS} \xleftarrow{\$} \text{Setup}(1^\lambda, n)$: On input the security parameter λ and number of parties n , Setup outputs a common reference string CRS .
- $(\text{sk}, \text{pk}, \text{hint}, \pi) \xleftarrow{\$} \text{KGen}(1^\lambda, \text{CRS})$: On input the CRS , KGen outputs a secret key sk , a public key pk , a hint, and a proof of knowledge π that the party knows sk corresponding to pk and hint .
- $1/0 \leftarrow \text{IsValid}(\text{CRS}, \text{pk}, \text{hint}, \pi)$: On input the CRS , a public key pk , a hint hint , and a proof π , IsValid outputs 1 if the public key is valid and 0 otherwise.
- $(\mathcal{K}, \pi, \text{tk}) \xleftarrow{\$} \text{Prep}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]})$: The Prep algorithm receives the CRS and all public keys and hints $\{\text{pk}_i, \text{hint}_i\}_{i \in [n]}$ as input. It outputs a preprocessing $\mathcal{K}$, a proof π , and a tracing key tk .

- $1/0 \leftarrow \text{PreVfy}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{K}, \pi)$: Given the CRS, n public keys and hints as well as a preprocessing $\mathcal{K}$ and corresponding proof π , PreVfy outputs 1 if the preprocessing proof is valid and 0 otherwise.
- $\text{ct} \xleftarrow{\$} \text{Enc}(\text{CRS}, \mathcal{K}, t, \text{m})$: On input the CRS, a preprocessing $\mathcal{K}$, a threshold t , and a message m , Enc outputs a ciphertext ct .
- $\text{ct} \xleftarrow{\$} \text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \text{m}, \mathcal{R})$: Given the CRS, the tracing key tk , a preprocessing $\mathcal{K}$, a threshold t , the message m , and a set of revoked parties $\mathcal{R} \subseteq [n]$, TrEnc outputs a ciphertext ct . Sometimes, we abuse notation and write $\text{TrEnc}(\mathcal{R})$ to refer to trace-encrypt with revoked set $\mathcal{R}$ whenever the remaining inputs are clear from the context.
- $\sigma \leftarrow \text{PartDec}(\text{sk}, \text{ct})$: On input a secret key sk and a ciphertext ct , PartDec outputs a partial decryption σ .
- $\text{m} \leftarrow \text{DecAggr}(\text{CRS}, \mathcal{K}, \text{ct}, \{\sigma_i\}_{i \in S})$: On input the CRS, the set of keys $\mathcal{K}$, a ciphertext ct and a set of partial decryptions $\{\sigma_i\}_{i \in S}$, DecAggr outputs a message m .

Correctness is standard, i.e., we require that every coalition $S \subseteq [n]$ with $|S| \geq t$ decrypts with overwhelming probability.

Definition 4.2 (Correctness for OSTE). We call an OSTE protocol correct if for all $n = \text{poly}(\lambda)$, $t < n$, all $S \subseteq [n]$ with $|S| \geq t$ it holds that

$$\Pr \left[\begin{array}{l} \text{DecAggr}(\text{CRS}, \mathcal{K}, \text{ct}, \{\sigma_i\}_{i \in S}) = \text{m} \\ \wedge \forall i \in [n]: \text{IsValid}(\text{CRS}, \text{pk}_i, \text{hint}_i, \pi_i) = 1 \\ \wedge \text{PreVfy}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{K}, \pi) = 1 \end{array} \right] \geq 1 - \text{negl}(\lambda),$$

where $\text{CRS} \xleftarrow{\$} \text{Setup}(1^\lambda, n)$, $(\text{sk}_i, \text{pk}_i, \text{hint}_i, \pi_i) \xleftarrow{\$} \text{KGen}(1^\lambda) \forall i \in [n]$, $(\mathcal{K}, \pi, \text{tk}) \xleftarrow{\$} \text{Prep}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]})$, $\text{ct} \xleftarrow{\$} \text{Enc}(\text{CRS}, \mathcal{K}, t, \text{m})$, and $\sigma_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct})$ for each $i \in [n]$.

We require three tracing-related properties named *indistinguishability*, *membership hiding*, and *message privacy*. To formally define these, we first introduce an additional parameter $\mathcal{X}$. For an OSTE scheme, $\mathcal{X}$ denotes the set of all possible sets of parties that can be revoked using TrEnc (hence $\mathcal{X} \subseteq \mathcal{P}([n])$). In standard PLBE, one can only revoke parties linearly (i.e. $\mathcal{X} = \{\emptyset, \{1\}, \{1, 2\}, \dots, [n]\}$). Looking ahead, we want to construct OSTE schemes that allow for more than one, or even all possible revocation orders ($\mathcal{X} = \mathcal{P}([n])$), which we will use to identify more than 1 and up to t traitors (see Section 5.2).

Goyal et al. [33] show that only decoder-based definitions of indistinguishability and membership hiding imply a (private) tracing scheme. In the following, we present definitions of distinguishing decoders as discussed in [33].

Definition 4.3 (Distinguishing Decoders [33]). For any $\delta \in [-1/2, 1/2]$, PPT algorithm $\mathcal{D}$, $n = \text{poly}(\lambda)$, $t < n$, any CRS, any message m , and any preprocessing $\mathcal{K}$ we say that

- $\mathcal{D}$ is $\delta\text{-Dist}^{\text{Enc}, \text{TrEnc}}$ if

$$\Pr \left[\mathcal{D}(\text{ct}_b) = b \mid \begin{array}{l} b \xleftarrow{\$} \{0, 1\}; \\ \text{ct}_0 \leftarrow \text{Enc}(\text{CRS}, \mathcal{K}, t, \text{m}); \\ \text{ct}_1 \leftarrow \text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \text{m}, \emptyset) \end{array} \right] \geq \frac{1}{2} + \delta.$$

- $\mathcal{D}$ is $\delta\text{-Dist}^{\mathcal{R}, \mathcal{R} \cup \{i\}}$ for revoked set $\mathcal{R} \subseteq [n]$ if

$$\Pr \left[\mathcal{D}(\text{ct}_b) = b \mid \begin{array}{l} b \xleftarrow{\$} \{0, 1\}; \\ \text{ct}_0 \leftarrow \text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \text{m}, \mathcal{R}); \\ \text{ct}_1 \leftarrow \text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \text{m}, \mathcal{R} \cup \{i\}) \end{array} \right] \geq \frac{1}{2} + \delta.$$

$\text{GameInd}_{\mathcal{A}}(1^\lambda, n, \delta):$	$\text{GameMH}_{\mathcal{A}}(1^\lambda, n, \mathcal{X}, \delta):$
$\text{CRS} \xleftarrow{\$} \text{Setup}(1^\lambda, n)$	$\text{CRS} \xleftarrow{\$} \text{Setup}(1^\lambda, n)$
$(\{\text{pk}_\ell, \text{hint}_\ell, \pi_\ell\}_{\ell \in [n]}, t, \text{st}) \xleftarrow{\$} \mathcal{A}(1^\lambda, \text{CRS})$	$(i, \text{st}) \xleftarrow{\$} \mathcal{A}(1^\lambda, \text{CRS})$
$(\mathcal{K}, \pi, \text{tk}) \xleftarrow{\$} \text{Prep}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]})$	$(\mathcal{R}, \{\text{pk}_\ell, \text{hint}_\ell, \pi_\ell\}_{\ell \in [n] \setminus \{i\}}, t, \text{st}) \xleftarrow{\$} \mathcal{A}(\text{pk}_i, \text{hint}_i, \pi_i, \text{st})$
$(\mathcal{D}, \text{m}) \xleftarrow{\$} \mathcal{A}(\mathcal{K}, \pi, \text{st})$	$(\mathcal{K}, \pi, \text{tk}) \xleftarrow{\$} \text{Prep}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]})$
return $\delta\text{-Dist}^{\text{Enc}, \text{TrEnc}}(\mathcal{D})$	return $\delta\text{-Dist}^{\mathcal{R}, \mathcal{R} \cup \{i\}}(\mathcal{D}) \wedge \mathcal{R} \in \mathcal{X} \wedge \mathcal{R} \cup \{i\} \in \mathcal{X} \wedge$ $i \in [n] \setminus \mathcal{R} \wedge \text{PreVfy}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{K}, \pi) \wedge$ $\forall i \in [n]: \text{IsValid}(\text{CRS}, \text{pk}_i, \text{hint}_i, \pi_i)$

Figure 2: The indistinguishability and membership hiding games for OSTE schemes

Indistinguishability. The indistinguishability notion captures that one cannot construct a decoder that has advantage δ for distinguishing if a given ciphertext ct was created using either Enc or $\text{TrEnc}(\mathcal{R} := \emptyset)$. In GameInd (Figure 2), the adversary can control *all* parties. The adversary wins the indistinguishability game if the decoder is $\delta\text{-Dist}^{\text{Enc}, \text{TrEnc}}$ according to Definition 4.3.

Definition 4.4 (δ -Indistinguishability). An OSTE scheme $\mathcal{E}$ is δ -indistinguishable for $\delta \in [-1/2, 1/2]$ if for all PPT adversaries $\mathcal{A}$, there exists a negligible function negl such that $\text{Adv}_{\mathcal{A}, \mathcal{E}}^{\text{GameInd}}(1^\lambda, n, \delta) \leq \text{negl}(\lambda)$, where $\text{GameInd}_{\mathcal{A}}$ is defined in Figure 2.

δ -Membership Hiding. The δ -membership hiding notion is a generalization of the index hiding property for PLBE [14]. Intuitively, our definition states that an adversary can not build a decoder that has advantage $\geq \delta$ of distinguishing whether a party i is revoked for a ciphertext or not, even if the adversary corrupts *all* parties other than i and freely chooses the remaining set of revoked parties $\mathcal{R}$. The only restriction is that both $\mathcal{R} \in \mathcal{X}$ and $\mathcal{R} \cup \{i\} \in \mathcal{X}$ (i.e., both sets *can* be revoked by the scheme).

Definition 4.5 (δ -Membership Hiding). An OSTE scheme $\mathcal{E}$ is δ -membership hiding for a subset $\mathcal{X} \subseteq \mathcal{P}([n])$ and $\delta \in [-1/2, 1/2]$ if for all PPT adversaries $\mathcal{A}$, there exists a negligible function negl such that $\text{Adv}_{\mathcal{A}, \mathcal{E}}^{\text{GameMH}}(1^\lambda, n, \mathcal{X}, \delta) \leq \text{negl}(\lambda)$, where $\text{GameMH}_{\mathcal{A}}$ is defined in Figure 2.

$\text{GameMP}_{\mathcal{A}}(1^\lambda, n, \mathcal{X}):$
$\text{CRS} \xleftarrow{\$} \text{Setup}(1^\lambda, n); b \xleftarrow{\$} \{0, 1\}$
$(\mathcal{T}, \text{st}) \xleftarrow{\$} \mathcal{A}(1^\lambda, \text{CRS})$
for $i \in [n], i \notin \mathcal{T}$ do
$(\text{sk}_i, \text{pk}_i, \text{hint}_i, \pi_i) \xleftarrow{\$} \text{KGen}(1^\lambda, \text{CRS})$
$(\{\text{pk}_i, \text{hint}_i, \pi_i\}_{i \in \mathcal{T}}, t, \mathcal{R}, \text{m}_0, \text{m}_1, \text{st}) \xleftarrow{\$} \mathcal{A}(\{\text{pk}_i, \text{hint}_i, \pi_i\}_{i \notin \mathcal{T}}, \text{st})$
$(\mathcal{K}, \pi, \text{tk}) \xleftarrow{\$} \text{Prep}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]})$
$\text{ct} \xleftarrow{\$} \text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \text{m}_b, \mathcal{R})$
$b' \xleftarrow{\$} \mathcal{A}^{\text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \cdot, \mathcal{R})}(\text{st}, \mathcal{K}, \pi, \text{ct})$
return $b = b' \wedge \mathcal{T} \setminus \mathcal{R} < t \wedge \mathcal{T} \subseteq [n] \wedge \mathcal{R} \in \mathcal{X} \wedge \text{PreVfy}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{K}, \pi)$ $\wedge \forall i \in \mathcal{T}: \text{IsValid}(\text{CRS}, \text{pk}_i, \text{hint}_i, \pi_i)$

Figure 3: The message privacy game for OSTE schemes

Message Privacy. For message privacy, we ensure that an adversary that holds less than t shares of *unrevoked* parties cannot decrypt the message. This is different from normal IND-CPA since the adversary

may corrupt more than t parties overall.

Definition 4.6 (Message Privacy for OSTE). An OSTE scheme $\mathcal{E}$ fulfills message privacy for a subset $\mathcal{X} \subseteq \mathcal{P}([n])$ if for all PPT adversaries $\mathcal{A}$, it holds that $\mathbf{Adv}_{\mathcal{A}, \mathcal{E}}^{\text{GameMP}}(1^\lambda, n, \mathcal{X}) \leq \text{negl}(\lambda)$, where $\text{GameMP}_{\mathcal{A}}$ is defined in Figure 3.

Security. Our security notion is a standard IND-CPA definition, i.e., an adversary who corrupts a set of parties $|\mathcal{T}| < t$ cannot decrypt ciphertexts. Note that in our game, the adversary generates the preprocessing $\mathcal{K}$ itself (it outputs $\mathcal{K}$ and π). This captures the requirement that the party that generates the preprocessing *is untrusted for security*, but only for traceability.

Definition 4.7 (IND-CPA security for OSTE). An OSTE scheme $\mathcal{E}$ is IND-CPA secure if for all PPT adversaries $\mathcal{A}$, there exists a negligible function $\text{negl}(\lambda)$ such that

$$\mathbf{Adv}_{\mathcal{A}, \mathcal{E}}^{\text{IND-CPA}}(1^\lambda, n) \leq \frac{1}{2} + \text{negl}(\lambda)$$

where $\text{IND-CPA}_{\mathcal{A}}$ is defined in Figure 4.

IND-CPA $_{\mathcal{A}}(1^\lambda, n)$:

CRS $\xleftarrow{\$}$ Setup($1^\lambda, n$)
 $(\mathcal{T}, \text{st}) \xleftarrow{\$} \mathcal{A}(1^\lambda, \text{CRS})$
for $i \in [n], i \notin \mathcal{T}$ **do**
 $(\text{sk}_i, \text{pk}_i, \text{hint}_i, \pi_i) \xleftarrow{\$} \text{KGen}(1^\lambda, \text{CRS})$
 $(\{\text{pk}_i, \text{hint}_i, \pi_i\}_{i \in \mathcal{T}}, t, \mathcal{K}, \pi, \text{m}_0, \text{m}_1, \text{st}) \xleftarrow{\$} \mathcal{A}(\{\text{pk}_i, \text{hint}_i, \pi_i\}_{i \notin \mathcal{T}}, \text{st})$
 $b \xleftarrow{\$} \{0, 1\}$
 $\text{ct} \xleftarrow{\$} \text{Enc}(\text{CRS}, \mathcal{K}, t, \text{m}_b)$
 $b' \xleftarrow{\$} \mathcal{A}(\text{ct}, \text{st})$
return $b = b' \wedge \mathcal{T} \subseteq [n] \wedge |\mathcal{T}| < t$
 $\wedge \forall i \in \mathcal{T}: \text{IsValid}(\text{CRS}, \text{pk}_i, \text{hint}_i, \pi_i)$
 $\wedge \text{PreVfy}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{K}, \pi)$

Figure 4: A game-based IND-CPA security definition for OSTE and TSTE schemes.

5 Traceable Silent Threshold Encryption

In this section, we define *traceable silent threshold encryption* (TSTE) and present a reduction from OSTE to TSTE.

Definition 5.1 (Traceable Silent Threshold Encryption). A traceable silent threshold encryption scheme consists of a tuple of algorithms (Setup, KGen, IsValid, Prep, PreVfy, Enc, PartDec, PartVfy, DecAggr, Trace), such that:

- $\text{ct} \xleftarrow{\$} \text{Trace}^{\mathcal{D}}(\text{CRS}, \text{tk}, \mathcal{K}, t, \epsilon)$: Given the CRS, the preprocessing $\mathcal{K}$, the tracing key tk , a bound on the quality of a decoder ϵ , and blackbox access to a decoder $\mathcal{D}$, Trace outputs a list of colluding parties $\mathcal{T}$.

The remaining algorithms are defined as for OSTE schemes (Definition 4.1).

The correctness and CPA-security of TSTE are defined similarly to OSTE.

GameTR_{$\mathcal{A}$}($1^\lambda, n, \epsilon$): CRS $\leftarrow$ Setup($1^\lambda, n, c$) $(\mathcal{T}, \text{st}) \leftarrow \mathcal{A}(1^\lambda, \text{CRS})$ for $i \notin \mathcal{T}$ do $(\text{sk}_i, \text{pk}_i, \text{hint}_i, \pi_i) \leftarrow \text{KGen}(1^\lambda, \text{CRS}, n)$ $(\{\text{pk}_i, \text{hint}_i, \pi_i\}_{i \in \mathcal{T}}, t, \text{st}) \leftarrow \mathcal{A}(\{\text{pk}_i, \text{hint}_i, \pi_i\}_{i \notin \mathcal{T}}, \text{st})$ $(\mathcal{K}, \pi, \text{tk}) \xleftarrow{\\$} \text{Prep}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \epsilon)$ $(\mathcal{D}, \text{m}_0, \text{m}_1) \xleftarrow{\\$} \mathcal{A}(\mathcal{K}, \pi, \text{st})$ $\mathcal{T}' \leftarrow \text{Trace}^{\mathcal{D}}(\text{CRS}, \text{tk}, \mathcal{K}, t, \epsilon)$ return $\epsilon\text{-Dist}^{\text{m}_0, \text{m}_1}(\mathcal{D}) \wedge (\mathcal{T}' = \emptyset \vee \mathcal{T}' \not\subseteq \mathcal{T})$ $\wedge \forall i \in \mathcal{T}: \text{IsValid}(\text{CRS}, \text{pk}_i, \text{hint}_i, \pi_i) \wedge \text{PreVfy}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{K}, \pi)$
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Figure 5: Decoder-based traceability definition (GameTR) for TSTE schemes. Note that the Prep algorithm is executed by a trusted party in GameTR, but is run by the adversary in IND-CPA.

Definition 5.2 (Correctness for TSTE). We call a TSTE scheme correct with respect to parameter λ_c if for all $n = \text{poly}(\lambda)$, $t < n$, all $S \subseteq [n]$ with $|S| \geq t$ and for

$$\begin{aligned}
& \text{CRS} \leftarrow \text{Setup}(1^\lambda, n); \\
& \forall i (\text{sk}_i, \text{pk}_i, \text{hint}_i, \pi_i) \xleftarrow{\$} \text{KGen}(1^\lambda); \\
& (\mathcal{K}, \pi, \text{tk}) \xleftarrow{\$} \text{Prep}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \epsilon); \\
& \text{ct} \xleftarrow{\$} \text{Enc}(\text{CRS}, \mathcal{K}, t, \text{m}); \\
& \forall i \sigma_i \leftarrow \text{PartDec}(\text{sk}_i, \text{ct})
\end{aligned}$$

it holds that

$$\Pr \left[\begin{array}{l} \text{DecAggr}(\text{CRS}, \mathcal{K}, \text{ct}, \{\sigma_i\}_{i \in S}) = \text{m} \\ \wedge \forall i \in [n]: \text{IsValid}(\text{CRS}, \text{pk}_i, \text{hint}_i, \pi_i) = 1 \\ \wedge \text{PreVfy}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{K}, \pi) = 1 \\ \wedge \forall i \in S: \text{PartVfy}(\text{ct}, \sigma_i, \text{pk}_i) = 1 \end{array} \right] \geq 1 - \text{negl}(\lambda_c).$$

Security. Like OSTE schemes, TSTE must fulfill a IND-CPA security definition. The security definitions for OSTE and TSTE definitions both concern the Enc function. A definition concerning Enc is sensible in the context of honest encryptions, while a consistent definition for both schemes simplifies our transformation between the two (Section 5.1).

Definition 5.3 (IND-CPA security for TSTE). We call a TSTEP scheme $\mathcal{E}$ IND-CPA secure, if for all PPT adversaries $\mathcal{A}$ and $n = \text{poly}(\lambda)$

$$\text{Adv}_{\mathcal{A}, \mathcal{E}}^{\text{IND-CPA}}(1^\lambda, n) \leq \text{negl}(\lambda),$$

where $\text{IND-CPA}_{\mathcal{A}}$ is defined in Figure 4.

Traceability. Our decoder-based definition of traceability makes use of distinguishing decoders.

Definition 5.4 (Distinguishing Decoders for TSTE [33]). For any $\epsilon \in [-\frac{1}{2}, \frac{1}{2}]$, PPT algorithm $\mathcal{D}$, $n = \text{poly}(\lambda)$, $t < n$, any CRS, and any preprocessing $\mathcal{K}$ and tracing key tk , we say $\mathcal{D}$ is $\epsilon\text{-Dist}^{\text{m}_0, \text{m}_1}$ for messages m_0, m_1 , if

$$\Pr \left[\mathcal{D}(\text{ct}_b) = b \mid \begin{array}{l} b \xleftarrow{\$} \{0, 1\}; \\ \text{ct}_b \leftarrow \text{Enc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \text{m}_b); \end{array} \right] \geq \frac{1}{2} + \epsilon.$$

We say that $\mathcal{D}$ is an ϵ -good decoder if $\mathcal{D}$ is $\epsilon\text{-Dist}^{\text{m}_0, \text{m}_1}$.

Definition 5.5 (Traceability for TSTE). We call a TSTE scheme $\mathcal{E}$ traceable for $\epsilon \in [-1/2, 1/2]$, if for all PPT adversaries $\mathcal{A}$ and $n = \text{poly}(\lambda)$ there exists a negligible function $\text{negl}(\lambda)$ such that $\text{Adv}_{\mathcal{A}, \mathcal{E}}^{\text{GameTR}}(1^\lambda, n, \epsilon) \leq \text{negl}(\lambda)$, where $\text{GameTR}_{\mathcal{A}}$ is defined in Figure 5.

5.1 From OSTE to TSTE

In this section, we show that an OSTE scheme can be used to construct a TSTE scheme. Let $\mathcal{E}$ be an OSTE scheme as defined in Definition 4.1. We construct a TSTE scheme Π_{TSTE} based on $\mathcal{E}$, where all TSTE algorithms defined for OSTE run the corresponding algorithm of $\mathcal{E}$. We present a Trace algorithm for Π_{TSTE} , with pseudocode in Figure 6. The Trace algorithm follows a similar approach to tracing algorithms in PLBE schemes [14]. In the threshold encryption setting, if the decoder shows a high success probability after revoking $n - t$ parties, the remaining t parties are considered colluders.

Remark 5.6 For simplicity, we assume the revocation order $\mathcal{L} = \{\mathcal{R}_i\}_{i \in [0, n]}$, where $\mathcal{R}_i = \{1, 2, \dots, i\}$ and $\mathcal{R}_0 = \emptyset$. However, the Trace algorithm does not require a specific order of revocation. A tracer can choose any valid order of revocation supported by the underlying OSTE construction (i.e., within the set $\mathcal{X}$ in Definition 4.5). In Section 5.2, we explore how this property enables tracing multiple traitors.

Trace ^{$\mathcal{D}, m_0, m_1$} (CRS, $\mathcal{K}$, tk, t , ϵ)

$W = 2^7 \cdot 3n^2\lambda/\epsilon$; $p_{-1} \leftarrow \epsilon$; $\mathcal{T}' \leftarrow \emptyset$

for $i \in [0, n - t]$ **do**

$a_i \leftarrow 0$

for $j \in [W]$ **do**

$b \xleftarrow{\$} \{0, 1\}$

$\text{ct} \xleftarrow{\$} \mathcal{E}.\text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, m_b, \mathcal{R}_i)$

if $\mathcal{D}(\text{ct}) = b$ **then** $a_i \leftarrow a_i + 1$

$\hat{p}_i \leftarrow a_i / (W)$

CONDITION 1. **if** $\hat{p}_{i-1} - \hat{p}_i \geq \frac{\epsilon}{4(n-t)}$ **then** $\mathcal{T}' \leftarrow \mathcal{T}' \cup \{i\}$

CONDITION 2. **if** $\hat{p}_{n-t} \geq \epsilon/4$ **then** $\mathcal{T}' \leftarrow \mathcal{T}' \cup \{n - t + 1, \dots, n\}$

return $\mathcal{T}'$

Figure 6: Π_{TSTE} : Blackbox tracing from an OSTE scheme, denoted by $\mathcal{E}$

Theorem 5.7 Let $\mathcal{E}$ be an OSTE scheme for the linear revocation order $\mathcal{X} = \mathcal{L}$ (Definition 4.1), satisfying $(\epsilon/16)$ -indistinguishability (Definition 4.4) and $(\epsilon/16n)$ -membership hiding (Definition 4.5) for $\epsilon = 1/\text{poly}(\lambda)$, with $\epsilon \in [-1/2, 1/2]$. Then, for every $n \in \text{poly}(\lambda)$ and $\lambda \in \mathbb{N}$, the scheme Π_{TSTE} (Figure 6) is a traceable silent threshold encryption scheme (Definition 5.1) with traceability parameter ϵ . Moreover, the secret share size, public key size, and message size match those of $\mathcal{E}$.

Establishing correctness and security is straightforward. The security of Π_{TSTE} is directly implied by the security of $\mathcal{E}$. Recall that both OSTE and TSTE schemes employ the same security definition. Further, our construction instantiates all functions of Π_{TSTE} with their counterpart in $\mathcal{E}$.

Next, we discuss traceability. The main challenge compared to classical PLBE schemes is that, in our setting, the adversary can distinguish between Enc and $\text{TrEnc}(\mathcal{R}_0)$ ciphertexts, as well as between $\text{TrEnc}(\mathcal{R}_{i-1})$ and $\text{TrEnc}(\mathcal{R}_i)$ with a non-negligible advantage. Despite this, we show that a TSTE scheme can be built from an OSTE scheme that is δ_1 -indistinguishable and δ_2 -membership hiding, per Definitions 4.4 and 4.5, where δ_1 and δ_2 are non-negligible and fixed in the proof.

That is, we prove the following lemma.

Lemma 5.8 Let $n \in \text{poly}(\lambda)$, $\lambda \in \mathbb{N}$, and $\epsilon = 1/\text{poly}(\lambda)$ with $\epsilon \in [-1/2, 1/2]$. For every PPT adversary $\mathcal{A}$, there exist PPT adversaries $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$ such that:

$$\begin{aligned} \text{Adv}_{\mathcal{A}, \Pi_{\text{TSTE}}}^{\text{GameTR}}(1^\lambda, n, \epsilon) &\leq \text{Adv}_{\mathcal{A}_1, \mathcal{E}}^{\text{GameInd}}(1^\lambda, n, \delta_1) + \text{Adv}_{\mathcal{A}_2, \mathcal{E}}^{\text{GameMP}}(1^\lambda, n) \\ &\quad + n \cdot \text{Adv}_{\mathcal{A}_3, \mathcal{E}}^{\text{GameMH}}(1^\lambda, n, \delta_2) + \text{negl}(\lambda), \end{aligned}$$

where $\delta_1 = \epsilon/16$, $\delta_2 = \epsilon/(16n)$, and $\text{negl}(\lambda)$ is a negligible function.

The full proof is given in Appendix B. We next discuss the ideas behind the proof.

Proof Sketch of Lemma 5.8. Recall that in the traceability game, the adversary provides a distinguishing decoder $\mathcal{D}^{\mathbf{m}_0, \mathbf{m}_1}$ that can distinguish encryptions of $\mathbf{m}_0$ from encryptions of $\mathbf{m}_1$, per Definition 5.4. Let p_i for $0 \leq i \leq n-t$ be the distinguishing advantage of $\mathcal{D}^{\mathbf{m}_0, \mathbf{m}_1}$ when it is queried with the ciphertext $\text{ct} = \text{TrEnc}(\mathbf{m}_b, \mathcal{R}_i)$. We show that the probability of winning the traceability game, as defined in Figure 5, is upper bounded by the probability of winning one of the three games of $(\epsilon/16)$ -indistinguishability, $(\epsilon/16n)$ -membership hiding, or message privacy, where the probability is over the random choice of b from $\{0, 1\}$. First, observe that the adversary wins the traceability game only in the following two cases:

- Case 1: $\mathcal{T}' = \emptyset$. This happens only if $\hat{p}_{i-1} - \hat{p}_i < \frac{\epsilon}{4(n-t)}$ for any $i \in [n-t]$ (Condition 1 in Π_{TSTE}) and $\hat{p}_{n-t} < \frac{\epsilon}{4}$ (Condition 2 in Π_{TSTE}).
- Case 2: $\mathcal{T}' \neq \emptyset$ and $\mathcal{T}' \not\subseteq T$. This happens only if $\hat{p}_{i-1} - \hat{p}_i \geq \frac{\epsilon}{4(n-t)}$ for some honest party $i \in [n-t]$ or when $\hat{p}_{n-t} \geq \frac{\epsilon}{4}$ and at least one of the parties $\{n-t+1, \dots, n\}$ is honest.

We show that if Case 1 holds, then $\hat{p}_0$ should be much smaller than ϵ , that is, $\epsilon - \hat{p}_0 > \epsilon/4$. Indeed, if $\hat{p}_0 \geq \frac{3\epsilon}{4}$, then either for at least one i it must hold that $\hat{p}_{i-1} - \hat{p}_i \geq \frac{\epsilon}{4(n-t)}$ or, otherwise, it holds that $\hat{p}_{n-t} \geq \frac{\epsilon}{4}$, so Case 1 will not hold. Hence, we show that if $\epsilon - \hat{p}_0 > \frac{\epsilon}{4}$ (i.e., Case 1 holds), then the adversary can construct a distinguishing decoder in the indistinguishability game with advantage $\delta_1 = \epsilon/16$. To show this, we first show by Chernoff bound that $p_0 - \hat{p}_0 < \frac{\epsilon}{8}$ with overwhelming probability. Thus, $\epsilon - p_0 > \frac{\epsilon}{8}$. The distinguishing decoder works as follows. Given a ciphertext ct , which is either $\text{Enc}(\mathbf{m}_b)$ or $\text{TrEnc}(\mathbf{m}_b, \mathcal{R}_0)$, it returns 0 if $\mathcal{D}^{\mathbf{m}_0, \mathbf{m}_1}(\text{ct}) = b$ and 1 otherwise. Observe that the distinguishing decoder wins with advantage ϵ if ct is encrypted using Enc and with advantage $1 - p_0$ if ct is encrypted using $\text{TrEnc}(\mathcal{R}_0)$ (since in this case, it returns 1 if and only if $\mathcal{D}$ fails). Therefore, the distinguishability advantage of the distinguishing decoder is $\frac{1}{2}\epsilon + \frac{1}{2}(1 - p_0) \geq \frac{1}{16}\epsilon$, as required.

Next, we show that if Case 2 holds, then:

- If $\hat{p}_{n-t} \geq \frac{\epsilon}{4}$ holds, then the adversary can break the message privacy of the scheme. Indeed, since in this case, the adversary can distinguish between $\text{TrEnc}(\mathbf{m}_0, \mathcal{R}_{n-t})$ and $\text{TrEnc}(\mathbf{m}_1, \mathcal{R}_{n-t})$ with less than t parties (since the first $n-t$ parties are revoked and, according to the condition, not all the remaining t parties are colluders).
- If $\hat{p}_{i-1} - \hat{p}_i \geq \frac{\epsilon}{4(n-t)}$ for some honest party, then the adversary can construct a distinguishing decoder in the membership hiding game with distinguishing advantage $\frac{\epsilon}{16(n-t)} > \frac{\epsilon}{16n}$. To see this, we first show using the Chernoff bound that $\hat{p}_i - p_i < \frac{\epsilon}{16(n-t)}$ with overwhelming probability. Thus, $p_{i-1} - p_i \geq \frac{\epsilon}{8(n-t)}$ with overwhelming probability. The distinguishing decoder works as follows. Given a ciphertext ct , which is either $\text{TrEnc}(\mathbf{m}_b, \mathcal{R}_{i-1})$ or $\text{TrEnc}(\mathbf{m}_b, \mathcal{R}_i)$ for $b \in \{0, 1\}$, it outputs 0 if $\mathcal{D}^{\mathbf{m}_0, \mathbf{m}_1}(\text{ct}) = b$ and 1 otherwise. Then, the distinguishing decoder wins with probability $\frac{1}{2}p_{i-1} + \frac{1}{2}(1 - p_i) \geq \frac{\epsilon}{16(n-t)} > \frac{\epsilon}{16n}$. This holds since the probability to output 0 when ct is encrypted to $\mathcal{R}_{i-1}$ is exactly the advantage of the decoder $\mathcal{D}^{\mathbf{m}_0, \mathbf{m}_1}$ in decrypting ct , which, by definition, happens with probability p_{i-1} . On the other hand, when ct is encrypted to $\mathcal{R}_i$, the distinguishing decoder returns 1 only when the decoder $\mathcal{D}$ fails to decrypt, which, by definition, happens with probability $1 - p_i$.

5.2 Generalizing Traceability to k Colluders

We now show how our Π_{TSTE} scheme can be extended to support k -traceability, meaning it can catch up to k traitors for any $1 \leq k \leq t$. The key idea is based on the tracing algorithm Trace (Figure 6),

which runs for at most $n - t$ iterations. In the i -th iteration, it revokes i parties, leaving at least t parties unrevoked throughout. Thus, if a traitor is found in Condition 1, it cannot be one of these t .

We assume (w.l.o.g.) that each execution of **Trace** returns a single traitor (except when Condition 2 applies, in which case t traitors are returned). To identify k traitors, we run **Trace** k times, each time excluding previously found traitors. Let $\mathcal{T}_i$ be the set of traitors found in the first i runs. For the $(i+1)$ -th run, we use only revocation sets that exclude all members of $\mathcal{T}_i$. Specifically, we define

$$S_{\mathcal{R}} = \{\emptyset, \{j_1\}, \{j_1, j_2\}, \dots, \{j_1, \dots, j_{n-t}\}\},$$

where the revocation order $[j_1, \dots, j_{n-t}]$ excludes all $j_v \in \mathcal{T}_i$. This ensures each execution finds a new traitor, resulting in k distinct traitors after k runs.

To enable this, the underlying OSTE scheme must support all revocation sets used in all executions. That is, all these revocation sets must be included in $\mathcal{X}$, as per Definition 4.5. We call such $\mathcal{X}$ a k -trace set.

Theorem 5.9 *Let $k \in \mathbb{N}$ with $1 \leq k \leq t \leq n$. If there exists an OSTE scheme satisfying $(\epsilon/16n)$ -membership hiding and $(\epsilon/16)$ -indistinguishability for a k -trace set $\mathcal{X}$, then there exists a traceable silent setup threshold encryption scheme with threshold t that can trace at least k traitors.*

As an instance of Theorem 5.9, in Section 6 we present an OSTE construction that satisfies membership hiding for $\mathcal{X} = \mathcal{P}([n])$, enabling tracing of up to t traitors.

6 The Baseline OSTE Construction

In this section, we present our Π_{OSTE} construction. Our construction is based on the silent threshold encryption of [30]. We start by presenting a high-level overview of the construction of [30]. Then, we discuss the modifications we make to achieve the properties of OSTE.

6.1 Overview of the STE Scheme

The Setting. The construction relies on a powers-of- τ trusted setup, i.e., the CRS is $([1]_{1,2}, [\tau]_{1,2}, \dots, [\tau^{n+1}]_{1,2})$, in an asymmetric pairing ensemble. Then, each party $i \in [n]$ samples its secret and public shares $(\text{sk}_i, \text{pk}_i)$, computes a $\text{hint}_i = ([\text{sk}_i \tau]_1, \dots, [\text{sk}_i \tau^n]_1)$ using the CRS, and publishes pk_i and hint_i . hint_i is an auxiliary information used during encryption and decryption. From $\{\text{pk}_i, \text{hint}_i\}_{i \in [n]}$, each entity can compute an encryption key $\text{ek} = \sum_{i \in [0, n]} [\text{sk}_i L_i(\tau)]_1$. Furthermore, an aggregation key ak that corresponds to ek is computed from the hints. Then, using ek , anyone can encrypt a message that is decrypted using t secret shares and ak . Note that both ek and ak are public.

The Intuition. The idea follows that of the Boneh-Franklin identity-based encryption scheme [9], which is essentially a witness-encryption to the statement “I know a BLS signature on the identity” [10]. In particular, in [30], the decryption key is basically a witness that the decryptor knows σ^* and aPK such that σ^* is verified as $[\gamma]_2 \circ \text{aPK} = [1]_1 \circ \sigma^*$, where σ^* is an aggregation of at least t partial signatures for a set of signers $S = \{i_1, \dots, i_t\}$, where $i_j \in [n]$, and aPK is an aggregated public key that combines $\{\text{pk}_0, \text{pk}_{i_1}, \dots, \text{pk}_{i_t}\}$.

Linear Constraints. In order to compute a witness for knowing such an aPK , the aggregator computes polynomials B, Q_x , and Q_Z that fulfill the sumcheck equation (Lemma 3.1) as follows:

$$\text{ek} \circ [B(\tau)]_2 = \text{aPK} \circ [1]_2 + [Q_x(\tau)]_1 \circ [\tau]_2 + [Q_Z(\tau)]_1 \circ [Z(\tau)]_2 \quad (1)$$

Notably, B is the polynomial extension of the set vector (i.e., $B(\omega^i) = 0$ where $i \notin S$); all evaluations can be computed using the hints and the CRS. This proves that if σ^* is verified, then aPK is indeed an inner product of ek and the set S , which means that aPK is an aggregation of the public keys in S . Additionally, three critical constraints need to be verified linearly: First, that B represents an authorized subset. That is, the polynomial B is non-zero in at least t positions, which is achieved by a degree check that $\deg(B) \leq n - t$. Second, the degree of Q_x is at most $|\mathbb{H}| - 2$, which is a requirement of the sumcheck lemma. Third, that $B(\omega^0) = 1$ (including a dummy party) to ensure that the trivial solution of setting

B to the zero-polynomial is not valid. For notation, we henceforth assume that $\text{sk}_0 = 1$ for the dummy party. We summarize how to verify these constraints linearly in Appendix C. Each one of the restrictions is represented as a linear equation in the matrix A , which we present next.

Encryption and Decryption. The encryption proceeds as follows:

- Given a message $m \in \mathbb{G}_T$, sample a random tag $\gamma \in \mathbb{Z}_p$ and compute:

$$A = \begin{pmatrix} \text{ek}(\tau) & [1]_2 & [Z(\tau)]_2 & [\tau]_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & [\tau]_2 & [1]_2 & 0 & 0 & 0 \\ 0 & [\gamma]_2 & 0 & 0 & 0 & [1]_1 & 0 & 0 \\ [\tau^{t+1}]_1 & 0 & 0 & 0 & 0 & 0 & [1]_2 & 0 \\ [1]_1 & 0 & 0 & 0 & 0 & 0 & 0 & [\tau - \omega^0]_2 \end{pmatrix} \begin{array}{l} \text{// aPK valid (sumcheck)} \\ \text{// } \deg(Q_x) \leq |\mathbb{H}| - 2 \\ \text{// } \sigma^* \text{ is valid} \\ \text{// } B \text{ is authorized} \\ \text{// dummy party} \end{array}$$

Note that all the entries in A are fixed (the same for all ciphertexts) except for $[\gamma]_2$, which must be unique for every ciphertext.

- Sample $s = (s_1, \dots, s_5) \xleftarrow{\$} \mathbb{Z}_p^5$ and compute:

$$\text{ct} = (\text{ct}_1, \text{ct}_2, \text{ct}_3) = ([\gamma]_2, s^\top \cdot A, s^\top \cdot b + m),$$

where $b = ([0]_T, [0]_T, [0]_T, [0]_T, [1]_T)^\top$.

For decryption, the aggregator collects partial decryptions from a set of parties $S \subseteq [n]$, where $|S| \geq t$. The partial signature of the i -th party is simply $\sigma_i \leftarrow \text{sk}_i \cdot [\gamma]_2$. Then, the aggregator computes

$$m = \text{ct}_3 - (\text{ct}_2 \circ w) = [s_5]_T + m - (s^\top A \circ w) = [s_5]_T + m - [s_5]_T = m,$$

where w is the witness vector such that $A \circ w = b$. Importantly, such w can be computed using the aggregation key ak as well as σ^* , and it guarantees that all constraints in A are fulfilled.

Rand(CRS, $\{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{R}$)

$\alpha \xleftarrow{\$} \mathbb{Z}_p^*$; $\forall i \in \mathcal{R}: \bar{\alpha}_i \xleftarrow{\$} \mathbb{Z}_p^*$

Let $r_i \leftarrow \alpha \ \forall i \in [0, n] \setminus \mathcal{R}$ // Randomize with α for unrevoked parties.

Let $r_i \leftarrow \bar{\alpha}_i \ \forall i \in \mathcal{R}$ // Randomize with $\bar{\alpha}_i$ for revoked parties i .

for $i = 0, \dots, n$ **do**

$\text{ak}_{i,0} \leftarrow r_i \text{pk}_i$; $\text{ak}_{i,1} \leftarrow [r_i \text{sk}_i (L_i(\tau) - L_i(0))]_1$; $\text{ak}_{i,2} \leftarrow \left[r_i \text{sk}_i \frac{L_i(\tau) - L_i(0)}{\tau} \right]_1$

$\text{ak}_{i,3} \leftarrow \left[r_i \text{sk}_i \frac{L_i(\tau)^2 - L_i(0)}{Z(\tau)} \right]_1$; $\text{ak}_{i,4} \leftarrow \left[\sum_{j \in [n], j \neq i} r_j \text{sk}_j \frac{L_i(\tau) L_j(\tau)}{Z(\tau)} \right]_1$

$\text{ek} \leftarrow \sum_{i \in [0, n]} [r_i \text{sk}_i L_i(\tau)]_1$

return $([\alpha]_1, \text{ek}, \text{ak} := \{(\text{ak}_{i,0}, \dots, \text{ak}_{i,4})\}_{i \in [0, n]})$

Figure 7: The rerandomization algorithm. Note that all group elements can be computed using the powers-of- τ CRS and hints. The ak and ek are the same as in [30] with the rerandomization highlighted in gray.

6.2 Description of the Construction

We first present a construction with ciphertext size linear in n , and later show in Sections 7 and 8 how to reduce it to $O(1)$.

In the STE scheme from [30], the aggregation key ak reveals which parties can decrypt. Looking at the encryption key $\text{ek} = \sum_{i \in [0, n]} [\text{sk}_i L_i(\tau)]_1$, suppose we want to revoke a set $\mathcal{R} \subseteq [n]$ (excluding the

dummy party). A natural idea is to randomize the public keys $\mathbf{pk}_j$ for $j \in \mathcal{R}$ and use *new* public keys in computing $\mathbf{ek}$ instead of $[\mathbf{sk}_j]_1$. Then, revoked parties lack the secret keys for the new public keys and cannot decrypt. However, for correctness, $\mathbf{ak}$ must be updated accordingly. This reveals which keys were changed, breaking membership hiding.

To address this, we rerandomize all public keys: sample a random α and a vector $\bar{\alpha} = (\bar{\alpha}_j)_{j \in \mathcal{R}}$, rerandomizing unrevoked parties' public keys with α , and those of $j \in \mathcal{R}$ with $\bar{\alpha}_j$. The $\mathbf{ek}$ is computed with respect to the *new* public keys, the hints are updated accordingly, and we attach the resulting $\mathbf{ak}$ in the ciphertext.

This preserves membership hiding: without knowing α , $\bar{\alpha}_i$, or $\mathbf{sk}_i$, but only $[\alpha]_1$ and $[\mathbf{sk}_i]_1$, an adversary cannot tell if $\mathbf{pk}'_i$ equals $\alpha \cdot \mathbf{pk}_i$ or $\bar{\alpha}_i \cdot \mathbf{pk}_i$. We reduce this to the XDH assumption in Lemma 6.2.

Correctness is preserved by verifying partial signatures against $[\alpha]_1$ instead of $[1]_1$, so unrevoked parties' partial signatures are accepted. For message privacy, the aggregator cannot include a signature from any $j \in \mathcal{R}$, since $\mathbf{aPK}$ must equal $\alpha \cdot \mathbf{aPK}$ (where $\mathbf{aPK}$ is the aggregate public key before the rerandomization), meaning only α -masked keys contribute.

Finally, we highlight that the baseline construction has linear ciphertext size in the worst case. We show in Section 7 how to achieve constant-size ciphertext with a preprocessing technique. In Section 8, we show how ciphertexts can be amortized to constant size, outlining an epoch-based approach.

Enc (CRS, $\{\mathbf{pk}_i, \text{hint}_i\}_{i \in [n]}, t, m$)	enc (CRS, $[\alpha]_1, \mathbf{ek}, \mathbf{ak}, t, m$)
$([\alpha]_1, \mathbf{ek}, \mathbf{ak}) \xleftarrow{\$} \text{Rand}(\text{CRS}, \{\mathbf{pk}_i, \text{hint}_i\}_{i \in [n]}, \emptyset)$	$\gamma \xleftarrow{\$} \mathbb{Z}_p$
return enc (CRS, $[\alpha]_1, \mathbf{ek}, \mathbf{ak}, t, m$)	Assemble the matrix A with the following values:
TrEnc (CRS, $\{\mathbf{pk}_i, \text{hint}_i\}_{i \in [n]}, t, m, \mathcal{R}$)	$(A_{0,0}, A_{2,1}, A_{2,5}) \leftarrow (\mathbf{ek}, [\gamma]_2, [\alpha]_1)$
$([\alpha]_1, \mathbf{ek}, \mathbf{ak}) \xleftarrow{\$} \text{Rand}(\text{CRS}, \{\mathbf{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{R})$	$s := (s_1, \dots, s_5) \xleftarrow{\$} \mathbb{Z}_p^5$
return enc (CRS, $[\alpha]_1, \mathbf{ek}, \mathbf{ak}, t, m$)	$\text{ct} \leftarrow ([\gamma]_2, \mathbf{ak}, s^\top \cdot A, s_5 + m)$
	return ct

Figure 8: The **Enc** and **TrEnc** algorithms of Π_{OSTE} . **enc** acts as helper algorithm to both **Enc** and **TrEnc**. In **enc**, we refer to matrix A from Section 6.1.

The Π_{OSTE} Construction. Our construction relies on a rerandomization algorithm (Figure 7) that realizes the rerandomization idea as discussed above. This algorithm is used in **Enc** and **TrEnc** to generate a new encryption key, which is then used to encrypt the message in a similar way to STE (as discussed in Section 6.1). The **Enc** and **TrEnc** algorithms are depicted in Figure 8. Partial decryption works as in [30] by computing $\sigma_i \leftarrow \mathbf{sk}_i \cdot [\gamma]_2$. Aggregation of partial decryption shares is also unchanged compared to [30], but the aggregator now uses the rerandomized aggregation key $\mathbf{ak}$ to compute $\mathbf{aPK}$ and evaluate the polynomials Q_x , Q_Z and other components of the witness w . As such, the aggregation is performed as if $\mathbf{sk}'_i = \alpha \mathbf{sk}_i$ for unrevoked parties and $\mathbf{sk}'_i = \bar{\alpha}_i \mathbf{sk}_i$ for revoked parties.

For completeness, we present the remaining algorithms **Setup**, **KGen**, **IsValid**, **Prep**, **PreVfy**, and **DecAggr**. Note that they are largely similar to those of STE in [30].

Setup. The setup algorithm samples a random $\tau \xleftarrow{\$} \mathbb{Z}_p$ and generate the CRS as

$$([1]_{1,2}, [\tau]_{1,2}, [\tau^2]_{1,2}, \dots, [\tau^{n+1}]_{1,2}).$$

Key Generation. In **KGen**, a party samples a random $\mathbf{sk} \xleftarrow{\$} \mathbb{Z}_p^*$ and computes the public key as $\mathbf{pk} \leftarrow [\mathbf{sk}]_1$ and $\text{hint} \leftarrow ([\mathbf{sk}\tau]_1, [\mathbf{sk}\tau^2]_1, [\mathbf{sk}\tau^n]_1)$. Additionally, it generates a proof in PS_{KGen} as $\pi \xleftarrow{\$} \text{PS}_{\text{KGen}}.\text{Prove}((\text{CRS}, \mathbf{pk}, \text{hint}), \mathbf{sk})$. To validate the correctness of the public key and hint, the **IsValid** algorithm runs $\text{PS}_{\text{KGen}}.\text{Verify}((\text{CRS}, \mathbf{pk}, \text{hint}), \pi)$.

Preprocessing. Our Π_{OSTE} construction has no preprocessing and supports public tracing. Hence we just set $\mathcal{K} := \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}$ and the tracing key $\text{tk} := \perp$. The PreVfy algorithm checks that all public keys are distinct, i.e., $\text{pk}_i \neq \text{pk}_j$ for all $i \neq j$. If so, it outputs 1.

DecAggr. The aggregation algorithm is identical to that of STE but uses the re-randomized aggregation key ak that is part of the ciphertext instead of a global one. The full algorithm is shown in Figure 9.

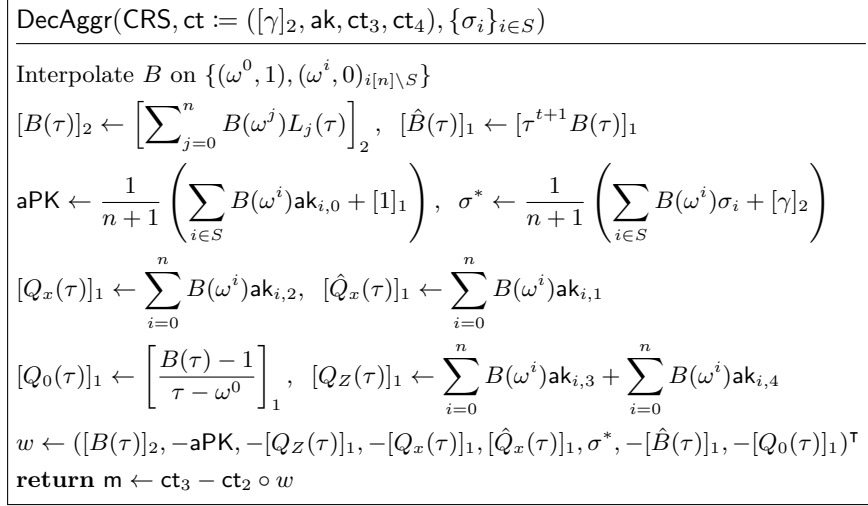


Figure 9: The aggregation algorithm of Π_{OSTE}

Public Tracing of t Traitors. We note that the rerandomization algorithm Rand supports arbitrary revoked sets $\mathcal{R} \in \mathcal{P}([n])$, which in turn allows us to use Π_{OSTE} to guarantee tracing t traitors (see Theorem 5.9). Our OSTE construction supports public tracing and does not have any preprocessing, so we omit the tracing key tk and set the preprocessing to $\mathcal{K} := \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}$. We next discuss the properties of the construction.

Correctness. To see correctness, consider how the aggregator combines t partial decryptions. The aggregator computes an aggregated key $\widehat{\text{aPK}} = \alpha \cdot \text{aPK}$, consistent with the new public keys, and computes the polynomials Q_x and Q_Z as in Equation 1, since the new public keys are consistent with the ek used for encryption. To validate the aggregated signature σ^* (so to satisfy the third restriction), we modify the signature verification equation to $[\alpha]_1 \circ [\sigma^*]_2 = \widehat{\text{aPK}} \circ [\gamma]_2$, replacing $[1]_1$ in $A_{2,5}$ with $[\alpha]_1$. Thus, correctness follows from this and the STE construction of Garg et al. [30].

Rushing Adversary. The construction is vulnerable to a rushing adversary in the membership hiding game GameMH (Figure 2). Specifically, the adversary can choose a corrupted party's key based on an honest party's key, i.e., it sets $\text{pk}_j = x \cdot \text{pk}_i$ and $\text{hint}_j = x \cdot \text{hint}_i$ for arbitrary $x \in \mathbb{Z}_p^*$, $x \neq 1$, with $\mathcal{R}_0 = \emptyset$. Upon receiving a ciphertext $\text{ct} = ([\gamma]_2, \text{ak}, \dots)$, it checks if $\text{ak}_{j,0} = x \cdot \text{ak}_{i,0}$ to determine if i is revoked. If i is not revoked, then $\text{ak}_{i,0} = \alpha \cdot \text{pk}_i$, and since $j \notin \mathcal{R}_0$ (i.e., not revoked), $\text{ak}_{j,0} = \alpha \cdot \text{pk}_j = \alpha x \cdot \text{pk}_i$.

To prevent this, each party must attach a NIZK proof of knowledge of sk to (pk, hint) and ensure public keys are unique. Formally, the proof system PS_{KGen} is defined via relation R_{KGen} over statements $\chi = (\text{CRS}, \text{pk}, \text{hint})$ and witness $w = \text{sk} \in \mathbb{Z}_p^*$, with $(\chi, w) \in \text{R}_{\text{KGen}}$ iff $\text{pk} = [\text{sk}]_1$ and $\text{hint} = ([\text{sk}\tau]_1, \dots, [\text{sk}\tau^n]_1)$. Further, we require that all public keys are distinct for the same reason, i.e. $\text{pk}_i \neq \text{pk}_j$ for all $i \neq j$. If we allow a rushing adversary to copy an honest party's public key, then he could replay the proof and carry out the above attack. Formally, this requirement is verified within PreVfy .

The following theorem summarizes our result for Π_{OSTE} .

Theorem 6.1 *The OSTE scheme Π_{OSTE} , described in Section 6.2, satisfies correctness, δ -indistinguishability, message privacy, δ' -membership hiding, and IND-CPA security for $n = \text{poly}(\lambda)$ and any $\delta, \delta' = 1/\text{poly}(\lambda)$ with $\delta, \delta' \in [-1/2, 1/2]$. The ciphertext size is $|\text{c}| = O(n)$, with amortized size $O(1)$.*

Discussion. The ciphertext size in our Π_{OSTE} construction is linear in the number of parties, primarily due to the aggregation key $\mathbf{ak}$ included in every ciphertext. One natural idea is to reuse the rerandomization triple $([\alpha]_1, \mathbf{ek}, \mathbf{ak})$ across multiple ciphertexts. However, doing so naively enables hardcoding attacks: a decoder may only work for a fixed tuple, rendering it untraceable, since tracing requires fresh rerandomizations.

To address this, we amortize the cost by reusing a single rerandomization across a batch of ciphertexts. In such a setting, we attach the $O(n)$ -sized aggregation key $\mathbf{ak}$ only once per batch, bringing the *amortized* ciphertext size down to $O(1)$. This reuse does not compromise IND-CPA security, even when the party generating the rerandomization is malicious (see Section 7). We further formalize an epoch-based setting where this technique is practical in Section 8.

Proof of Theorem 6.1. We next prove the membership hiding of the scheme. We present a formal proof for the rest of the properties required in Theorem 6.1 in Appendix D.

Lemma 6.2 (Membership Hiding of Π_{OSTE}). *Let $n = \text{poly}(\lambda)$. Our Π_{OSTE} construction, as described in Section 6.2, is membership hiding for arbitrary $\delta = \frac{1}{\text{poly}(\lambda)}$ with $\delta \in [-1/2, 1/2]$ if the XDH assumption (Definition 3.2) holds and PS_{KGen} is zero-knowledge and simulation-extractable (Definition A.1).*

At its core, our membership-hiding proof works by embedding an XDH challenge $[x]_1, [y]_1, T$ into the rerandomization for a ciphertext ct such that x corresponds to $\mathbf{sk}_i$, and y corresponds to α . If the challenge $T = [xy]_1 \equiv [\alpha \mathbf{sk}_i]_1$, then party i is not revoked for ct and if $T = [v]_1$ for a random $v \equiv \bar{\alpha}_i \cdot \mathbf{sk}_i$, then party i is revoked for ct . We query the decoder on ct and a reference ciphertext, where we randomly choose to revoke i or not. We compare the results to determine whether $T = [xy]_1$ or $T = [v]_1$.

Next, we present a formal proof of Lemma 6.2.

Proof. Intuitively, our OSTE construction is membership hiding because an adversary that only learns $[\alpha]_1$ and $[\mathbf{sk}_i]_1$ for some i should be unable to distinguish between $[\alpha \cdot \mathbf{sk}_i]_1$ (reflecting the case where party i is not revoked by TrEnc) and $[\bar{\alpha}_i \cdot \mathbf{sk}_i]_1$ (representing the case where party i is revoked), which is reminiscent of the XDH problem in bilinear groups (Definition 3.2). The same argument applies to the re-randomized $\mathbf{ak}$.

We formally prove this through a series of game hops with a final reduction to the XDH assumption in $\mathbb{G}_1$. Let Game_0 be the membership hiding experiment for the Π_{OSTE} construction $\text{GameMH}^{\Pi_{\text{OSTE}}}$ (Figure 2). We proceed with the first game hop, where we simulate the proof π_i on behalf of the honest party.

Game₁: In this game, we modify how the challenger computes the proof π_i that is attached to $\mathbf{pk}_i$ and hint_i on behalf of the honest party at position i . In particular, we replace π_i by a simulated proof as $\pi_i \xleftarrow{\$} \text{Sim}(\text{td}, \mathbf{pk}_i, \text{hint}_i)$, where Sim is the simulator for the proof system PS_{KGen} .

Claim 6.3 If the proof system PS_{KGen} is zero-knowledge (Definition A.1), then Game_0 and Game_1 are computationally indistinguishable.

Proof. The computational indistinguishability follows directly from the computational indistinguishability of a real proof and a simulated proof as stated by the zero-knowledge property of PS_{KGen} . ■

In our final reduction to XDH, we will need to know the secret keys that are chosen by the adversary for the parties at positions $[n] \setminus \{i\}$ so we can properly embed the XDH challenge into their re-randomized public keys and hints. We use the simulation-extractability of PS_{KGen} to extract $\mathbf{sk}_\ell$ from π_ℓ for all $\ell \in [n] \setminus \{i\}$.

Game₂: In Game_2 , we use the extractor of PS_{KGen} to extract $\{\mathbf{sk}_\ell\}_{\ell \in [n] \setminus \{i\}}$ from the adversary's proof as $\mathbf{sk}_\ell \xleftarrow{\$} \text{PS}_{\text{KGen}}.\text{Ext}(\text{td}, (\text{CRS}, \mathbf{pk}_\ell, \text{hint}_\ell), \pi_\ell)$. We add an additional abort condition, where Game_2 outputs 0, if there exists an ℓ such that $\mathbf{pk}_\ell \neq [\mathbf{sk}_\ell]_1$ or $\text{hint}_\ell \neq ([\mathbf{sk}_\ell \tau]_1, \dots, [\mathbf{sk}_\ell \tau^n]_1)$.

Claim 6.4 For all $n = \text{poly}(\lambda)$ and all PPT adversaries $\mathcal{A}$ there exists a PPT adversary $\mathcal{B}$ such that

$$\begin{aligned} & |\Pr[\text{Game}_{1,\mathcal{A}}(1^\lambda, n, \mathcal{X}, \delta) = 1] - \Pr[\text{Game}_{2,\mathcal{B}}(1^\lambda, n, \mathcal{X}, \delta) = 1]| \\ & \leq (n-1) \cdot \text{Adv}_{\mathcal{B}, \text{PS}_{\text{KGen}}}^{\text{sim-extract}}(1^\lambda). \end{aligned}$$

Proof. First, observe that for all $\mathcal{A}$ it holds that

$$\Pr[\text{Game}_{1,\mathcal{A}}(1^\lambda, n, \mathcal{X}, \delta) = 1] \geq \Pr[\text{Game}_{2,\mathcal{A}}(1^\lambda, n, \mathcal{X}, \delta) = 1],$$

because we only add an additional abort condition. Let $\Pr[\text{Game}_{1,\mathcal{A}}(1^\lambda, n, \mathcal{X}, \delta) = 1] - \Pr[\text{Game}_{2,\mathcal{A}}(1^\lambda, n, \mathcal{X}, \delta) = 1] = \varepsilon$. Then, with probability ε , the adversary $\mathcal{A}$ hits the additional abort condition in Game_2 . We construct a PPT reduction $\mathcal{B}$ that runs in the simulation-extractability game of PS_{KGen} . Given $\text{CRS}_{\text{PS}_{\text{KGen}}}$, $\mathcal{B}$ embeds it in the global CRS and simulates Game_2 to $\mathcal{A}$ as follows. $\mathcal{B}$ runs $\mathcal{A}(1^\lambda, \text{CRS})$ and receives $i \in [n]$. Then, the reduction samples a random $(\text{sk}_i, \text{pk}_i, \text{hint}_i, _) \xleftarrow{\$} \text{KGen}(1^\lambda, \text{CRS})$ and simulates the proof π_i using the $\text{Sim}(\text{td}, (\text{CRS}, \text{pk}_i, \text{hint}_i))$ oracle provided by the simulation-extractability game and sends $(\text{pk}_i, \text{hint}_i, \pi_i)$ to $\mathcal{A}$. In return $\mathcal{B}$ receives $\{\text{pk}_\ell, \text{hint}_\ell, \pi_\ell\}_{\ell \in [n] \setminus \{i\}}$. $\mathcal{B}$ chooses a random $\ell^* \in [n] \setminus \{i\}$ and outputs $(\chi^*, \pi^*) = ((\text{CRS}, \text{pk}_{\ell^*}, \text{hint}_{\ell^*}), \pi_{\ell^*})$ to the simulation-extractability game.

Observe that $\chi^* = (\text{CRS}, \text{pk}_{\ell^*}, \text{hint}_{\ell^*}) \notin Q = \{(\text{CRS}, \text{pk}_i, \text{hint}_i)\}^2$, whenever either $\text{Game}_{1,\mathcal{A}}$ or $\text{Game}_{2,\mathcal{A}}$ would output 1 because of the check in PreVfy that $\text{pk}_i \neq \text{pk}_\ell$ for all $i \neq \ell$. Further, it holds that with probability ε there exists an $\ell' \in [n] \setminus \{i\}$ such that $(\chi_{\ell'}, \text{sk}_{\ell'}) \notin \text{R}_{\text{KGen}}$. $\mathcal{B}$ chooses $\ell^* = \ell'$, hence $\mathcal{B}$ wins the simulation-extractability game with probability at least $\varepsilon/(n-1)$. ■

```

SimRand ( $[x]_1, [y]_1, T, \tau, \{\text{sk}_\ell\}_{\ell \in [n] \setminus \{i\}}, \mathcal{R}, i$ )
  Sample  $\bar{\alpha}_\ell \xleftarrow{\$} \mathbb{Z}_p^* \forall \ell \in \mathcal{R}$ 
  for  $\ell \in [0, n] \setminus \mathcal{R}$  with  $\ell \neq i$  do
     $\text{ak}_{\ell,0} \leftarrow \text{sk}_\ell \cdot [y]_1 \quad (= [\alpha \text{sk}_\ell]_1)$ 
     $\text{ak}_{\ell,1} \leftarrow \text{sk}_\ell (L_\ell(\tau) - L_\ell(0)) \cdot [y]_1 \quad (= [\alpha \text{sk}_\ell (L_\ell(\tau) - L_\ell(0))]_1)$ 
     $\text{ak}_{\ell,2} \leftarrow \text{sk}_\ell \frac{L_\ell(\tau) - L_\ell(0)}{\tau} \cdot [y]_1$ 
     $\text{ak}_{\ell,3} \leftarrow \text{sk}_\ell \frac{L_\ell(\tau)^2 - L_\ell(\tau)}{Z(\tau)} \cdot [y]_1$ 
     $\text{ak}_{\ell,4} \leftarrow \sum_{j \in [0, n] \setminus \mathcal{R}, j \neq \ell, j \neq i} \text{sk}_j \frac{L_\ell(\tau) L_j(\tau)}{Z(\tau)} \cdot [y]_1$ 
     $\quad + \sum_{j \in \mathcal{R}} \bar{\alpha}_j \text{sk}_j \frac{L_\ell(\tau) L_j(\tau)}{Z(\tau)} + \frac{L_\ell(\tau) L_i(\tau)}{Z(\tau)} \cdot T$ 
  for  $\ell \in \mathcal{R}$  do
     $\text{ak}_{\ell,0} \leftarrow [\bar{\alpha}_\ell \text{sk}_\ell]_1$ 
    ...
   $\text{ak}_{i,0} \leftarrow T$ 
   $\text{ak}_{i,1} \leftarrow (L_i(\tau) - L_i(0)) \cdot T$ 
  ...
   $\text{ek} \leftarrow \sum_{\ell \in [0, n] \setminus \mathcal{R}, \ell \neq i} \text{sk}_\ell L_\ell(\tau) [y]_1 + \sum_{\ell \in \mathcal{R}} [\bar{\alpha}_\ell \text{sk}_\ell L_\ell(\tau)]_1 + L_i(\tau) \cdot T$ 
  return ( $[y]_1, \text{ek}, \text{ak}$ )

```

Figure 10: The simulation of the rerandomization algorithm for the challenge ciphertext, where α corresponds to y . Note that the reduction can compute all values because it knows τ and $\{\text{sk}_\ell\}_{\ell \in [n] \setminus \{i\}}$.

We conclude our proof with a reduction from Game_2 to the XDH assumption (Definition 3.2). Let $\mathcal{A}$ be a PPT adversary against Game_2 . We construct a PPT reduction $\mathcal{B}$ that internally uses $\mathcal{A}$ to break the XDH assumption. Let b_{XDH} be the internal bit of the XDH experiment. Initially, $\mathcal{B}$ receives an XDH challenge $([x]_1, [y]_1, T)$, where either $T = [xy]_1$ for $(b_{\text{XDH}} = 0)$ or $T = [v]_1$ for $(b_{\text{XDH}} = 1)$.

First, $\mathcal{B}$ samples $\tau \xleftarrow{\$} \mathbb{Z}_p$ and generates the CRS. It queries $\mathcal{A}(1^\lambda, \text{CRS})$ to receive $i \in [n]$. $\mathcal{B}$ now embeds the XDH challenge into the public key and hint that it generates on behalf of party i . In particular,

²With Q , we refer to the set of statements that the reduction requests a simulated proof for (see Definition A.1).

$\mathcal{B}$ sets $\mathbf{pk}_i \leftarrow [x]_1$ and $\text{hint}_i \leftarrow (\tau[x]_1, \dots, \tau^n[x]_1)$. Observe that, in doing so, $\mathcal{B}$ simulates for $\mathbf{sk}_i = x$ without knowing x itself. After sending $\mathbf{pk}_i, \text{hint}_i$ and the simulated proof π_i to $\mathcal{A}$, the reduction receives $\{\mathbf{pk}_\ell, \text{hint}_\ell\}_{\ell \in [n] \setminus \{i\}}$ and the extracted $\{\mathbf{sk}_\ell\}_{\ell \in [n] \setminus \{i\}}$ as well as the set $\mathcal{R}$, and the threshold t . As the preprocessing for Π_{OSTE} is just equal to the public keys and hints of all parties, $\mathcal{B}$ sends those again to $\mathcal{A}$ and receives a decoder $\mathcal{D}$ as well as a message $\mathbf{m}$. In the following, we write $\text{TrEnc}(\mathcal{R})$ as a shorthand for $\text{TrEnc}(\text{CRS}, \text{tk}, \mathcal{K}, t, \mathbf{m}, \mathcal{R})$ as well as $\mathcal{R}^0 = \mathcal{R}$, and $\mathcal{R}^1 = \mathcal{R} \cup \{i\}$.

Next, $\mathcal{B}$ samples a random bit β and computes $\text{ct} \xleftarrow{\$} \text{TrEnc}(\mathcal{R}_\beta)$. Additionally, $\mathcal{B}$ samples a challenge ciphertext c' , embedding the XDH challenge such that, in the rerandomization, α is equivalent to y . Further, T is equivalent to $[\alpha \mathbf{sk}_i]_1$ if $b_{\text{XDH}} = 0$, or $[\bar{\alpha}_i \mathbf{sk}_i]_1$ if $b_{\text{XDH}} = 1$. The details are shown in Figure 10.

Next, $\mathcal{B}$ computes ct' as $\text{enc}(\text{CRS}, [y]_1, \text{ek}, \text{ak}, \mathbf{m})$. Observe that now $\text{ct} = \text{TrEnc}(\mathcal{R}^{b_{\text{XDH}}})$, i.e., if $T = [xy]_1$, then $\mathcal{B}$ simulates $\text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R})$, and if $T = [v]_1$, then $\mathcal{B}$ simulates $\text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R} \cup \{i\})$. Finally, $\mathcal{B}$ outputs $b = \beta$, if $\mathcal{D}(\text{ct}') = \mathcal{D}(\text{ct})$ and $b = 1 - \beta$ otherwise. If $\mathcal{A}$ would lose the game on some technicality, say, duplicate two honest party's public keys, then $\mathcal{B}$ just outputs a random bit. Let p_w be the probability that $\mathcal{B}$ wins under the condition that $\mathcal{A}$ outputs a $\delta - \text{Dist}^{\mathcal{R}, \mathcal{R} \cup i}$ decoder $\mathcal{D}$. Then, the following holds:

$$\begin{aligned}
p_w &= \Pr \left[b = b_{\text{XDH}} \left| \begin{array}{l} \beta \xleftarrow{\$} \{0, 1\}; \text{ct} \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^\beta); \\ b_{\text{XDH}} \xleftarrow{\$} \{0, 1\}; \text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^{b_{\text{XDH}}}); \\ b \leftarrow \beta \text{ if } \mathcal{D}(\text{ct}') = \mathcal{D}(\text{ct}) \text{ else } b \leftarrow 1 - \beta \end{array} \right. \right] \\
&= \Pr \left[\mathcal{D}(\text{ct}') = b_{\text{XDH}} \left| \begin{array}{l} b_{\text{XDH}} \xleftarrow{\$} \{0, 1\}; \\ \text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^{b_{\text{XDH}}}) \end{array} \right. \right] \cdot \Pr \left[\mathcal{D}(\text{ct}) = \beta \left| \begin{array}{l} \beta \xleftarrow{\$} \{0, 1\}; \\ \text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^\beta) \end{array} \right. \right] \\
&+ \Pr \left[\mathcal{D}(\text{ct}') \neq b_{\text{XDH}} \left| \begin{array}{l} b_{\text{XDH}} \xleftarrow{\$} \{0, 1\}; \\ \text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^{b_{\text{XDH}}}) \end{array} \right. \right] \cdot \Pr \left[\mathcal{D}(\text{ct}) \neq \beta \left| \begin{array}{l} \beta \xleftarrow{\$} \{0, 1\}; \\ \text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^\beta) \end{array} \right. \right] \\
&= \Pr \left[\mathcal{D}(\text{ct}) = \beta \left| \begin{array}{l} \beta \xleftarrow{\$} \{0, 1\}; \\ \text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^\beta) \end{array} \right. \right]^2 + \Pr \left[\mathcal{D}(\text{ct}) \neq \beta \left| \begin{array}{l} \beta \xleftarrow{\$} \{0, 1\}; \\ \text{ct}' \xleftarrow{\$} \text{TrEnc}(\mathcal{R}^\beta) \end{array} \right. \right]^2 \\
&= \left(\frac{1}{2} + \delta \right)^2 + \left(\frac{1}{2} - \delta \right)^2 \\
&= \frac{1}{4} + \delta + \delta^2 + \frac{1}{4} - \delta + \delta^2 \\
&= \frac{1}{2} + 2\delta^2
\end{aligned}$$

We get the following advantage for $\mathcal{B}$ against XDH.

$$\begin{aligned}
&\frac{1}{2} + \mathbf{Adv}_{\mathcal{B}}^{\text{XDH}}(1^\lambda, G) \\
&\geq \mathbf{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, \delta) \cdot p_w + \left(1 - \mathbf{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, \delta) \right) \left(\frac{1}{2} \right) \\
&= \mathbf{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, \delta) \left(\frac{1}{2} + 2\delta^2 \right) + \left(1 - \mathbf{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, \delta) \right) \left(\frac{1}{2} \right) \\
&= \frac{1}{2} + \mathbf{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, \delta) \cdot 2\delta^2
\end{aligned}$$

Summarizing, we conclude that for all adversaries $\mathcal{A}$ there exist PPT reductions $\mathcal{B}_1, \mathcal{B}_2$, and $\mathcal{B}_3$ such that

$$\begin{aligned}
&\mathbf{Adv}_{\mathcal{A}}^{\text{GameMH}^{\Pi_{\text{OSTE}}}}(1^\lambda, n, \mathcal{P}([n]), \delta) \leq \\
&\mathbf{Adv}_{\mathcal{B}_1}^{\text{PSKGen-zk}}(1^\lambda) + (n-1) \cdot \mathbf{Adv}_{\mathcal{B}_2}^{\text{PSKGen-sim-ext}}(1^\lambda) + \frac{1}{2\delta^2} \mathbf{Adv}_{\mathcal{B}_3}^{\text{XDH}}(1^\lambda, \mathcal{G}).
\end{aligned}$$

■

7 The Preprocessing-based OSTE Construction

In this section, we present our main OSTE construction, denoted by Π_{OSTEP} , in which we add a preprocessing phase to reduce the size of the ciphertext of the Π_{OSTE} construction. We summarize the model using the following definition.

Definition 7.1 (OSTE with Preprocessing (OSTEP)). An OSTE scheme is said to be an *OSTE scheme with preprocessing* (OSTEP) if, after all parties publish their public keys and auxiliary setup information (e.g., hints), a *preprocessing phase* computes and publishes a polynomial-size set of encryption keys and their corresponding aggregation keys. The encryption procedures **Enc** and **TrEnc** are then restricted to using only these pre-generated encryption keys. No further encryption keys are generated after the preprocessing phase.

Let $\mathcal{L} = \{\emptyset, \{1\}, \{1, 2\}, \dots, [n]\}$ be the linear revocation order. Throughout this section, we imply $\mathcal{X} = \mathcal{L}$ when referring to membership hiding and message privacy.

The following theorem summarizes the main result of this section.

Theorem 7.2 *The Π_{OSTEP} construction is an OSTEP scheme with $(\epsilon/16)$ -indistinguishability (Definition 4.4) and $(\epsilon/16n)$ -membership hiding for $\mathcal{X} = \mathcal{L}$ (Definition 4.5), where $n = \text{poly}(\lambda)$, $\lambda \in \mathbb{N}$, and $\epsilon = 1/\text{poly}(\lambda)$ with $\epsilon \in [-1/2, 1/2]$. The secret key size is 1 group element per party, public key size $O(n^4)$, and ciphertext size $O(1)$.*

As a corollary of Theorem 5.7 and Theorem 7.2, we achieve a traceable silent threshold encryption scheme (per Definition 5.1), where ϵ is the decoder's advantage in the traceability game (Figure 5).³

7.1 The Π_{OSTEP} Construction

Overview. Let $\mathcal{R}_j = [j]$ denote the set of revoked parties up to j , with $\mathcal{R}_0 = \emptyset$. A ciphertext **ct** *revokes* a set $\mathcal{R}$ if no $p_i \in \mathcal{R}$ can contribute a valid decryption share. An encryption key **ek** *revokes* $\mathcal{R}$ if all ciphertexts under **ek** revoke $\mathcal{R}$.

We preprocess multiple rerandomizations for different $\mathcal{R}$ sets. During encryption, a key **ek** is randomly chosen from the preprocessed set. Since **ek** may revoke some parties, the message is encrypted under multiple keys to amplify decryption success. For tracing, the tracer uses designated keys from the same set.

The main challenge is designing **Prep**, **Enc**, and **TrEnc** to ensure indistinguishability, membership hiding, and message privacy. The Π_{OSTEP} construction appears in Figures 11 and 12. We now describe its main components.

Preprocessing. The preprocessing algorithm **Prep** (Figure 11) uses the rerandomization algorithm from Section 6.2 (Figure 7) to generate $2N$ encryption keys and their corresponding aggregation keys. Of these, N keys (grouped in K_{TR}) are for tracing: for each $0 \leq j \leq n$, we create $N/(n+1)$ keys that revoke $\mathcal{R}_j$. Another N keys, in K_{NORM} , revoke no party (i.e., they revoke $\mathcal{R}_0$) and are not used for tracing. Thus, keys revoking $\mathcal{R}_0$ appear in both K_{TR} and K_{NORM} ; the ones in K_{TR} are used by **TrEnc**($\cdot, \mathcal{R}_0$) queries (see Figure 6). The full key set is $K_{\text{TR}} \cup K_{\text{NORM}}$.

To hide key types, we publish them in random order: $\mathcal{K} \xleftarrow{\$} \text{Shuffle}(K_{\text{TR}} \cup K_{\text{NORM}})$. The original order of K_{TR} and K_{NORM} is kept as the private tracing key. Sets are treated as sequences, so indices are stored (e.g., the permutation from **Shuffle**).

A malicious entity could generate and inject known keys. To prevent this, the rerandomization algorithm must include a NIZK proof of knowledge of $w = (\mathcal{R}, \alpha, \bar{\alpha} := \{\bar{\alpha}_\ell\}_{\ell \in \mathcal{R}})$ to the statement $\chi = (\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, [\alpha]_1, \text{ek}, \text{ak})$ for every $([\alpha]_1, \text{ek}, \text{ak}) \in \mathcal{K}$. The relation $\mathbf{R}_{\text{Rand}}$ for corresponding proof system PS_{Rand} (Definition A.1) is defined as

$$\mathbf{R}_{\text{Rand}} = \left\{ (\chi, w) : ([\alpha]_1, \text{ek}, \text{ak}) = \text{Rand}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}; \alpha, \bar{\alpha}) \right\}.$$

We define **RandAndProve**, extending **Rand** (Figure 7), to attach this proof. This ensures the adversary cannot inject its own public keys to break semantic security or message privacy. We show this formally in the proofs of semantic security and message privacy in Appendix D.

³Our construction requires knowing the decoder's advantage ϵ in advance. This requirement can be eliminated using the Zhandry's compiler [46], without affecting the parameters' sizes.

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Prep(CRS,  $\{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \epsilon$ )


---


 $N \leftarrow 1024 \cdot (n+1) \lambda \lambda_c^2 n^2 / \epsilon^2$ ;  $d \leftarrow N/(n+1)$ 
for  $j = 0, \dots, n$  do
  for  $\ell \in [d]$  do
     $([\alpha]_1, \text{ek}, \text{ak}, \pi) \xleftarrow{\$} \text{RandAndProve}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{R}_j)$ 
     $K_{\text{TR}}[j, \ell] \leftarrow ([\alpha]_1, \text{ek}, \text{ak}, \pi)$ 
  Denote by  $K_{\text{TR}}[j]$  the set  $\{K_{\text{TR}}[j, \ell]\}_{\ell \in [d]}$  of tracing keys that revoke the set  $\mathcal{R}_j$ .
for  $\ell \in [N]$  do
   $([\alpha]_1, \text{ek}, \text{ak}, \pi) \xleftarrow{\$} \text{RandAndProve}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{R}_0)$ 
   $K_{\text{NORM}}[\ell] \leftarrow ([\alpha]_1, \text{ek}, \text{ak}, \pi)$ 
 $\mathcal{K} \xleftarrow{\$} \text{Shuffle}(K_{\text{NORM}}, K_{\text{TR}})$ 
return  $(\mathcal{K}, \text{tk} := (K_{\text{NORM}}, K_{\text{TR}}), \pi := \{\pi_\ell\}_{\ell \in [|\mathcal{K}|]})$ 

```

Figure 11: The preprocessing of the Π_{OSTEP} construction. The **Shuffle** algorithm computes a random permutation. The sets K_{NORM} , K_{TR} , and $\mathcal{K}$ are interpreted as sequences and the **RandAndProve** algorithm is as defined above.

Enc (CRS, $\mathcal{K}, t, m$)	TrEnc (CRS, $\text{tk} := (K_{\text{NORM}}, K_{\text{TR}}), \mathcal{K}, t, m, \mathcal{R}_j$)
for $\ell \in [\lambda_c]$ do $\text{idx} \xleftarrow{\$} [\mathcal{K}]$ without repetitions $([\alpha]_1, \text{ek}, \text{ak}) \leftarrow \mathcal{K}[\text{idx}]$ (c_1, c_2, c_3, c_4) $\leftarrow \text{enc}(\text{CRS}, [\alpha]_1, \text{ek}, \text{ak}, m)$ $\text{// Replace } c_2 = \text{ak, with idx.}$ $\text{ct}_\ell \leftarrow (c_1, \text{idx}, c_3, c_4)$ return $\text{ct} \leftarrow (\text{ct}_1, \dots, \text{ct}_{\lambda_c})$	for $\ell \in [\lambda_c]$ do Sample as follows without repetitions: With prob. $1/2 + (j+1)/(2n+2)$ do $k := ([\alpha]_1, \text{ek}, \text{ak}) \xleftarrow{\$} K_{\text{TR}}[j]$ With prob. $(n-j)/(2n+2)$ do $k := ([\alpha]_1, \text{ek}, \text{ak}) \xleftarrow{\$} \bigcup_{v \in \{j+1, \dots, n\}} K_{\text{TR}}[v]$ $\text{idx} \leftarrow \text{Index of } k \text{ in } \mathcal{K}$ $(c_1, c_2, c_3, c_4) \leftarrow \text{enc}(\text{CRS}, [\alpha]_1, \text{ek}, \text{ak}, m)$ $\text{ct}_\ell \leftarrow (c_1, \text{idx}, c_3, c_4)$ $\text{// Replace } c_2 = \text{ak, with idx.}$ return $\text{ct} \leftarrow (\text{ct}_1, \dots, \text{ct}_{\lambda_c})$

Figure 12: The **Enc** and **TrEnc** algorithms of Π_{OSTEP} with respect to correctness parameter λ_c . The helper function **enc** is as defined for Π_{OSTE} in Figure 8.

Encryption. The encryption algorithms are presented in Figure 12. In normal encryption (**Enc**), we randomly select λ_c keys from $\mathcal{K}$ and encrypt under each. We replace the **ak** part with the corresponding index, making ciphertext size constant in n . Since some keys revoke parties, encrypting under multiple keys is required to amplify the decryption's correctness.

In **TrEnc**, we encrypt the message m with respect to the revoked set $\mathcal{R}_j$. Note that the execution of **TrEnc** requires the tracing key $\text{tk} = (K_{\text{NORM}}, K_{\text{TR}})$, i.e., the tracer needs to know which keys in $\mathcal{K}$ revoke which parties. Denote by $K_{\text{TR}}[j]$ the set of keys that revoke the set $\mathcal{R}_j$.

A naive approach to encrypt with λ_c keys sampled from $K_{\text{TR}}[j]$ ensures message privacy but fails at indistinguishability. Indeed, normal encryptions would likely include keys that revoke $\mathcal{R}_j$ for $j > 0$. An adversary with $\mathcal{K}$, the indices of the λ_c keys, and some secret keys could then detect if revocations occurred. Thus, it can distinguish $\text{TrEnc}(\mathcal{R}_0)$ from a normal ciphertext encrypted using **Enc**. To prevent this, $\text{TrEnc}(\mathcal{R}_0)$ samples keys from all $K_{\text{TR}}[j]$ sets using the following distribution. For all k_i sampled by $\text{TrEnc}(\mathcal{R}_0)$, it holds that:

- $k_i \in K_{\text{TR}}[0]$ with probability $\frac{1}{2} + \frac{1}{2n+2}$,
- $k_i \in K_{\text{TR}}[j]$ with probability $\frac{1}{2n+2}$ for all $1 \leq j \leq n$.

This mimics the distribution generated by Enc , where a key revokes $\mathcal{R}_0$ with probability $\frac{1}{2} + \frac{1}{2n+2}$ (i.e., $k_i \in K_{\text{NORM}} \cup K_{\text{TR}}[0]$), and revokes any other $\mathcal{R}_j$ with probability $\frac{1}{2n+2}$. These distributions are indistinguishable to the adversary due to the indistinguishability guarantees of the underlying OSTE scheme: if k_j revokes no party, it is unclear whether $k_j \in K_{\text{TR}}[0]$ or $k_j \in K_{\text{NORM}}$.

This now introduces the same problem in membership hiding: by a similar argument, an adversary that holds some keys but not sk_1 , can distinguish between $\text{TrEnc}(\mathcal{R}_0)$ and $\text{TrEnc}(\mathcal{R}_1)$. Then, we extend this technique to ensure membership hiding: when encrypting with $\text{TrEnc}(\mathcal{R}_j)$, we also sample keys from $K_{\text{TR}}[j']$ for $j' > j$, mimicking the distribution of keys in $\text{TrEnc}(\mathcal{R}_{j-1})$. By the membership hiding property of the underlying Π_{OSTE} scheme, an adversary lacking sk_j cannot distinguish between keys in $K_{\text{TR}}[j]$ and $K_{\text{TR}}[j-1]$, ensuring key distributions remain indistinguishable up to negligible differences.

7.2 Proof of Theorem 7.2

The correctness of the scheme is straightforward given the correctness of the Π_{OSTE} construction. The correctness is lower bounded by $1 - \frac{1}{2^{\lambda_c}}$ since the probability of sampling a key that does not revoke any party is at least $1/2$. Given a set of partial decryptions $\{\sigma_u\}_{u \in S}$, we can easily check if party u is unrevoked for key k_i by checking whether $[\gamma_i]_2 \circ \text{pk}_u = [\alpha_i]_1 \circ \sigma_u$. Then, we aggregate and decrypt any of the ciphertexts for which we have sufficiently many unrevoked partial decryptions. We show that Π_{OSTEP} satisfies IND-CPA, $(\epsilon/16)$ -indistinguishability (Definition 4.4), $(\epsilon/16n)$ -membership hiding (Definition 4.5), and message privacy (Definition 4.6). The message privacy and IND-CPA security proofs appear in Appendix E. We next prove indistinguishability and membership hiding.

$(\epsilon/16)$ -Indistinguishability. Let $c_0 = \text{Enc}(m)$ and $c_1 = \text{TrEnc}(m, \mathcal{R}_0)$, using encryption keys EK_0 and EK_1 , respectively. Since keys in $K_{\text{TR}}[0]$ are indistinguishable from those in K_{NORM} , distinguishing c_0 from c_1 requires a statistical distance of at least $\epsilon/16$ between EK_0 and EK_1 . Using the Chernoff bound, we show that this occurs only with negligible probability. The full proof is given in Lemma 7.3

$(\epsilon/(16n))$ -Membership Hiding. The proof proceeds in two steps. First, we show that an adversary without sk_i cannot distinguish preprocessing keys in $K_{\text{TR}}[i-1]$ from those in $K_{\text{TR}}[i]$. Then, using Chernoff bound, we show that the adversary cannot distinguish $\text{TrEnc}(\mathcal{R}_{i-1})$ from $\text{TrEnc}(\mathcal{R}_i)$ with advantage greater than $\epsilon/(16n)$. The full proof is given in Lemma 7.4.

Lemma 7.3 *The Π_{OSTEP} construction, as presented in Section 7.1, is $(\epsilon/16)$ -indistinguishable for $\epsilon = 1/\text{poly}(\lambda)$, per Definition 4.4, for some $d = O(\lambda_c^2 \lambda n^2 / \epsilon^2)$.*

Proof. Consider $\delta = \frac{\epsilon}{16}$ for an $\epsilon \in \frac{1}{\text{poly}(\lambda)}$. Let $n = \text{poly}(\lambda)$ and let $\mathcal{A}$ be a PPT adversary. We show for construction Π_{OSTEP} that $\text{Adv}_{\mathcal{A}}^{\text{GameInd}(1^\lambda, n, \delta)} \leq \text{negl}(\lambda)$, where $\text{GameInd}_{\mathcal{A}}$ is defined in Figure 2.

First, note that in order to win $\text{GameInd}(1^\lambda, n, \delta)$, the adversary $\mathcal{A}$ is required to output a distinguishing decoder box $\mathcal{D}$ which is a δ - $\text{Dist}^{\text{Enc}, \text{TrEnc}}$ for $\delta = \epsilon/16$, per Definition 4.3. As we consider only stateless decoders, $\mathcal{D}$ cannot rely on runtime information, such as how often a key has occurred. Furthermore, note that Π_{OSTEP} ensures that the same encryption key cannot be used twice in the same ciphertext. Consequently, $\mathcal{D}$ cannot assess the size of the keyspace from which keys are sampled.

Also, recall that the keys in K_{NORM} are distributed identically to the keys in $K_{\text{TR}}[0]$. As the preprocessing $\mathcal{K}$ is randomly permuted according to tk , an adversary who does not know tk can only guess whether a given key that revokes $\mathcal{R}_0$ either belongs to K_{NORM} or $K_{\text{TR}}[0]$. Essentially, by construction, the only difference between Enc ciphertexts and $\text{TrEnc}(\mathcal{R}_0)$ ciphertexts is that the latter does not use keys from K_{NORM} . Therefore, to break indistinguishability, $\mathcal{D}$ can only try to use some strategy that utilizes the keys in $K_{\text{NORM}} \cup K_{\text{TR}}[0]$. I.e., its behavior would depend on the occurrence of one or more of these keys in the ciphertext ct . Let us assume that the adversary chooses a subset of keys $K \subseteq K_{\text{NORM}} \cup K_{\text{TR}}[0]$ for which $\mathcal{D}$ fixes a behavior. We show that the probabilities of drawing a key from K when, in the indistinguishability game, $b = 0$ or $b = 1$ fall apart by less than $\epsilon/16$ with overwhelming probability. As a result, we get that the probability that $\mathcal{D}$ behaves significantly differently in the two cases, and as such is $\frac{\epsilon}{16} \cdot \text{Dist}^{\text{Enc}, \text{TrEnc}}$, is negligible. That is, we show that $\Pr[\mathcal{D}(\text{ct}_b) = b] < \frac{1}{2} + \frac{\epsilon}{16}$ with overwhelming probability.

To see this, consider the subset of keys K and let $L_0 = |K \cap K_{\text{NORM}}|$ and $L_1 = |K \cap K_{\text{TR}}[0]|$. Furthermore, let ct be a ciphertext that is either an Enc ciphertext or a $\text{TrEnc}(\mathcal{R}_0)$ ciphertext and let $k_1, \dots, k_{\lambda_c}$ denote the keys used to encrypt ct , that is, the j -th component of ct is encrypted under k_j . By construction, when $b = 0$ (i.e., normal encryption), for every key $1 \leq j \leq \lambda_c$, it holds that $\Pr[k_j \in K_{\text{NORM}}] = 1/2$ and $\Pr[k_j \in K_{\text{TR}}[0]] = \frac{1}{2n+1}$. When $b = 1$ (i.e., $\text{TrEnc}(\mathcal{R}_0)$), these probabilities

are 0 and $\frac{1}{2} + \frac{1}{2(n+1)}$, respectively. Let y_0, y_1 be the expected size of $K \cap \{k_1, \dots, k_{\lambda_c}\}$ when $b = 0$ and $b = 1$, respectively. For $\mathcal{D}$ to break indistinguishability, it must behave sufficiently differently for $b = 0$ and $b = 1$. Then, since the adversary cannot distinguish which keys are from L_0 and which are from L_1 , it must hold that $|y_0 - y_1| \geq \epsilon/16$ with overwhelming probability. By construction, it holds that $y_0 = \lambda_c \cdot (\frac{1}{2} \frac{L_0}{d(n+1)} + \frac{1}{2(n+1)} \frac{L_1}{d})$ and $y_1 = \lambda_c \cdot (\frac{1}{2} \frac{L_1}{d} + \frac{1}{2(n+1)} \frac{L_1}{d})$. By the requirement that $|y_0 - y_1| \geq \epsilon/16$, we get

$$|L_0 - L_1(n+1)| \geq \frac{2d\epsilon(n+1)}{16\lambda_c}. \quad (2)$$

W.l.o.g, assume that $L_0 > L_1$ and let μ denote the expected size of L_0 . Since $|K_{\text{NORM}}| = (n+1) \cdot |K_{\text{TR}}[0]|$ and $\mathcal{A}$ picks keys from $K_{\text{NORM}} \cup K_{\text{TR}}[0]$ at random, with the same distribution for K_{NORM} and $K_{\text{TR}}[0]$ (since he can not distinguish between the two sets), then the expected size of L_1 is $\frac{\mu}{n+1}$. Furthermore, it holds that $L_0 - \mu = \frac{\mu}{n+1} - L_1$. Thus, for inequality 2 to hold, it must hold that:

$$\begin{aligned} |L_0 - \mu + (L_0 - \mu)(n+1)| &\geq \frac{2d\epsilon(n+1)}{16\lambda_c} \\ |L_0 - \mu| &\geq \frac{2d\epsilon(n+1)}{16\lambda_c(n+2)}. \end{aligned}$$

By the Chernoff bound, for $\delta_c = \frac{2d\epsilon(n+1)}{16\lambda_c(n+2)\mu}$, we obtain

$$\Pr[|L_0 - \mu| > \delta_c \mu] \leq 2e^{-\mu\delta_c^2/3}. \quad (3)$$

As $\mu \leq |K_{\text{NORM}}| = d(n+1)$, it holds that for some $d = O(\frac{\lambda_c^2 \lambda n^2}{\epsilon^2})$, the probability in Equation 3 is negligible. Therefore, inequality 2 could hold only with negligible probability. Hence, constructing a distinguishing decoder $\mathcal{D}$ with advantage $\frac{\epsilon}{16}$ could happen only with negligible probability. The probability of winning the indistinguishability game is negligible for any subset K chosen by the adversary. By union bound, the probability is also negligible if the adversary chooses a polynomial number of sets K . ■

Lemma 7.4 *The Π_{OSTEP} construction, as presented in Section 7.1, is membership hiding for $\delta = \epsilon/16n$ and $\epsilon = 1/\text{poly}(\lambda)$, per Definition 4.5, for some $d = O(\lambda\lambda_c^2 n^2/\epsilon^2)$.*

Proof. Let $n = \text{poly}(\lambda)$, $\delta = \frac{\epsilon}{16n}$, and $\epsilon \in \frac{1}{\text{poly}(\lambda)}$. We show that

$$\text{Adv}_{\mathcal{A}, \Pi_{\text{OSTEP}}}^{\text{GameMH}}(1^\lambda, n, \delta) \leq \text{negl}(\lambda)$$

Our proof proceeds in two steps: First, we show that the keys in $K_{\text{TR}}[i-1]$ are indistinguishable from those in $K_{\text{TR}}[i]$ for any adversary that does not know sk_i . Intuitively, this follows from the membership hiding property of the underlying Π_{OSTE} scheme. Then, in the second step, relying on the fact that the only difference between $\text{TrEnc}(\cdot, \mathcal{R}(i-1))$ and $\text{TrEnc}(\cdot, \mathcal{R}(i))$ is that the former uses also keys from $K_{\text{TR}}[i-1]$ while the latter does not, we show that the adversary can not artificially choose a set of keys K from the set of $K_{\text{TR}}[i-1] \cup K_{\text{TR}}[i]$ that allows distinguishing between ciphertexts that use keys in $K_{\text{TR}}[i]$ from ciphertexts that do not. In combination, the indistinguishability of keys in $K_{\text{TR}}[i]$ and $K_{\text{TR}}[i-1]$ and the limited advantage an adversary can gain based on their occurrence imply membership hiding.

Proving that $K_{\text{tr}}[i-1] \approx_c K_{\text{tr}}[i]$. Ultimately, we aim to show that $\text{GameMH}^{\Pi_{\text{OSTEP}}}$ (Figure 2) is computationally close to a variant of the game, where in the preprocessing, we sample all keys in $K_{\text{TR}}[i-1]$ by revoking $\mathcal{R}_i$ instead of $\mathcal{R}_{i-1}$, i.e., equivalently to $K_{\text{TR}}[i]$. We do so through a series of game hops involving the zero-knowledge and simulation-extractability property of the proof system PS_{KGen} and the zero-knowledge property of PS_{Rand} (Definition A.1). We reduce the final step, where we substitute the keys for $K_{\text{TR}}[i-1]$, to the XDH assumption (Definition 3.2).

Before our reduction to XDH, we need to simulate the KGen proof on behalf of party i , extract the secret keys from the KGen proofs the adversary sends on behalf of all other parties, and simulate the Rand

proofs for the preprocessing keys in $K_{\text{TR}}[i-1]$. These steps are necessary because, in our reduction to XDH, we embed the XDH challenge into the public key of party i . The reduction needs to know the secret keys of the honest parties to simulate the preprocessing keys with respect to the XDH challenge, which is why we need to extract them. Further, the reduction does not know sk_i , as well as α and possibly $\bar{\alpha}_{i-1}$ for $K_{\text{TR}}[i-1]$, so these proofs must be simulated.

Let Game_0 be the membership hiding game $\text{GameMH}^{\Pi_{\text{OSTEP}}}$ for Π_{OSTEP} .

Game₁: In the first game hop, we replace the proof π_i in PS_{KGen} that the challenger attaches to the public keys and hints of the honest party with a simulated one

$$\pi_i \xleftarrow{\$} \text{PS}_{\text{KGen}}.\text{Sim}(\text{td}_{\text{KGen}}, \text{CRS}, \text{pk}_i, \text{hint}_i).$$

Claim 7.5 If the proof system PS_{KGen} is zero-knowledge (Definition A.1), then Game_0 and Game_1 are computationally indistinguishable.

Proof. The claim follows directly from the zero-knowledge property of PS_{KGen} . In particular, it holds that

$$(\text{CRS}, \text{pk}_i, \text{hint}_i, \pi_i) \approx_c (\text{CRS}, \text{pk}_i, \text{hint}_i, \text{PS}_{\text{KGen}}.\text{Sim}(\text{td}_{\text{KGen}}, \text{CRS}, \text{pk}_i, \text{hint}_i))$$

for $(\text{pk}_i, \text{hint}_i, \pi_i) \xleftarrow{\$} \text{KGen}(1^\lambda)$. ■

Game₂: In Game_2 , we use the extractor of PS_{KGen} to extract $\{\text{sk}_\ell\}_{\ell \in [n] \setminus \{i\}}$ from the proofs attached to the adversary's public keys and hints as $\text{sk}_\ell \xleftarrow{\$} \text{PS}_{\text{KGen}}.\text{Ext}(\text{td}_{\text{KGen}}, (\text{CRS}, \text{pk}_\ell, \text{hint}_\ell), \pi_\ell)$. Additionally, Game_2 aborts if there exists an $\ell \in [n] \setminus \{i\}$ such that $\text{pk}_\ell \neq [\text{sk}_\ell]_1$ or $\text{hint}_\ell \neq ([\text{sk}_\ell \tau]_1, \dots, [\text{sk}_\ell \tau^n]_1)$.

Claim 7.6 For all $n = \text{poly}(\lambda)$ and all PPT adversaries $\mathcal{A}$ there exists a PPT adversary $\mathcal{B}$ such that

$$\begin{aligned} & |\Pr[\text{Game}_{1,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1] - \Pr[\text{Game}_{2,\mathcal{B}}(1^\lambda, n, \mathcal{L}, \delta) = 1]| \\ & \leq (n-1) \mathbf{Adv}_{\mathcal{B}, \text{PS}_{\text{KGen}}}^{\text{sim-extract}}(1^\lambda). \end{aligned}$$

The proof of the above claim is identical to that in the membership hiding proof for Π_{OSTE} (Lemma 6.2).

Game₃: In Game_3 , we replace the proofs attached to re-randomizations in $K_{\text{TR}}[i-1]$ with simulated ones. In particular, we set

$$k.\pi \xleftarrow{\$} \text{PS}_{\text{Rand}}.\text{Sim}(\text{td}_{\text{Rand}}, (\text{CRS}, \{\text{pk}_\ell, \text{hint}_\ell\}_{\ell \in [n]}, k.[\alpha]_1, k.\text{ek}, k.\text{ak}))$$

for all $k \in K_{\text{TR}}[i-1]$.

Claim 7.7 For all PPT adversaries $\mathcal{A}$ there exists a PPT adversary $\mathcal{B}$ such that

$$\begin{aligned} & |\Pr[\text{Game}_{2,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1] - \Pr[\text{Game}_{3,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1]| \\ & \leq d \cdot \mathbf{Adv}_{\mathcal{B}, \text{PS}_{\text{Rand}}}^{\text{zero-knowledge}}(1^\lambda), \end{aligned}$$

where $d = N/(n+1)$.

Proof. The claim follows from a simple hybrid argument over the $N/(n+1)$ proofs in $K_{\text{TR}}[i-1]$. ■

Game₄: In Game_4 , we finally modify how the keys in $K_{\text{TR}}[i-1]$ are sampled. In particular, we sample each key in $k \in K_{\text{TR}}[i-1]$ by re-randomizing with respect to the revoked set $\mathcal{R}_i$ instead of $\mathcal{R}_{i-1}$, i.e., for all $k \in K_{\text{TR}}[i-1]$, we sample $k \xleftarrow{\$} \text{Rand}(\text{CRS}, \{\text{pk}_j, \text{hint}_j\}_{j \in [n]}, \mathcal{R}_i)$. Note that in Game_3 , the corresponding proofs are already simulated.

Claim 7.8 For all PPT adversaries $\mathcal{A}$ there exists a PPT adversary $\mathcal{B}$ and a negligible function negl such that

$$\begin{aligned} & |\Pr[\text{Game}_{3,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1] - \Pr[\text{Game}_{4,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1]| \\ & \leq d \cdot \mathbf{Adv}_{\mathcal{B}, \mathcal{G}}^{\text{XDH}}(1^\lambda) + \text{negl}(\lambda), \end{aligned}$$

where $d = N/(n+1)$.

Proof. We prove the claim through a hybrid argument. Let H_0 be Game_3 , H_d be Game_4 and H_u , for $0 < u \leq d$ be the game, where we replace the first u keys of $K_{\text{TR}}[i-1]$ with keys that revoke $\mathcal{R}_i$. We can upper-bound the success probability of any PPT adversary $\mathcal{A}$ to distinguish between Game_3 and Game_4 by that of an adversary $\mathcal{A}'$ who tries to distinguish H_{u-1} from H_u for arbitrary $0 < u \leq d$:

$$\begin{aligned} & |\Pr[\text{Game}_{3,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1] - \Pr[\text{Game}_{4,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1]| \\ & \leq d \cdot |\Pr[H_{u-1,\mathcal{A}'}(1^\lambda, n, \mathcal{L}, \delta) = 1] - \Pr[H_{u,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1]| \end{aligned}$$

To conclude the proof, we show that we can further upper bound the success probability of $\mathcal{A}'$ by reducing to XDH.

Let $\mathcal{A}'$ be a PPT adversary that distinguishes between H_{u-1} and H_u . We construct a PPT reduction $\mathcal{B}$ that internally uses $\mathcal{A}'$ to break the XDH assumption (Definition 3.2) for the pairing ensemble $\mathcal{G}$.

First, $\mathcal{B}$ receives an XDH challenge $[x]_1, [y]_1, T$, where either $T = [xy]_1$ or $T = [v]_1$ for a random $v \xleftarrow{\$} \mathbb{Z}_p^*$. The reduction $\mathcal{B}$ samples a random $\tau \xleftarrow{\$} \mathbb{Z}_p$ and generates the CRS. Then, it runs $\mathcal{A}'(1^\lambda, \text{CRS})$ to receive the party's index i . It simulates a public key pk_i and hint hint_i for i by embedding the XDH challenge as follows:

$$\text{pk}_i \leftarrow [x]_1, \text{ and } \text{hint}_i \leftarrow (\tau[x]_1, \dots, \tau^n[x]_1).$$

Observe that $\mathcal{B}$ therefore simulates $\text{sk}_i = x$ without knowledge of x . $\mathcal{B}$ further simulates a proof $\pi_i^{\text{PS}_{\text{KGen}}}$ according to both H_{u-1} and H_u and runs $\mathcal{A}'(\text{pk}_i, \text{hint}_i, \pi_i^{\text{PS}_{\text{KGen}}})$. As a response, $\mathcal{B}$ receives a revoked set $\mathcal{R}^4$ as well as the corrupted public keys and hints with attached proof $\{\text{pk}_\ell, \text{hint}_\ell, \pi_\ell^{\text{PS}_{\text{KGen}}}\}_{\ell \in [n] \setminus \{i\}}$. Again, according to both games, $\mathcal{B}$ extracts the secret keys sk_ℓ from the proofs $\pi_\ell^{\text{PS}_{\text{KGen}}}$, aborting if the extraction is unsuccessful for any ℓ (in case either game aborts, $\mathcal{B}$ outputs 0).

Next, $\mathcal{B}$ simulates the preprocessing as follows. $\mathcal{B}$ generates all preprocessing keys in K_{NORM} and $K_{\text{TR}}[j]$ for $j \neq i-1$ honestly. To generate tracing keys for $K_{\text{TR}}[i-1, q]$ where $q \in [d]$, $\mathcal{B}$ does the following:

- **For $q < u$** , generate tracing keys that revoke the set $\mathcal{R}_i$.
- **For $q > u$** , generate tracing keys that revoke the set $\mathcal{R}_{i-1}$.
- **For $q = u$** , embed the XDH challenge into the re-randomization by executing

$$([\alpha]_1, \text{ek}, \text{ak}) \leftarrow \text{SimRand}([x]_1, [y]_1, [z]_1, T, \tau, \{\text{sk}_\ell\}_{\ell \in [n] \setminus \{i\}}, \mathcal{R}_{i-1}, i),$$

where SimRand is the same simulation algorithm that we use in the membership hiding proof of Π_{OSTE} (Figure 10). $\mathcal{B}$ can also simulate the corresponding proof in PS_{Rand} , as required by both hybrids, because the statement is part of $\mathcal{L}_{\text{Rand}}$.

Let b_{XDH} be the internal bit of the XDH challenge. If $b_{\text{XDH}} = 0$, then $T = [xy]_1 \equiv [\alpha \text{sk}_i]_1$ (i.e., party i is unrevoked in $K_{\text{TR}}[i-1, u]$) and we simulate the preprocessing for H_{u-1} . If $b_{\text{XDH}} = 1$, then $T = [v]_1 \equiv [\bar{\alpha}_i \text{sk}_i]_1$ (i.e., party i is revoked in $K_{\text{TR}}[i-1, u]$) and we simulate the preprocessing for H_u .

The reduction assembles the preprocessing and sends it to $\mathcal{A}'$, receiving a decoder $\mathcal{D}$ and a message m . Finally, $\mathcal{B}$ checks if $\mathcal{A}'$ wins either game (both games have the same win condition) by checking if $\mathcal{D}$ is $\delta\text{-Dist}^{\mathcal{R}_{i-1}, \mathcal{R}_i}$ according to Definition 4.3⁵ and whether the remaining winning conditions apply. If $\mathcal{A}'$ wins, then $\mathcal{B}$ outputs 1. Otherwise, $\mathcal{B}$ outputs 0.

Further, recall that if $b_{\text{XDH}} = 0$, then $\mathcal{B}$ simulates H_{u-1} as the key for $K_{\text{TR}}[i-1, u]$ does not revoke party i . If $b_{\text{XDH}} = 1$, then $\mathcal{B}$ simulates H_u as party i is revoked in $K_{\text{TR}}[i-1, u]$. Hence, we can bound the probability of $\mathcal{A}'$ to distinguish as follows:

$$\begin{aligned} & |\Pr[H_{u-1,\mathcal{A}'}(1^\lambda, n, \mathcal{L}, \delta) = 1] - \Pr[H_{u,\mathcal{A}'}(1^\lambda, n, \mathcal{L}, \delta) = 1]| \\ & \leq |\Pr[\mathcal{B}(1^\lambda, E, [x]_1, [y]_1, [xy]_1) = 1] - \Pr[\mathcal{B}(1^\lambda, E, [x]_1, [y]_1, [v]_1) = 1]| + \text{negl}(\lambda) \\ & \leq \text{Adv}_{\mathcal{B}, \mathcal{G}}^{\text{XDH}}(1^\lambda) + \text{negl}(\lambda), \end{aligned}$$

where $E \xleftarrow{\$} \mathcal{G}(1^\lambda)$ and $x, y, v \xleftarrow{\$} \mathbb{Z}_p^*$.

⁴As Π_{OSTE} only supports the linear revocation order, it must hold that $\mathcal{R} = \mathcal{R}_{i-1}$. Otherwise, either game will output 0.

⁵Note that while $\mathcal{B}$ can not compute the exact probability, it can approximate it up to negligible error $\text{negl}(\lambda)$ using a polynomial number of queries by applying a Chernoff bound.

Overall, we can bound the success probability of any PPT adversary $\mathcal{A}$ to distinguish between Game_3 and Game_4 by

$$\begin{aligned} & |\Pr[\text{Game}_{3,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1] - \Pr[\text{Game}_{4,\mathcal{A}}(1^\lambda, n, \mathcal{L}, \delta) = 1]| \\ & \leq d \cdot \text{Adv}_{\mathcal{B},\mathcal{G}}^{\text{XDH}}(1^\lambda) + \text{negl}(\lambda), \end{aligned}$$

for some negligible function negl . $\blacksquare$

Proving the negligibility of $\text{Adv}_{\mathcal{A},\Pi_{\text{OSTEP}}}^{\text{GameMH}}$. First, we note that in order to win $\text{GameMH}^{\Pi_{\text{OSTEP}}}(\delta)$, the adversary $\mathcal{A}$ is required to return a distinguishing decoder $\mathcal{D}$ which is a δ - $\text{Dist}^{\mathcal{R},\mathcal{R} \cup \{i\}}$ (per Definition 4.3) for $\delta = \epsilon/(16n)$. Similarly to our argument in Lemma 7.3, we need only to prove that the adversary cannot find a subset of keys K from $K_{\text{TR}}[i-1] \cup K_{\text{TR}}[i]$ by which the probability of drawing a key from K when $b = 0$ in the membership hiding game falls apart from the probability when $b = 1$ by more than $\frac{\epsilon}{16n}$. To see this, let $L_{i-1} = |K \cap K_{\text{TR}}[i-1]|$ and $L_i = |K \cap K_{\text{TR}}[i]|$. Further, let ct be the challenge ciphertext that is either $\text{TrEnc}(\mathcal{R}_{i-1})$ or $\text{TrEnc}(\mathcal{R}_i)$ and let $k_1, \dots, k_{\lambda_c}$ be the encryption keys from $K_{\text{NORM}} \cup K_{\text{TR}}$ that are used in the encryption of ct . By construction, it holds that $K_{\text{TR}}[i-1] = K_{\text{TR}}[i] = d$ and when $b = 0$, i.e., ct is $\text{TrEnc}(\mathcal{R}_{i-1})$, then it holds that $\Pr[k_j \in K_{\text{TR}}[i-1]] = \frac{1}{2} + \frac{i}{2(n+1)}$ and $\Pr[k_j \in K_{\text{TR}}[i]] = \frac{1}{2(n+1)}$. When $b = 1$, i.e., $\text{ct} = \text{TrEnc}(\mathcal{R}_i)$, then these probabilities are 0 and $\frac{1}{2} + \frac{i+1}{2(n+1)}$, respectively. Let y_0, y_1 be the expected size of $K \cap \{k_1, \dots, k_{\lambda_c}\}$ when $b = 0$ and $b = 1$, respectively. Similarly to Lemma 7.3, we argue that the adversary can win the membership hiding game only if it holds that $|y_0 - y_1| > \epsilon/(16(n-t))$. This is required since, as we showed, the adversary can not distinguish between keys from L_{i-1} and keys from L_i . Thus, its strategy would depend only on the occurrence of the keys in $L_{i-1} \cup L_i$. We next show that

$$|y_0 - y_1| < \frac{\epsilon}{16n} \quad (4)$$

with overwhelming probability when we choose the appropriate d as in the lemma.

First, note that:

$$y_0 = \lambda_c \left(\left(\frac{1}{2} + \frac{i}{2(n+1)} \right) \frac{L_{i-1}}{d} + \frac{1}{2(n+1)} \frac{L_i}{d} \right)$$

and

$$y_1 = \lambda_c \left(\frac{1}{2} + \frac{i+1}{2(n+1)} \right) \frac{L_i}{d}.$$

Hence, Equation 4 would not hold if and only if

$$|L_{i-1} - L_i| \geq \frac{\epsilon d}{16n\lambda_c} \left(\frac{2(n+1)}{n+i+1} \right). \quad (5)$$

W.l.g, assume that $L_{i-1} > L_i$ and let μ denote the expected size of L_{i-1} . Since $|K_{\text{TR}}[i-1]| = |K_{\text{TR}}[i]|$ and $\mathcal{A}$ picks keys from $K_{\text{TR}}[i-1] \cup K_{\text{TR}}[i]$ at random, with the same distribution for $K_{\text{TR}}[i-1]$ and $K_{\text{TR}}[i]$, then it holds that the expected size of L_i is also μ . Therefore, $L_{i-1} - \mu = \mu - L_i$. Hence, from Equation 5, we get it must hold that

$$|L_{i-1} - \mu| > \frac{\epsilon d}{32n\lambda_c}.$$

By the Chernoff bound, for $\delta_c = \frac{\epsilon d}{32n\lambda_c\mu}$, it holds that:

$$\Pr \left[|L_{i-1} - \mu| > \frac{\epsilon d}{32n\lambda_c} \right] \leq 2e^{-\mu\delta_c^2/3},$$

which is negligible if

$$\lambda \leq \mu\delta_c^2/3 = \frac{\epsilon^2 d^2}{(32)^2 3 n^2 \lambda_c^2 \mu}.$$

Thus, since $\mu \leq d/2$, it must hold that $d = O(\frac{\lambda_c^2 \lambda n^2}{\epsilon^2})$. Given this number of keys, the probability that a subset K can influence the success probability of $\mathcal{D}$ by $\delta = \frac{\epsilon}{16n}$ is only negligible. As the adversary can try only a polynomial number of sets K , then by union bound, the probability of winning the game will be negligible. $\blacksquare$

8 The Epoch-Based OSTE Construction

Instead of pre-generating encryption and tracing keys as in Π_{OSTE} , we now elaborate on a technique to amortize ciphertext size where new keys are generated regularly and used only for a fixed time, balancing traceability guarantees with efficiency.

The Epoch Setting. We consider an encryption model that proceeds in epochs. Each epoch is identified by an ID eid , and at the beginning of each epoch, a public epoch key k_{eid} is published. Users encrypt under k_{eid} for the duration of the epoch, after which a new key $k_{\text{eid}+1}$ is published for the next epoch.

This setting directly captures our motivating use case of mempool privacy. We extend OSTE to this model by introducing the following changes:

- **Epoch Key Generation.** We define a new algorithm

$$(k_{\text{eid}}, \pi_{\text{eid}}) \xleftarrow{\$} \text{GenEpochKey}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]})$$

which generates an epoch public key k_{eid} for epoch eid as well as a proof π_{eid} .

- **Epoch Key Verification.** We define an algorithm

$$1/0 \leftarrow \text{VerifyEpochKey}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, k_{\text{eid}}, \pi_{\text{eid}})$$

to validate the epoch key before encryption.

- **Encryption.** The encryption algorithms Enc and TrEnc now take k_{eid} as input.
- **Epoch Leader Election.** We require a mechanism to select an epoch leader who generates $(k_{\text{eid}}, \pi_{\text{eid}})$.
- **Good Decoder.** The decoder $\mathcal{D}$ now additionally receives an epoch id eid , an epoch key k_{eid} , as well as a proof π_{eid} . We say that a decoder is *good* for an epoch, if it is ϵ -good when fixing the inputs eid and k_{eid} for appropriate ϵ according to Definition 5.4.

Tracing. During tracing, we simply pick an arbitrary future epoch eid that the decoder is *good* for, and run the tracing algorithm as in Figure 6. Note that we send to the decoder the modified output of TrEnc .

The Π_{Epoch} Construction. We instantiate our epoch-based construction by applying the following modifications to the baseline Π_{OSTE} .

- **GenEpochKey:** We can instantiate epoch key generation with

$$(k_{\text{eid}}, \pi_{\text{eid}}) \xleftarrow{\$} \text{RandAndProve}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \mathcal{R} := \emptyset).$$

The proof ensures semantic security even against maliciously generated epoch keys.

- **Enc:** We add eid to the ciphertext instead of $([\alpha]_1, \text{ek}, \text{ak})$, reducing the ciphertext size to be constant in n .
- **TrEnc($\mathcal{R}$):** TrEnc proceeds as in OSTE with the change that we do not include the triple $([\alpha]_1, \text{ek}, \text{ak})$ in the ciphertext, and instead we return $k_{\text{eid}} = ([\alpha]_1, \text{ek}, \text{ak})$ separately as well as a proof in PS_{Rand} .

Attacks and Mitigations. We briefly discuss the following attacks and mitigations:

Hardcoding Attacks. Let us assume for now that the epoch leader is fixed and that he colludes with a subset of shareholders to construct a decoder $\mathcal{D}$, hardcoding the list of epoch keys $\mathbf{k}_{>\text{eid}} = (k_{\text{eid}+1}, k_{\text{eid}+2}, \dots)$ that he will generate for all (or a large quantity of) future epochs. They construct the decoder such that it refuses to decrypt whenever the given epoch key k is not in this list $k \notin \mathbf{k}_{>\text{eid}}$. This decoder can clearly not be traced, since with overwhelming probability any rerandomization k_{trace} generated during tracing will not be in the list $k_{\text{trace}} \notin \mathbf{k}_{>\text{eid}}$. Hence, we propose to instantiate **EpochLeader** with a *single* leader election protocol [5] to ensure that for any given epoch, the epoch leader is dynamically selected from a

large pool of candidates, and gets revealed only in the beginning of the epoch. This mitigates the risk of colluders being chosen as epoch leaders. As a result, decoders that only work during epochs where a colluder is the epoch leader are of very limited use.

Disrupting Decryption. The proof of knowledge π_{eid} is sufficient to ensure IND-CPA security even against adversarially generated epoch keys. However, a malicious epoch leader could still disrupt correctness, for instance, to mount a denial-of-service attack. Indeed, the leader could generate an epoch key in which certain parties are revoked by setting $\mathcal{R} \neq \emptyset$ when executing **Rand** in **GenEpochKey**, as the proof of knowledge π_{eid} does not constrain the choice of $\mathcal{R}$. To mitigate this, we require the epoch leader to put up some stake at the beginning of the epoch. Then, at the end of the epoch, we force him to publish $\alpha, \bar{\alpha}$ and slash his stake if $k_{\text{eid}} \neq \text{Rand}(\text{CRS}, \{\text{pk}_i, \text{hint}_i\}_{i \in [n]}, \emptyset)$.

Properties of Π_{Epoch} . To conclude, the epoch-based scheme Π_{Epoch} achieves *constant-size ciphertext* and *t-traceability* with *public tracing* and *without trusted dealer* against decoders that are good for even a *single future epoch*.

9 Discussion and Future Work

In this paper we present two traceable threshold encryption schemes without a trusted dealer. We next discuss interesting extensions, limitations, and open problems left by our work.

Multiverse Threshold Encryption. As noted in [30], their silent threshold encryption implies a threshold encryption construction in the multiverse setting [1], where each universe is defined by a tuple (U, t_U) , with $U \subseteq [n]$ and t_U being the threshold. Only parties in U can decrypt. In the multiverse, we support arbitrary universes, so each party publishes a single (sk, pk) pair that can be used in any universe. Our constructions inherit this feature from [30]. This is another advantage over generic MPC solutions, where a party must be involved in the setup execution for each universe, while our approach only requires each party to publish a single public key (independent on the number of universe).

Broadcast Encryption. Our constructions also imply a traitor tracing scheme in the broadcast encryption setting (when $t = 1$) without a central authority, similar to [17]. We achieve constant-size ciphertexts, while they achieve $O(\sqrt{n})$ ciphertexts. However, their scheme relies on a transparent setup for generating the CRS, whereas ours, like [30], uses a structured CRS similar to the KZG polynomial commitment scheme [36]. Eliminating the need for a non-transparent CRS remains an interesting future direction.

CCA Security. Our preprocessing-based construction assumes the adversary has no access to a decryption oracle, consistent with prior work such as [12]. Traceability does not hold in this construction if the adversary can query a decryption oracle, as this may reveal information about the tracing key (e.g., which encryption keys decrypt correctly). A key open question is how to ensure traceability under such access. One possible direction is to limit the adversary to a polynomial number of decryption queries, fixed at setup, as suggested in [22].

More Traitors. Our baseline and epoch constructions catch t traitors, while the preprocessing construction catches only one. A future direction is adapting our technique to trace more than one traitor with the preprocessing construction.

Efficiency and Practicality. Our epoch-based construction is practical for use cases such as encrypted mempools, while our preprocessing-based construction serves as a proof of concept for a dealer-less scheme with constant-size ciphertexts. These results represent a significant step forward in enabling more efficient and scalable threshold encryption schemes for decentralized systems, particularly in blockchain applications. A key direction for future work is improving preprocessing efficiency, particularly by reducing its dependency on n .

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Appendices

A Preliminary Definitions

Definition A.1 (NIZK-PoK). Let R be an NP-relation of statement-witness pairs $(\chi, w) \in R$ a non-interactive zero-knowledge proof system of knowledge PS_R for R is a tuple of algorithms $\text{PS}_R = (\text{Setup}, \text{Prove}, \text{Verify}, \text{Sim}, \text{Ext})$ with the following syntax:

- $(\text{CRS}, \text{td}) \xleftarrow{\$} \text{Setup}(1^\lambda)$. Given the security parameter λ , the **Setup** algorithm outputs a common reference string CRS and a simulation and extraction trapdoor td . The CRS is implicit input to all subsequent algorithms.
- $\pi \xleftarrow{\$} \text{Prove}(\chi, w)$. Given a statement-witness pair (χ, w) , the **Prove** algorithm outputs a proof π .
- $1/0 \leftarrow \text{Verify}(\chi, \pi)$. Given a statement χ and a proof π , the **Verify** algorithm outputs 1, iff π is valid for χ .
- $\pi \xleftarrow{\$} \text{Sim}(\text{td}, \chi)$. Given the trapdoor td and a statement χ , the simulation algorithm **Sim** outputs a simulated proof π .
- $w \xleftarrow{\$} \text{Ext}(\text{td}, \chi, \pi)$. Given the trapdoor td , a statement χ and a proof π , the extractor **Ext** returns a witness w .

A proof system PS_R must fulfill the following properties.

- **Correctness**: For all $(\chi, w) \in R$ it holds that $\text{Verify}(\chi, \text{Prove}(\chi, w)) = 1$.
- **Simulation-Extractability**: For all PPT adversaries $\mathcal{A}$, negligible function negl such that

$$\Pr \left[\begin{array}{l} \text{Verify}(\chi, \pi) = 1 \wedge (\text{CRS}, \text{td}) \xleftarrow{\$} \text{Setup}(1^\lambda) \\ (\chi, w) \notin R \wedge (\chi, \pi) \xleftarrow{\$} \mathcal{A}^{\text{Sim}(\text{td}, \cdot)}(1^\lambda, \text{CRS}) \\ \chi \notin Q \quad \quad \quad w \xleftarrow{\$} \text{Ext}(\text{td}, \chi, \pi) \end{array} \right] \leq \text{negl}(\lambda),$$

where Q is the set of $\mathcal{A}$'s queries to the $\text{Sim}(\text{td}, \cdot)$ oracle.

- **Zero-Knowledge**: For all $(\chi, w) \in R$ it holds that

$$(\text{CRS}, \chi, \text{Prove}(\chi, w)) \approx_c (\text{CRS}, \chi, \text{Sim}(\text{td}, \chi)),$$

where $(\text{CRS}, \text{td}) \xleftarrow{\$} \text{Setup}(1^\lambda)$.

Definition A.2 (Independence [43]). Let $p \in \mathbb{N}$ be a prime, let $m, t_1, t_2, t_t \in \mathbb{N}$, let $\mathbf{l}_1 \in \mathbb{F}_p[X_1, \dots, X_m]^{t_1}, \mathbf{l}_2 \in \mathbb{F}_p[X_1, \dots, X_m]^{t_2}, \mathbf{l}_t \in \mathbb{F}_p[X_1, \dots, X_m]^{t_t}$ be tuples of polynomials such that $l_1[0] = l_2[0] = l_t[0] = 1$, and let $f \in \mathbb{F}[X_1, \dots, X_m]$ be a polynomial. We say that f is dependent on $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_t)$ if there exist integers $\{k_i\}_{i \in [t]}, \{k'_i\}_{i \in [t]}, \{k''_i\}_{i \in [t]}$ such that

$$f = \sum_{i,j \in [t]} k_i \cdot l_1[i] \cdot k'_j \cdot l_2[j] + \sum_{i \in [t]} k''_i \cdot t[i]$$

If f is not dependent on $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_t)$, we say that it is independent of $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_t)$.

We require the Master Theorem to prove the security of our construction(s).

Theorem A.3 (Master Theorem [30]). Let $p \in \mathbb{N}$ be a prime, let $m, t_1, t_2, t_3 \in \mathbb{N}$, let $\mathbf{l}_1 \in \mathbb{F}_p[X_1, \dots, X_m]^{t_1}, \mathbf{l}_2 \in \mathbb{F}_p[X_1, \dots, X_m]^{t_2}, \mathbf{l}_t \in \mathbb{F}_p[X_1, \dots, X_m]^{t_3}$ be tuples of polynomials of maximum degree d_r, d_s and d_t respectively. Let $f \in \mathbb{F}[X_1, \dots, X_m]$ be a polynomial of degree at most d_f . Denote $d = \max\{d_r, d_s, d_t, d_f\}$ and $t = t_1 + t_2 + t_3$. If f is independent of $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_t)$, then for any generic ppt adversary $\mathcal{A}$ that makes at most q group oracle queries

$$\left| \Pr \left[\mathcal{A} \left(\begin{array}{c} p, h_1[\mathbf{l}_1(x)] \\ h_2[\mathbf{l}_2(x)], \\ h_T[f(x)] \end{array} \right) = 1 \right] - \Pr \left[\mathcal{A} \left(\begin{array}{c} p, h_1[\mathbf{l}_1(x)] \\ h_2[\mathbf{l}_2(x)], \\ h_T[r] \end{array} \right) = 1 \right] \right| \leq \frac{(q + t + 2)^2 \cdot d}{2p}$$

where h_1, h_2 and h_T denote handles for groups $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$ and the probabilities are taken over the choices of $x \xleftarrow{\$} \mathbb{Z}_p^n$ and $r \xleftarrow{\$} \mathbb{Z}_p$.

B Full Traceability Proof for Π_{TSTE}

In the following, we prove Lemma 5.8.

Proof. Our proof takes inspiration from [14] but considers the decoder-based definitions of indistinguishability and membership-hiding. Compared to tracing schemes building on classical PLBE, adversaries against TSTE schemes can use the preprocessing output. This allows them to build decoders with a small but non-negligible advantage in the indistinguishability and membership hiding games. Consequently, our proof does not require the indistinguishability advantage δ_1 or the membership hiding advantage δ_2 to be negligible. This is a major difference to classical PLBE constructions and a requirement when building a TSTE scheme from our construction in the preprocessing model. We show that it is possible to achieve traceability nonetheless by imposing upper bounds of $\delta_1 \leq \frac{\epsilon}{16}$ and $\delta_2 \leq \frac{\epsilon}{16n}$.

Consider a decoder box $\mathcal{D}$ in the traceability game **GameTR** (Figure 5) and a fixed order of revocation of parties. Let p_i denote the advantage that $\mathcal{D}$ has in distinguishing between the encryption of two messages m_0 and m_1 for $\mathcal{R}_i$, meaning that the first i parties of the revocation order have been revoked. Let $\hat{p}_i$ denote the approximation of p_i measured by the **Trace** algorithm using W queries and let $\mathcal{T}'$ be the set returned by the algorithm. In the following, denote the number of iterations of the tracing algorithm by W . For Π_{TSTE} , we set $W = 2^7 \cdot 3n^2 \frac{\lambda}{\epsilon}$.

In any **GameTR** execution that is won by the adversary, the **Trace** algorithm either does not trace any party ($\mathcal{T}' = \emptyset$) or traces an honest party ($\mathcal{T}' \not\subseteq \mathcal{T}$). In the first case, by the conditions of the **Trace** algorithm (Figure 6) presented in Figure 6, it must hold that $\hat{p}_{i-1} - \hat{p}_i < \frac{\epsilon}{4(n-t)}$ for all $i \in [n-t]$ (Condition 1) and $\hat{p}_{n-t} < \frac{\epsilon}{4}$ (Condition 2). As a result of the small success probability remaining after $n-t$ revocation steps and the bounded difference between success probabilities in each step, the initial advantage $\hat{p}_0$ must be small. For the quality of the decoder ϵ , it holds that

$$\begin{aligned} \epsilon - \hat{p}_0 &= \epsilon - \hat{p}_{n-t} - \sum_{i=1}^{n-t} (\hat{p}_{i-1} - \hat{p}_i) \\ &> \epsilon - (n-t) \frac{\epsilon}{4(n-t)} - \frac{\epsilon}{4} > \frac{\epsilon}{4} \end{aligned}$$

If $\mathcal{D}$ wins the traceability game by framing an honest party, it either holds that $\hat{p}_{n-t} \geq \frac{\epsilon}{4}$ and one of the last t parties to be revoked is honest, or $\hat{p}_{i-1} - \hat{p}_i \geq \frac{\epsilon}{4(n-t)}$ for an honest i . If $\mathcal{D}$ is successful, at least one of the following statements must hold:

1. $\mathcal{T}' = \emptyset$ and $\epsilon - \hat{p}_0 > \frac{\epsilon}{4}$.
2. $\mathcal{T}' \not\subseteq \mathcal{T}$ and $\hat{p}_{n-t} \geq \frac{\epsilon}{4}$.
3. $\mathcal{T}' \not\subseteq \mathcal{T}$ and $\hat{p}_{i-1} - \hat{p}_i \geq \frac{\epsilon}{4(n-t)}$ for an honest i .

In the rest of the proof, we treat cases (1), (2), and (3) separately and show that they impose a contradiction to the indistinguishability, message privacy, and membership hiding properties of $\mathcal{E}$. For each case, we use the traceability adversary $\mathcal{A}$ to construct an adversary that breaks one of the properties if $\mathcal{A}$ has non-negligible advantage.

Consider case (1). Given an adversary $\mathcal{A}$ in **GameTR**, we build an adversary $\mathcal{A}_1$ playing the indistinguishability game $\text{GameInd}_{\mathcal{A}_1}^{\text{OSTE}}(1^\lambda, n, \delta_1)$ as defined in Figure 2. We then show that $\mathcal{A}_1$ breaks indistinguishability if $\mathcal{A}$ has non-negligible advantage. Given a decoder $\mathcal{D}$ output by $\mathcal{A}$, $\mathcal{A}_1$ builds a decoder box $\mathcal{D}'$ that distinguishes between **Enc** encryptions and **TrEnc** encryptions for $\mathcal{R} = \emptyset$ with advantage at least δ_1 . $\mathcal{A}_1$ proceeds as follows:

- Obtain CRS from the challenger, generate hints, relay the necessary information to $\mathcal{A}$ and obtain hints, and t in return, obtain $\mathcal{K}$ and relay it to $\mathcal{A}$.
- Sample $b \xleftarrow{\$} \{0, 1\}$.
- Obtain traceability decoder $\mathcal{D}$ and messages m_0, m_1 from $\mathcal{A}$.
- Output $(\mathcal{D}', m_b)$ where $\mathcal{D}'(\text{ct}) = \begin{cases} 0 & \text{if } \mathcal{D}(\text{ct}) = b \\ 1 & \text{else} \end{cases}$.

To analyze the success probability of $\mathcal{A}_1$, consider that $\mathcal{A}$ outputs an $\epsilon\text{-Dist}^{\mathbf{m}_0, \mathbf{m}_1}$ decoder with probability $\text{Adv}_{\mathcal{A}, \Pi_{\text{TSTE}}}^{\text{GameTR}}(1^\lambda, n, \epsilon)$. In case (1), it holds that $\epsilon - \hat{p}_0 > \frac{\epsilon}{4}$. Using the Chernoff bound, we find that

$$\Pr \left[p_0 - \hat{p}_0 > \frac{\epsilon}{8} \right] \leq 2e^{-W \frac{\epsilon^2}{8^2 \cdot 3 \cdot p_0}} \leq 2e^{-\lambda}$$

which is negligible. It holds that $\epsilon - p_0 \geq \frac{\epsilon}{8}$ with overwhelming probability. Given an **Enc** ciphertext, $\mathcal{D}'$ outputs the correct result with probability ϵ . For ciphertexts computed using **TrEnc** this probability is $(1 - p_0)$, as $\mathcal{D}'$ only outputs 1 if the result of $\mathcal{D}$ is incorrect. This results in an overall success probability of $\frac{1}{2}\epsilon + \frac{1}{2}(1 - p_0) = \frac{1}{2} + \frac{\epsilon - p_0}{2}$. With overwhelming probability, it holds that $\frac{1}{2} + \frac{\epsilon - p_0}{2} - \frac{1}{2} \geq \frac{\epsilon}{16} \geq \delta_1$. Overall, we find that:

$$\text{Adv}_{\mathcal{A}, \Pi_{\text{TSTE}}}^{\text{GameTR}}(1^\lambda, n, \epsilon) \leq \text{Adv}_{\mathcal{A}_1, \mathcal{E}}^{\text{GameInd}}(1^\lambda, n, \delta_1) + \text{negl}(\lambda)$$

We have shown by reduction that the first case contradicts the fact that $\mathcal{E}$ fulfills indistinguishability with parameter δ_1 .

Consider now case (2). Using $\mathcal{A}$ we build an adversary $\mathcal{A}_2$ playing the message privacy game $\text{GameMP}_{\mathcal{A}_2}(1^\lambda, n)$ as defined in Figure 3. We show that $\mathcal{A}_2$ breaks the message privacy of $\mathcal{E}$ if $\mathcal{A}$ has non-negligible probability. $\mathcal{A}_2$ proceeds as follows:

- Obtain CRS from the challenger, relay information to $\mathcal{A}$, and obtain $\mathcal{T}$ and hints for compromised parties in return. Obtain $\mathcal{K}$ from the challenger and relay it to $\mathcal{A}$.
- Obtain traceability decoder $\mathcal{D}$ and messages $\mathbf{m}_0, \mathbf{m}_1$ from $\mathcal{A}$.
- Sample a bit β .
- Obtain the encryption ct' of $\mathbf{m}_\beta$ from the **TrEnc** oracle. Obtain the challenge ciphertext ct .
- Return $b' \leftarrow \begin{cases} \beta & \text{if } \mathcal{D}(\text{ct}) = \mathcal{D}(\text{ct}') \\ 1 - \beta & \text{else} \end{cases}$.

In case (2), it holds that $\hat{p}_{n-t} > \frac{\epsilon}{4}$. $\mathcal{A}_2$ wins **GameMP** if it has a non-negligible advantage in distinguishing between encryptions of $\mathbf{m}_0$ and $\mathbf{m}_1$. By the Chernoff bound, it holds that

$$\Pr \left[p_{n-t} - \hat{p}_{n-t} > \frac{\epsilon}{8} \right] \leq 2e^{-W \frac{\epsilon^2}{8^2 \cdot 3 \cdot p_0}} \leq 2e^{-\lambda}$$

whereby $p_{n-t} > \frac{\epsilon}{8}$ with overwhelming probability. If $\mathcal{D}$ is a good decoder, $\mathcal{A}_2$ wins the message privacy game if both calls to $\mathcal{D}$ yield the correct result or if both are wrong. This occurs with a probability of $\left(\frac{1}{2} + \frac{\epsilon}{8}\right)^2 + \left(\frac{1}{2} - \frac{\epsilon}{8}\right)^2 = \frac{1}{2} + 2\left(\frac{\epsilon}{8}\right)^2$. As $\frac{\epsilon}{8}$ is non-negligible, this results in a non-negligible advantage of breaking message privacy. $\mathcal{A}_2$ produces a good decoder with probability $\text{Adv}_{\mathcal{A}, \Pi_{\text{TSTE}}}^{\text{GameTR}}(1^\lambda, n, \epsilon)$, which in turn is non-negligible. Note that in case (2), we assume that one of the last t parties to be revoked is honest and is framed by the adversary. Because of this, $\mathcal{A}_2$ wins **GameMP** if it sets $\mathcal{R} = [n - t]$ as less than t of its shares are used for decryption. We find that

$$\text{Adv}_{\mathcal{A}, \Pi_{\text{TSTE}}}^{\text{GameTR}}(1^\lambda, n, \epsilon) \leq \text{Adv}_{\mathcal{A}_2, \mathcal{E}}^{\text{GameMP}}(1^\lambda, n) + \text{negl}(\lambda)$$

We have shown a contradiction to the message privacy of $\mathcal{E}$ in the second case.

Finally, consider case (3). We build an adversary $\mathcal{A}_3$ from $\mathcal{A}$ that plays the membership hiding game $\text{GameMH}_{\mathcal{A}_3}^{\text{OSTE}}(1^\lambda, n, \delta_2)$, depicted in Figure 2. We then show that $\mathcal{A}_3$ breaks membership hiding if $\mathcal{A}$ has a non-negligible advantage. To do so, $\mathcal{A}_3$ produces a decoder box $\mathcal{D}'$ that is a $\delta_2\text{-Dist}^{\mathcal{R}, \mathcal{R} \cup \{i\}}$ as follows

- Obtain CRS from the challenger, relay information to $\mathcal{A}$, obtain hints for compromised parties from $\mathcal{A}$, choose i as one of the honest indices, and sample the remaining hints.
- Relay $\mathcal{K}$ to $\mathcal{A}$ and obtain $\mathcal{D}$, $\mathbf{m}_0$, $\mathbf{m}_1$ in return.
- Sample $b \xleftarrow{\$} \{0, 1\}$.

- Return $(\mathcal{D}', \mathbf{m}_b)$ where $\mathcal{D}'(\text{ct}) = \begin{cases} 0 & \text{if } \mathcal{D}(\text{ct}) = b \\ 1 & \text{else} \end{cases}$.

In case (3) it holds that $\hat{p}_{i-1} - \hat{p}_i \geq \frac{\epsilon}{4(n-t)}$ for an honest i . Using the Chernoff bound, we find that $p_{i-1} - p_i \geq \frac{\epsilon}{8(n-t)}$ with overwhelming probability, as

$$\begin{aligned} \Pr \left[p_{i-1} - p_i < \frac{\epsilon}{8(n-t)} \right] &\leq \Pr \left[p_i - \hat{p}_i \geq \frac{\epsilon}{16(n-t)} \right] \\ &\leq e^{-W \frac{2\epsilon^2}{(16(n-t))^2 \cdot 3 \cdot p_i}} \leq \text{negl}(\lambda) \end{aligned}$$

The probability that this statement holds for the same i that $\mathcal{A}_3$ chooses in **GameMH** is $\frac{1}{n-t}$. Then, $\mathcal{D}'$ returns 0 if i is not revoked with probability p_{i-1} and 1 for revoked i with probability $1 - p_i$, since it only outputs i if $\mathcal{D}'$'s output is incorrect. Overall, $\mathcal{D}'$ returns the correct result with probability $\frac{1}{2}p_{i-1} + \frac{1}{2}(1-p_i) = \frac{1}{2} + \frac{p_{i-1}-p_i}{2}$. With overwhelming probability, it holds that $\frac{1}{2} + \frac{p_{i-1}-p_i}{2} - \frac{1}{2} \geq \frac{\epsilon}{16(n-t)} \geq \delta_2$. In consequence:

$$\mathbf{Adv}_{\mathcal{A}, \Pi_{\text{TSTE}}}^{\text{GameTR}}(1^\lambda, n, \epsilon) \leq (n-t) \mathbf{Adv}_{\mathcal{A}_3, \mathcal{E}}^{\text{GameMH}}(1^\lambda, n, \delta_2) + \text{negl}(\lambda)$$

We have shown a contradiction to the membership hiding property of $\mathcal{E}$ in case (3).

Combining all three results, we find that any one of the three statements that are implied by the existence of a traceability adversary $\mathcal{A}$ with non-negligible success probability implies a contradiction to the indistinguishability, message privacy or membership hiding properties of the underlying **OSTE** scheme $\mathcal{E}$. It holds that

$$\begin{aligned} \mathbf{Adv}_{\mathcal{A}, \Pi_{\text{TSTE}}}^{\text{GameTR}}(1^\lambda, n, \epsilon) &\leq \mathbf{Adv}_{\mathcal{A}_1, \mathcal{E}}^{\text{GameInd}}(1^\lambda, n, \delta_1) + \mathbf{Adv}_{\mathcal{A}_2, \mathcal{E}}^{\text{GameMP}}(1^\lambda, n) \\ &\quad + (n-t) \mathbf{Adv}_{\mathcal{A}_3, \mathcal{E}}^{\text{GameMH}}(1^\lambda, n, \delta_2) + \text{negl}(\lambda). \end{aligned}$$

■

C Remaining linear constraints of **STE** [30]

We briefly elaborate on the remaining constraints and how Garg et al. linearly verify them.

Verifying that $\deg(Q_x) \leq |\mathbb{H}| - 2$. To comply with the sumcheck lemma (Lemma 3.1), one must ensure that Q_x has degree at most $|\mathbb{H}| - 2$. To prove that Q_x has degree at most $|\mathbb{H}| - 2$ for $|\mathbb{H}| = n + 1$ linearly, we require the aggregator to evaluate a polynomial $\hat{Q}_x(X) = Q_x(X) \cdot X$ at position τ . The corresponding equation is

$$[Q_x(\tau)]_1 \circ [\tau]_2 = [\hat{Q}_x(\tau)]_1 \circ [1]_2.$$

This ensures that Q_x is of maximum degree $|\mathbb{H}| - 2 = n - 1$ because if Q_x were of higher degree, say n , then $\hat{Q}_x$ would be of degree $n + 1$, for which the aggregator can not interpolate at position τ since the hints only reaches up to $[\text{sk}_i \tau^n]$. We expand on why this argument is sufficient in the full security proof of our construction. Before we can understand the check that the set B is authorized, we mention that we always include a dummy party at position 0 and assume that its secret key is also 0.

Forcing non-trivial witnesses. Furthermore, we force the aggregator to interpolate $B(\omega^0) = 1$ (i.e., a dummy party must be included) to ensure that B is non-zero. This requirement can be linearly verified through a KZG opening proof, i.e., we force the verifier to compute $[Q_0(\tau)]_1$ for a polynomial Q_0 such that $B(X) - 1 = Q_0(X)(X - \omega^0)$. The verification is the linear KZG verification:

$$[1]_1 \circ [B(\tau)]_2 = [Q_0(\tau)]_1 \circ [\tau - \omega^0]_2 + [1]_T$$

Ensuring that B is authorized. To ensure that B is authorized (i.e., the threshold is met), Garg et al. again use a degree check. In particular, they force the aggregator to compute $[\hat{B}(\tau)]_1$ such that $\hat{B}(X) = B(X) \cdot X^{t+1}$. The reasoning behind this is that the aggregator has to set $B(\omega^i) = 0$ for every party i that it does not know a partial decryption of, as the sumcheck equation enforces to aggregate the public keys correctly, and the signature check ensures that the aggregated signature is valid under aPK. Further, the aggregator has to set $B(\omega^0) = 1$ to compute the opening proof for the dummy party. Assuming that the aggregator has an insufficient amount of partial signatures, say $t - 1$, then B would have to be 0 in $n - (t - 1)$ positions and 1 at position ω^0 , which means that the aggregator has to interpolate B with respect to $n - t + 1$ positions. The degree of B is therefore $n - t + 1$, which means the degree of $\hat{B}$ is $n + 2$. Hence, the aggregator with insufficient partial signatures is unable to evaluate $[\hat{B}(\tau)]_1$, as the CRS only contains powers-of- τ up to $[\tau^{n+1}]_{1,2}$. An aggregator with t partial signatures, however, would have a polynomial B of degree $n - t$, which means it is able to evaluate $[\hat{B}(\tau)]_1$, as $\hat{B}$ is of degree $n + 1$. The corresponding linear verification equation is

$$[B(\tau)]_2 \circ [\tau^{t+1}]_1 = [\hat{B}(\tau)]_1 \circ [1]_2.$$

D Proof of Theorem 6.1

Next, we prove the correctness and security of our Π_{OSTE} construction (Theorem 6.1). To do so, we separately consider correctness, indistinguishability, membership hiding, message privacy, and semantic security.

D.1 Correctness of Π_{OSTE}

Lemma D.1 *Our Π_{OSTE} construction (Section 6.2) is correct according to definition 4.2.*

Proof. Consider the Π_{OSTE} scheme. It differs from the original silent setup construction by a rerandomization step of the secret keys of all parties and randomization of the secret keys of all parties outside the universe. In particular, Π_{OSTE} uses αsk_i for $\alpha \xleftarrow{\$} \mathbb{Z}_p$, instead of sk_i to aggregate key material. Notice that the aggregation key ak contains values $a_0, \dots, a_4$ that use the rerandomized instances of secret keys instead of the original keys. Therefore, any decrypting party will have access to the rerandomized ak and can compute a decryption similarly to the STE construction of [30]. Because of this, the correctness of Π_{OSTE} follows from the correctness of the original STE scheme for the rerandomized key material instead of the original sk_i . Note that our definition of correctness (Definition 4.2) includes the correctness of both Enc and TrEnc encryptions. The correctness for TrEnc ciphertexts mirrors that of Enc ciphertexts since both are constructed similarly, but TrEnc considers revoked parties not to be in the universe. If a party is revoked, any decryption attempt using the share of this party will fail. However, an adversary with sufficiently many shares can still decrypt by simply ignoring a portion of its shares. Therefore, TrEnc ciphertexts can be decrypted with the same probability as Enc ciphertexts.

The rerandomization of secret keys for parties outside the universe does not influence correctness. Because the updated secret keys are also included in the aggregation key, both terms cancel out when evaluating the first row of the matrix. ■

D.2 Indistinguishability of Π_{OSTE}

Lemma D.2 (Indistinguishability of Π_{OSTE}). *For $n = \text{poly}(\lambda)$, it holds that Π_{OSTE} is indistinguishable (Definition 4.4) for arbitrary $\delta = 1/\text{poly}(\lambda)$ where $\delta \in [-1/2, 1/2]$.*

For Π_{OSTE} , indistinguishability is trivial, as Enc and $\text{TrEnc}(\mathcal{R} = \emptyset)$ are identical.

D.3 Message Privacy of Π_{OSTE}

Lemma D.3 (Message Privacy of Π_{OSTE}). *For $n = \text{poly}(\lambda)$ and $\mathcal{X} = \mathcal{P}([n])$, Π_{OSTE} fulfills message privacy (Definition 4.6), if PS_{KGen} is zero-knowledge (Definition A.1).*

For message privacy, we prove that an adversary who corrupts less than t *unrevoked* parties cannot decrypt a ciphertext. Our proof is carried out in the GGM, using the master theorem (Theorem A.3).

Intuitively, message privacy holds since the signature verification is done against $[\alpha]_1$, and the aggregated public key $\widehat{\mathbf{aPK}}$ needs to be consistent with $\mathbf{ek}$ to fulfill the sumcheck restriction. Hence, $\widehat{\mathbf{aPK}}$ must equal $\alpha \mathbf{aPK}$ (where $\mathbf{aPK}$ is the aggregation before the randomization). Thus, $\mathbf{aPK}$ cannot use public keys not masked with α (i.e., keys of revoked parties). If the adversary tries to use such keys, it will need to find a linear combination such that the sum of these keys in $\widehat{\mathbf{aPK}}$ is zero. To find such a combination without knowing α_i , the adversary must fix points on the polynomial B , resulting in a degree larger than $n - t$. So, the degree check of B would not pass.

We note that for the Π_{OSTE} construction IND-CPA-security can be considered as a special case of message privacy, where $\mathcal{R} = \emptyset$, because we define $\text{Enc}(\cdot) = \text{TrEnc}(\cdot, \mathcal{R} = \emptyset)$ for Π_{OSTE} . In addition, the fact that the adversary is trusted for preprocessing in IND-CPA is irrelevant for Π_{OSTE} . Therefore, message privacy implies IND-CPA security of Π_{OSTE} .

In the following, we present the detailed proof of Lemma D.3.

Proof. Our proof resembles the proof presented in the original paper by Garg et al. [30]. For completeness, we give the full details of the proof, accounting for changes in the encryption scheme, the definition, and the additional group elements the adversary is presented with. We show the lemma via a reduction to the master theorem as presented in Theorem A.3. The master theorem holds in the generic group model [30], and we present a reduction to it for generic adversaries. We present a series of game hops, simplifying the message privacy game before finally reducing it to the master theorem. In particular, we consider the following games:

Game₀: We define $\text{Game}_0 \leftarrow \text{GameMP}$.

Game₁: Game_1 is equivalent to Game_0 , but the adversary does not get access to a TrEnc oracle.

Game₂: Game_2 is equivalent to Game_1 , but we replace the proofs π attached to the hints by simulated proofs as $\text{Sim}(\text{td}, \text{pk}_i, \text{hint}_i)$ where Sim is the simulator of the proof system PS , executed on a trapdoor td .

We show

$$\begin{aligned} \text{Adv}_{\mathcal{A}}^{\text{GameMP}}(1^\lambda, n) &= \text{Adv}_{\mathcal{A}}^{\text{Game}_0}(1^\lambda, n) \\ &\stackrel{D.4}{\leq} \text{Adv}_{\mathcal{A}}^{\text{Game}_1}(1^\lambda, n) + \text{negl}(\lambda) \\ &\stackrel{D.5}{\leq} \text{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, n) + \text{negl}(\lambda) \\ &\stackrel{D.6, D.7}{\leq} \text{negl}(\lambda) \end{aligned}$$

In the final step of our proof, we present our reduction to the master theorem (Lemma D.6) and prove that our parameters for the master theorem are linearly independent as per Definition A.2 (Lemma D.7). ■

Lemma D.4 *For all PPT adversaries $\mathcal{A}$, for $n = \text{poly}(\lambda)$, and considering scheme Π_{OSTE} , there exists a PPT adversary $\mathcal{B}$ such that*

$$\text{Adv}_{\mathcal{A}, \Pi_{\text{OSTE}}}^{\text{Game}_0}(1^\lambda, n) = \text{Adv}_{\mathcal{B}, \Pi_{\text{OSTE}}}^{\text{Game}_1}(1^\lambda, n)$$

Note that there is no private tracing key in the context of Π_{OSTE} . Therefore, any adversary can compute TrEnc on their own. More formally, for any adversary $\mathcal{A}$ in Game_0 , there exists an adversary $\mathcal{B}$ in Game_1 which performs the same steps as $\mathcal{A}$ but evaluates the TrEnc function on its own instead of calling the oracle. As both obtain the same result, their success probability must be the same.

Lemma D.5 *For a PPT adversary $\mathcal{A}$, for $n = \text{poly}(\lambda)$, and considering scheme Π_{OSTE} instantiated with a zero-knowledge proof system PS_{Rand} , it holds that Game_1 and Game_2 are computationally indistinguishable.*

The computational indistinguishability follows directly from the computational indistinguishability of a real and a simulated proof as stated by the zero-knowledge property of PS_{Rand} .

Lemma D.6 *Consider a generic PPT adversary $\mathcal{A}$ that makes up to q queries to the group oracle. Let $n = \text{poly}(\lambda)$ and consider construction Π_{OSTE} defined in Section 6. Let $\mathcal{R} \in \mathcal{X}$ be a set of revoked parties, and let $\mathcal{T}$ be the set of parties compromised by the adversary. We show that for each choice of $\mathcal{R}$ and $\mathcal{T}$, the following statement holds:*

$$\text{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, n) \leq \text{negl}(\lambda)$$

Proof. Consider an adversary $\mathcal{A}$ in Game_2 . Recall the master theorem as defined in Theorem A.3 for the following choice of polynomial lists:

$$\begin{aligned}\mathbf{l}_1 &= \left(\alpha, s_1 \mathbf{ek} + s_4 \tau^{t+1} + s_5, s_3 \alpha, [\tau^i]_{i \in [0, n+1]}, [H_i]_{i \in [0, n]} \right) \\ \mathbf{l}_2 &= (s_1 + s_3 \gamma, s_1 Z(\tau), s_1 \tau + s_2 \tau, s_2, s_4, s_5 Z_0, \gamma, [\tau^i]_{i \in [0, n+1]}) \\ \mathbf{l}_T &= (1) \\ f &= s_5\end{aligned}$$

where $\mathbf{l}_1, \mathbf{l}_2$ and $\mathbf{l}_T$ are lists of polynomials in $y = (\tau, \gamma, \alpha, \bar{\alpha}_1, \dots, \bar{\alpha}_n, \mathbf{sk}_1, \dots, \mathbf{sk}_n, s_1, \dots, s_5)$, $\mathcal{V} = [n] \setminus \mathcal{R}$ is the set of non-revoked (valid) parties and $\mathcal{H} = [n] \setminus \mathcal{T}$ is the set of honest parties. To simplify notation, we abbreviate the hint terms as

$$H_i = \left[[\mathbf{sk}_j \tau^i]_{j \in [n]}, [\alpha \mathbf{sk}_j \tau^i]_{j \in \mathcal{V}}, [\bar{\alpha}_j \mathbf{sk}_j \tau^i]_{j \in \mathcal{R}} \right].$$

Note that the polynomial lists implicitly also depend on $\mathcal{R}$ and $\mathcal{T}$, yet these values are known to the adversary. We consider $\mathbf{ek}$ with respect to specific $\mathcal{R}$ and $\mathcal{T}$ in our proof. We show, by reduction, that an adversary $\mathcal{A}$ in Game_2 can be used to distinguish between f and random input in the master theorem, thereby upper-bounding $\mathcal{A}$'s success probability by a negligible function. We construct a distinguisher $\mathcal{B}$ that uses $\mathcal{A}$ as a subroutine to achieve that.

First, we motivate our choices of $\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T$, and f for our instantiation of the master theorem. In the context of our reduction, $\mathcal{B}$ wants to use $\mathcal{A}$ to gain significant advantage in distinguishing $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T, f)$ from $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T, r)$ for the instantiations given above. To do so, $\mathcal{B}$ obtains a challenge and uses it to simulate Game_2 to $\mathcal{A}$ perfectly. All information that is required to do so is included in the polynomial lists. While the first polynomials in lists $\mathbf{l}_1$ and $\mathbf{l}_2$ correspond to the components of the ciphertext itself, the latter polynomials correspond to the information in the CRS, hints, and rerandomized hints.

Adversary $\mathcal{B}$ simulates Game_2 as follows: Given the challenge $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T)$, $\mathcal{B}$ relays labels for elements $[\tau^i]_1$ for $i \in [0, n+1]$ to $\mathcal{A}$ as the CRS. $\mathcal{B}$ also gives a list of labels for the public keys and hints of honest parties, $[\mathbf{sk}_i \tau^j]_{i \in \mathcal{H}, j \in [0, n]}$ to $\mathcal{A}$. As defined in Game_2 , $\mathcal{B}$ simulates the honest parties' hints, using the simulator of proof system PS_{KGen} . $\mathcal{B}$ then outputs a list of labels for the hints of corrupted parties. As $\mathcal{A}$ can control the group oracle, they can assign the labels of the adversary to the specific hints given in $\mathbf{l}_1$. Then, $\mathcal{A}$ computes the preprocessing output, including the encryption key in $\mathbf{l}_1$ and its corresponding aggregation key. Note that both these keys can be computed knowing $\mathcal{R}, \mathcal{T}$, and the elements of $\mathbf{l}_1$, as described in the Rand function (Figure 7).

Finally, after obtaining the preprocessing output, $\mathcal{A}$ returns two messages $\mathbf{m}_0, \mathbf{m}_1$ to be encrypted. $\mathcal{B}$ samples a random bit β and encrypts message $\mathbf{m}_\beta$ as $([\gamma]_2, \mathbf{ak}, s^\top \cdot A, s_5 + \mathbf{m}_\beta)$. The components of $s^\top A$ are taken from $\mathbf{l}_1$ and $\mathbf{l}_2$, $[\gamma]_2$ is taken from $\mathbf{l}_2$ and $\mathbf{ak}$ refers to the corresponding $\mathbf{ak}$ computed during preprocessing.

Finally, the reduction outputs 0 if $\mathcal{A}$ produces the correct result and 1 otherwise. As $\mathcal{B}$ perfectly simulates Game_2 to $\mathcal{A}$, $\mathcal{B}$ has at least the same success probability as $\mathcal{A}$ when distinguishing f from random. However, the master theorem states that for independent polynomial lists, a polynomial number of group oracle queries, polynomial size, and order of the polynomial lists, the advantage of any adversary in distinguishing the two cases should only be negligible. We show that the polynomial lists $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T)$ are independent in Lemma D.7. This implies that $\text{Adv}_{\mathcal{A}, \Pi_{\text{OSTE}}}^{\text{Game}_2}(1^\lambda, n)$ can only be negligible. ■

Lemma D.7 *Let*

$$\begin{aligned}\mathbf{l}_1 &= \left(\alpha, s_1 \mathbf{ek} + s_4 \tau^{t+1} + s_5, s_3 \alpha, [\tau^i]_{i \in [0, n+1]}, [H_i]_{i \in [0, n]} \right) \\ \mathbf{l}_2 &= (s_1 + s_3 \gamma, s_1 Z(\tau), s_1 \tau + s_2 \tau, s_2, s_4, s_5 Z_0, \gamma, [\tau^i]_{i \in [0, n+1]}) \\ \mathbf{l}_T &= (1) \\ f &= s_5\end{aligned}$$

be lists of polynomials in $\mathbb{Z}_p$ where

$$H_i = \left[[\mathbf{sk}_j \tau^i]_{j \in [n]}, [\alpha \mathbf{sk}_j \tau^i]_{j \in \mathcal{V}}, [\bar{\alpha}_j \mathbf{sk}_j \tau^i]_{j \in \mathcal{R}} \right].$$

It holds that $\mathbf{l}_1, \mathbf{l}_2$ and $\mathbf{l}_T$ are independent of f , according to Definition A.2.

Proof. This proof proceeds similarly to the independence proof for the original silent setup scheme in [30]. Towards a contradiction, we assume that a linear combination of group elements exists that allows the adversary to compute $f = s_5$. We then extract requirements on the values of some of the combinations' coefficients to reach a contradiction to the univariate sumcheck, as detailed in Lemma 3.1. In detail, this involves using partial derivatives to extract the five equations detailed in Section 6.1 from the linear combination that need to be fulfilled for a correct decryption.

Intuitively, these equations imply an upper bound on the degree of polynomial B that an adversary multiplies with the first component of the ciphertext. This limitation expresses the lower bound of t parties required for decryption. At the same time, the univariate sumcheck, in combination with the five equations, imposes a lower bound on the degree of B , expressing the constraint of the number of valid shares the adversary actually has. The lower bound is larger than the upper bound, which yields a contradiction.

More formally, let $\mathbf{1}^n$ denote the n -dimensional unit vector and let $\mathbf{1} = \mathbf{1}^{3+(2n+1)(n+1)}$ and $\mathbf{1}' = \mathbf{1}^{7+(n+1)}$. Furthermore, let $\odot$ denote the Hadamard product (point-wise multiplication) for matrices and let $a_{i,*}$ and $a_{*,j}$ denote the row i and column j of a matrix A , respectively. Assume towards contradiction that the polynomial lists $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T, \mathbf{f})$ are linearly dependent. Then there exist coefficients $K = (k_{i,j}) \in (\mathbb{Z}_p^{3+(2n+1)(n+1)} \times \mathbb{Z}_p^{7+(n+1)})$, $k_T \in \mathbb{Z}_p$, such that

$$s_5 = \mathbf{1}(K \odot \mathbf{l}_1^\top \cdot \mathbf{l}_2) \mathbf{1}'^\top + k_T \cdot \mathbf{l}_T$$

This linear combination includes all possible combinations of group elements in $\mathbf{l}_1$ and $\mathbf{l}_2$ that might be used to compute s_5 . Consider now the differential:

$$\frac{\partial}{\partial s_i} s_5 = \frac{\partial}{\partial s_i} \mathbf{1}(K \odot \mathbf{l}_1^\top \cdot \mathbf{l}_2) \mathbf{1}'^\top + k_T \cdot \mathbf{l}_T$$

for $i \in [1, \dots, 5]$. Differentiating with respect to $s_1, \dots, s_5$ eliminates all terms that do not contain the variable, simplifying the expression to the following five equations (compare Section 6.1):

$$0 = k_{2,*} \mathbf{l}_2^\top \mathbf{e} k + k_{*,1} \mathbf{l}_1^\top + k_{*,2} \mathbf{l}_1^\top Z(\tau) + k_{*,3} \mathbf{l}_1^\top \tau \quad (1)$$

$$0 = k_{*,3} \mathbf{l}_1^\top \tau + k_{*,4} \mathbf{l}_1^\top \quad (2)$$

$$0 = k_{*,1} \mathbf{l}_1^\top \gamma + k_{3,*} \mathbf{l}_2^\top \alpha \quad (3)$$

$$0 = k_{2,*} \mathbf{l}_2^\top \tau^{t+1} + k_{*,5} \mathbf{l}_1^\top \quad (4)$$

$$1 = k_{2,*} \mathbf{l}_2^\top + k_{*,6} \mathbf{l}_1^\top Z_0 \quad (5)$$

The adversary can compute s_5 only if there are coefficients such that all the above equations are fulfilled. Observe that $k_{i,j} = 0$ for all summands that consist of a product of variables unique in their equation. For instance, $k_{i,j} = 0$ for a term $k_{i,j} \alpha \gamma \tau$ if $\alpha \gamma \tau$ appears nowhere else in the equation. Such unique combinations of variables cannot be expressed as a linear combination of other summands. The same might be observed by considering the derivative of the equation in all combination variables.

With this observation, we impose constraints on the coefficient vectors, eliminating terms from the equations one after the other. For each equation, we mark summands that are eliminated by constraints of the observed equation in **blue** and eliminations by constraints of previously observed equations in **grey**. Expanding equation (2), we find:

$$0 = k_{*,3} \tau \left(\begin{array}{c} \alpha, \text{blue } s_1 \mathbf{e} k + s_4 \tau^{t+1} + s_5, \text{blue } s_3 \alpha, [\tau^i]_{i \in [0, \text{grey } n-1]}, [H_i]_{i \in [0, \text{grey } n-1]} \end{array} \right)^\top + k_{*,4} \left(\begin{array}{c} \alpha, \text{blue } s_1 \mathbf{e} k + s_4 \tau^{t+1} + s_5, \text{blue } s_3 \alpha, [\tau^i]_{i \in [0, n+1]}, [H_i]_{i \in [0, n]} \end{array} \right)^\top,$$

where $k_{*,3} \mathbf{l}_1^\top$ can only contain factors of τ^i where $i < n$. Summands that contain τ^{n+1} are unique in this equation and must have a coefficient of 0. This captures the degree-check property this equation aims to enforce. Expanding equation (3) yields:

$$0 = k_{*,1} \gamma \left(\begin{array}{c} \alpha, \text{blue } s_1 \mathbf{e} k + s_4 \tau^{t+1} + s_5, \text{blue } s_3 \alpha, [\tau^i]_{i \in [0, n+1]}, [H_i]_{i \in [0, n]} \end{array} \right)^\top + k_{3,*} \alpha \left(\begin{array}{c} s_1 + s_3 \gamma, s_1 Z(\tau), s_1 \tau + s_2 \tau, s_2, s_4, \text{blue } s_5 Z_0, \gamma, [\tau^i]_{i \in [0, n+1]} \end{array} \right)^\top.$$

An adversary can only use terms containing $\gamma\alpha$ and constants to solve this equation. This captures the validity of the aggregated signature. Notably, they cannot validate an aggregated signature using the public keys and hints of parties they do not control. From equation (4), it follows that:

$$0 = k_{2,*}\tau^{t+1} \left(\begin{array}{c} s_1 + s_3\gamma, s_1Z(\tau), s_1\tau + s_2\tau, s_2, s_4, s_5Z_0, \gamma, [\tau^i]_{i \in [0, \frac{n-t}{n+1}]} \end{array} \right)^\top \\ + k_{*,5} \left(\begin{array}{c} \alpha, s_1\mathbf{ek} + s_4\tau^{t+1} + s_5, s_3\alpha, [\tau^i]_{i \in [0, n+1]}, [H_i]_{i \in [0, n]} \end{array} \right),$$

again limiting the maximum degree of τ^i terms that may be used to fulfill this equation. In particular, $k_{2,*}\mathbf{l}_2^\top$ cannot contain τ^i terms for $i > n - t$, enforcing a degree check on $k_{2,*}\mathbf{l}_2^\top$. This ensures that an adversary must use at least t shares to decrypt correctly.

By equation (5) it holds that $1 = k_{2,*}\mathbf{l}_2^\top + k_{*,6}(\tau - 1)\mathbf{l}_1^\top$. This implies that $k_{2,*}\mathbf{l}_2^\top = 1$ if $\tau = 1$. Therefore, $k_{2,*}\mathbf{l}_2^\top$ cannot be the zero polynomial. The same may be observed by expanding the terms of the equation. Finally, it holds for equation (1), that:

$$0 = k_{2,*}\mathbf{ek} \left(\begin{array}{c} s_1 + s_3\gamma, s_1Z(\tau), s_1\tau + s_2\tau, s_2, s_4, s_5Z_0, \gamma, [\tau^i]_{i \in [0, \frac{n-t}{n+1}]} \end{array} \right)^\top \\ + k_{*,1} \left(\begin{array}{c} \alpha, s_1\mathbf{ek} + s_4\tau^{t+1} + s_5, s_3\alpha, [\tau^i]_{i \in [0, n+1]}, [H_i]_{i \in [0, n]} \end{array} \right)^\top \\ + k_{*,2}Z(\tau) \left(\begin{array}{c} \alpha, s_1\mathbf{ek} + s_4\tau^{t+1} + s_5, s_3\alpha, [\tau^i]_{i \in [0, n+1]}, [H_i]_{i \in [0, n]} \end{array} \right)^\top \\ + k_{*,3}\tau \left(\begin{array}{c} \alpha, s_1\mathbf{ek} + s_4\tau^{t+1} + s_5, s_3\alpha, [\tau^i]_{i \in [0, n]}, [H_i]_{i \in [0, n-1]} \end{array} \right)^\top \\ = \mathbf{ek}g_1 + g_2 + g_3Z(\tau) + g_4\tau$$

where we define g_1, g_2, g_3 and g_4 such that the equation above holds. Note that our constraints imply, as discussed above, that $\deg(g_1) \leq n - t$. Consider that by definition

$$\mathbf{ek} = \sum_{i \in [n] \setminus (T \cup \mathcal{R})} \alpha \mathbf{sk}_i L_i(\tau) + \sum_{i \in \mathcal{R} \setminus \mathcal{T}} \bar{\alpha}_i \mathbf{sk}_i L_i(\tau) \\ + \sum_{i \in \mathcal{T} \cup \{0\} \setminus \mathcal{R}} \alpha \kappa_i L_i(\tau) + \sum_{i \in \mathcal{T} \cap \mathcal{R}} \bar{\alpha}_i \kappa_i L_i(\tau),$$

where $\mathbf{sk}_i, \alpha$, and $\bar{\alpha}_j$ for $j \in [n]$ are variables in $\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T$ and f and κ_i for $i \in \mathcal{T}$ are keys known to the adversary (constants in $\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T$). Now, consider the Lagrange basis form of polynomial $g_1(y) = \sum_{i \in [n]} b_i L_i(\tau)$, where b_i denotes the function value of g_1 at positions ω^i . The univariate sumcheck in the context of equation (1) equates to:

$$\mathbf{ek} \cdot g_1 \stackrel{3.1}{=} \sum_{i \in [n] \setminus (T \cup \mathcal{R})} \alpha \mathbf{sk}_i b_i + \sum_{i \in \mathcal{R} \setminus \mathcal{T}} \bar{\alpha}_i \mathbf{sk}_i b_i + \sum_{i \in \mathcal{T} \cup \{0\} \setminus \mathcal{R}} \alpha \kappa_i b_i + \sum_{i \in \mathcal{T} \cap \mathcal{R}} \bar{\alpha}_i \kappa_i b_i \\ + Q_x(\tau) \cdot \tau + Q_Z(\tau) \cdot Z(\tau).$$

By equation (1), it also holds that:

$$\stackrel{(1)}{=} - (g_2 + g_3Z(\tau) + g_4\tau)$$

Recall that all unique combinations of variables must have a coefficient of 0 and $Z(\tau) = \tau^{n+1} - 1$. As all hint terms in g_4 are of degree at most $n - 1$ and $Q_x(\tau)$ is of the same degree by definition, the only polynomials of containing hint terms of degree $n + 1$ or larger are $Q_Z(\tau)Z(\tau)$ and $g_3Z(\tau)$. Assume that polynomial $(g_3 - Q_Z(\tau))(\tau^{n+1} - 1)$ contains a term involving $\mathbf{sk}_i$ for some $i \in [n]$, and a factor of τ^j where $j \leq n$. Then the equation must include a term involving the same $\mathbf{sk}_i$ with a factor of τ^{j+n+1} and inverted coefficient. As $(g_3 - Q_Z(\tau))(\tau^{n+1} - 1)$ is the only polynomial with τ coefficients of degree larger than n this leads to a contradiction. Thereby, $(g_3 - Q_Z(\tau))(\tau^{n+1} - 1)$ cannot contain terms involving $\mathbf{sk}_i$.

In extension, $(g_4 - Q_X(\tau))\tau$ cannot contain terms involving sk_i , as $g_4 - Q_X(\tau)$ cannot be divisible by τ and terms involving sk_i and τ^j for $j > 0$ are unique.

As a result, terms αsk_i , $\bar{\alpha}_i \text{sk}_i$, and $\bar{\alpha}_i \kappa_i$ can appear only in the first two as well as the fourth term originating from the sumcheck. It follows that $b_i = 0$ for each $i \in [n] \setminus (\mathcal{T} \setminus \mathcal{R})$. Given its Lagrange basis form, g_1 must have at least $n - |\mathcal{T} \setminus \mathcal{R}|$ roots. As g_1 cannot be the zero polynomial, this implies that $\deg(g_1) \geq n - |\mathcal{T} \setminus \mathcal{R}|$. Since $|\mathcal{T} \setminus \mathcal{R}| < t$ by virtue of the message privacy game, it holds that $\deg(g_1) > n - t$. However, by the constraints imposed by the equation system, it holds that $n - t \geq \deg(g_1)$. We arrive at a contradiction. Therefore, $(f, \mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_T)$ must be linearly independent. ■

E Message Privacy and Security for Π_{OSTEP}

Lemma E.1 *Consider an arbitrary PPT adversary $\mathcal{A}$ and let $n = \text{poly}(\lambda)$. If PS_{Rand} is zero-knowledge, Π_{OSTEP} as presented in Section 7 fulfills message privacy.*

Observe that the differences between Π_{OSTE} and Π_{OSTEP} are the added preprocessing and the restriction of the keyspace to the pre-generated keys. However, the challenger is responsible for preprocessing in GameMP . Message privacy can, therefore, be proven in exactly the same fashion as for Π_{OSTE} . The lemma follows immediately from Lemma D.3.

Lemma E.2 *Consider an arbitrary PPT adversary $\mathcal{A}$ and let $n = \text{poly}(\lambda)$. If PS_{Rand} is sound and zero-knowledge, Π_{OSTEP} as presented in Section 7 is IND-CPA secure.*

In contrast to Π_{OSTE} , IND-CPA security of Π_{OSTEP} does not follow immediately from its message privacy. In IND-CPA, the adversary is trusted with computing the preprocessing output. For Π_{OSTEP} , this implies that the rerandomization values α and $\bar{\alpha}_j$ are known to the adversary. In the following, we present a proof sketch for IND-CPA security of construction Π_{OSTEP} .

IND-CPA follows similarly to the message privacy proof by a reduction to the master theorem. Since the values of α and $\bar{\alpha}_j$ are now known to the adversary, the polynomial lists of the master theorem simplify to

$$\begin{aligned} \mathbf{l}_1 &= (s_1 \text{ek} + s_4 \tau^{t+1} + s_5, s_3, [\tau^i]_{i \in [0, n+1]}, [\text{sk}_j \tau^i]_{i, j \in [n]}) \\ \mathbf{l}_2 &= (s_1 + s_3 \gamma, s_1 Z(\tau), s_1 \tau + s_2 \tau, s_2, s_4, s_5 Z_0, \gamma, [\tau^i]_{i \in [0, n+1]}) \\ \mathbf{l}_T &= (1) \\ f &= s_5. \end{aligned}$$

Similarly to the message privacy proof, we present two game hops before reducing the final game hop to the master theorem.

Game₀: We define $\text{Game}_0 \leftarrow \text{IND-CPA}$.

Game₁: Game_1 is equivalent to Game_0 , but the adversary merely chooses α , $\{\bar{\alpha}_j\}_{j \in \mathcal{R}}$ and sends them to the challenger who instead computes the preprocessing output.

Game₂: Game_2 is equivalent to Game_1 , but we replace the proofs π attached to the hints by simulated proofs as $\text{PS}_{\text{KGen}}.\text{Sim}(\text{td}, \text{pk}_i, \text{hint}_i)$ where Sim is the simulator of the proof system PS_{KGen} , executed on a trapdoor td .

We then show that

$$\text{Adv}_{\mathcal{A}}^{\text{Game}_0}(1^\lambda, n) \leq \text{Adv}_{\mathcal{A}}^{\text{Game}_1}(1^\lambda, n) + \text{negl}(\lambda)$$

by a reduction to the soundness of proof system PS_{Rand} (soundness is implied by simulation-extractability). As long as PS_{Rand} is sound, an adversary is forced to compute the preprocessing honestly, only learning $\alpha, \{\bar{\alpha}_j\}_{j \in \mathcal{R}}$ in the process.

Further, we prove that

$$\text{Adv}_{\mathcal{A}}^{\text{Game}_1}(1^\lambda, n) \leq \text{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, n) + \text{negl}(\lambda)$$

by a reduction to the zero-knowledge property of PS_{Rand} . This reduction is equivalent to the proof of Lemma D.5.

Finally, we prove that

$$\mathbf{Adv}_{\mathcal{A}}^{\text{Game}_2}(1^\lambda, n) \leq \text{negl}(\lambda)$$

by a reduction to the master theorem. The reduction is built in exactly the same way as for the proof of Lemma D.6, but using the modified polynomial lists above. Since the reduction has knowledge of α and $\{\bar{\alpha}_j\}_{j \in \mathcal{R}}$, it can simulate Game_2 to $\mathcal{A}$ in the same way as for message privacy. Using the master theorem requires the polynomial lists $(\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_t, f)$ to be independent. As the polynomial lists used in this proof consider the rerandomization to be known, they are the same used in the security proof of the original STE construction of [30]. The independence of these lists of polynomials has already been proven in [30], concluding our proof.